FUNCTIONAL MATHEMATICS: AN INTERACTIVE LEARNING AID FOR STUDENTS WITH SPECIAL NEEDS

A Capstone Project presented to the Faculty, College of Computer Studies
Laguna State Polytechnic University
Los Baños Campus
Los Baños, Laguna

In partial fulfillment of the requirements for the degree
Bachelor of Science in Information Technology
Specialized in Animation and Motion Graphics

JOHN REI L. DOMALANTA
AUBREY MAE R. SUAREZ
ILEANA MARIE P. OLAN

MERARDO ARENAS CAMBA, JR.

JUNE 2026

APPROVAL SHEET

The Capstone Project entitled **"FUNCTIONAL MATHEMATICS: AN INTERACTIVE LEARNING AID FOR STUDENTS WITH SPECIAL NEEDS"** prepared and submitted by **John Rei L. Domalanta, Aubrey Mae R. Suarez, and Ileana Marie P. Olan** in partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology specialized in Animation and Motion Graphics, is hereby recommended for approval and acceptance.

MERARDO ARENAS CAMBA, JR.
Adviser

Approved and Accepted by the Committee on Oral Examination with a grade of ____.

ROWAN N. ELOMINA
Member

SHERWIN B. SAPIN
Member

MYRA V. PAICAGLINO
Member

JEFFERSON L. LERIOS
Member

Accepted in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology specialized in Animation and Motion Graphics.

LOYD S. ECHALAR
Associate Dean, College of Computer Studies

OFELIA B. MANINGAS, Ed.D.
Chairperson, Research and Development Office

_____
Date Signed

RESEARCH CONTRIBUTION NO.	

ACKNOWLEDGMENT

The researchers would like to express their gratitude to all the people whose contributions have been invaluable to the completion of this project.

First and foremost, the researchers are deeply thankful to their families for providing unending support and offering encouragement. The love and faith of these families have been a constant source of strength and motivation.

To the adviser, Merardo Arenas Camba, Jr., the researchers are grateful for the guidance, patience, and encouragement provided throughout this journey. This mentorship and the efforts to enhance the study helped the team to complete this project.

To the panelists, Myra V. Paicaglino, Sherwin B. Sapin, and Rowan N. Elomina, and to the research coordinators, Jonardo R. Asor and Jefferson L. Lerios, the researchers extend appreciation for the time, expertise, and constructive feedback provided, which helped to strengthen the project.

The researchers are grateful to Los Baños National High School-Poblacion for the permission to conduct this study. Deep appreciation is extended to Rhea P. Barboza for the valuable insights and support provided in shaping the project, and to the SNED students whose participation was vital to the accomplishment of this research.

And lastly, the researchers thank the Almighty God for granting protection, strength, and wisdom that guided the team along the way.

THE RESEARCHERS

DEDICATION

This work is wholeheartedly dedicated to Almighty God, whose strength, wisdom, and guidance have been the researchers' constant support throughout this journey.

To researchers' beloved parents and grandparents:

Mr. Renato V. and Mrs. Jean L. Domalanta

Mr. Angelito and Mrs. Maura R. Suarez

Mr. Rommel and Dr. Ma. Jonelea P. Olan, Mrs. Josefina G. Padilla

Their unwavering love, encouragement, and support have been the researchers' greatest source of strength. This accomplishment would not have been possible without your presence and faith in them.

To researchers' cherished pets:

Awrei, Poochi, Pipper, Amery, Shiro, Ellie, Nougat, and Toffee

Whose joy, comfort, and companionship have lightened the researchers' burdens and brought happiness throughout this journey.

To the researchers' relatives and friends, whose encouragement and support have uplifted them during the academic challenges, and to all those who have offered guidance, kindness, and inspiration, your presence has been an invaluable part of this achievement.

J.R.L.D

A.M.R.S

I.M.P.O

ABSTRACT

This study encompasses the development of a desktop-based web application that will help students develop a deeper understanding of the subject Functional Mathematics through visual learning and interactivity. The project aims to address the learning difficulties of students enrolled in the Special Needs Education (SNED) program by incorporating interactive and accessible features, along with animated skits, into a platform aligned with the Functional Mathematics curriculum. The assessment was conducted using two frameworks which are: the Technology Acceptance Model (TAM) for the evaluation of user acceptability, perceived usefulness, and participation, and the ISO 25010 Software Quality Model for the assessment of system characteristics like functionality, usability, and reliability. In general, the results from teachers and students obtained an acceptability rate of 3.76, which is verbally interpreted as "highly acceptable," meaning the learning aid is both effective and easy to use by the target users. These findings confirm that integrating animation and interactivity enhances comprehension, focus, and retention among learners. This study concludes that the learning aid satisfies educational and software quality standards and is therefore suitable for SNED instruction. It is recommended that future enhancements include extending accessibility for other types of disabilities and consideration of its potential as a standalone mobile application.

Keywords: *Functional Mathematics, Special Needs Education (SNED), Interactive Learning Aid, Animation and Motion Graphics, Speech Recognition, Click-to-Answer, Web-Based Application, Technology Acceptance Model (TAM), ISO 25010*

TABLE OF CONTENTS

Preliminaries	***Page***
Title Page	i
Approval Sheet	ii
Acknowledgement	iii
Dedication	iv
Abstract	v
Table of Contents	vi
List of Tables	vii
List of Figures	viii
Definition of Terms	ix
CHAPTER	
I INTRODUCTION	
Project Context	1
Project Objectives	3
Project Purpose	3
Scope and Limitation of the Study	5
II THEORETICAL FRAMEWORK OF THE PROJECT	
Review of Related Literature, Studies, and Systems	7
Teaching Philosophy for Special Needs Education	7
Classification of Students with Special Needs	8
Teaching Mathematics to Students with Special Needs	10
Numerical Comprehension of Students with Special Needs	11
Implementing Interactive Learning Aids in Special Education	12
Impact of Interactive Learning Aids on Cognitive and Behavioral Growth	13
Synthesis and Relevance of the Study	13
Literature Matrix	14
Conceptual Framework	16
III METHODOLOGY	
Project Design	19
Software Development Life Cycle	19
Planning	20
Defining	21
Designing	21
Building	26
Testing	27
Deployment	28
Materials	29
Hardware Specification	30
Software Specification	31
Evaluation Procedure	32
Sampling Design	32
Data Gathering Instrument and Validation	33

Population of the Study 36
Statistical Treatment of Data 36
Ethical Consideration 37

IV RESULTS AND DISCUSSIONS
Results by Objectives of the Study 39
Animated Content Effectiveness Evaluation 43
Software Quality Evaluation 52

V SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS
Summary of Findings 64
Conclusions 66
Recommendations 68

REFERENCES
APPENDICES
Appendix A. Interface and Storyboard
Appendix B. Character Design
Appendix C. Scripts, Curriculum-Based Lesson Plan and Learning Materials
Appendix D. Animation and Interactive Design
Appendix E. Approval Letters and Parent Consent
Appendix F. Survey Questionnaire
Appendix G. Level of Acceptance
Appendix H. Publishable Paper
Appendix I. Student Curriculum Vitae
Appendix J. Grammar Rate
Appendix K. Similarity Report

LIST OF TABLES

No.	Title	Page
1	Matrix of Related Literature	15
2	Hardware Specification of Laptop 1	30
3	Hardware Specification of Laptop 2	30
4	Hardware Specification of Pen Tablet	31
5	Software Specification	32
6	Likert Scale	36
7	Reliability Rest Results for TAM	37
8	Reliability Rest Results for ISO 25010	37
9	Perceived Usefulness based on Respondents' Response	44
10	Perceived Ease of Use based on Respondents' Response	45
11	Attitude toward Using based on Respondents' Response	46
12	Behavioral Intention to Use based on Respondents' Response	47
13	Perceived Cost based on Respondents' Response	49
14	Information Awareness based on Respondents' Response	50
15	Social Influence based on Respondents' Response	51
16	Summary Mean Table of TAM	52

17	Functional Suitability based on Respondents' Response	53
18	Performance Efficiency based on Respondents' Response	54
19	Compatibility based on Respondents' Response	55
20	Interaction Capability based on Respondents' Response	56
21	Reliability based on Respondents' Response	57
22	Security based on Respondents' Response	58
23	Maintainability based on Respondents' Response	59
24	Flexibility based on Respondents' Response	61
25	Safety based on Respondents' Response	62
26	Summary Mean Table of ISO 25010	63

LIST OF FIGURES

No.	Title	Page
1	Conceptual Framework	18
2	Software Development Life Cycle	20
3	Sample Character Design	22
4	Sample Interface	24
5	Sample Storyboard	24
6	Sample Animation Process	25
7	Interactive Features	26
8	Materials Used	29
9	Technology Acceptance Model	34
10	ISO 25010 Software Quality Model	35
11	Sample Activities from Curriculum-Based Lesson Plans	40
12	Sample Interface of Interactive Learning Aid	42

DEFINITION OF TERMS

Adobe Animate	A software application used to design vector graphics and animation. In this study, it is utilized as a tool for creating animated visual elements, as well as 2D animated short skits, to guarantee dynamic graphic presentation and encourage learner interaction.
Adobe Photoshop	It is a digitally based software used for modifying, creating illustrations, images, and other graphic elements. In this learning aid, it was primarily utilized for character designing and for the creation of character puppets that were later integrated into Adobe Animate for streamlined animation purposes.
Adobe Illustrator	It is a vector-based graphic design software that has been used in this project to create the visual elements in the interactive learning aid, which are the icons, buttons, and individual clipart required for this project for activities and other instructional materials.
Animation and Motion Graphics	A technique in visual communication through the means of moving illustrations, text, and graphical components that deliver content interestingly and dynamically. In this study, it was integrated to simplify the explanation of mathematical concepts and create skits that facilitate the understanding and retention of topics among students in Special Needs Education.
Attitude Towards Using	A component of the Technology Acceptance Model (TAM) that reflects students' perception regarding the enjoyment, engagement, and benefit of the interactive learning tool in terms of learning Functional Mathematics.
Behavioral Intention to Use	Regarding the Technology Acceptance Model (TAM), it is an individual's propensity or preference for accepting and retaining the usage of a technology in the future. It is significantly established based on the perceived advantage as well as perceived ease in utilizing it, and therefore, it is a crucial indicator in ascertaining the adoption of the interactive learning tool.

CapCut	An integrated video editing software application that assists the users in developing and editing videos and other graphics. In this study, it was primarily utilized in compilation of animation sequences, integration of sound designs, and performing other necessary editing needed to finalize the multimedia output and achieve the overall cohesiveness of the animated contents.
Click-to-Answer	An interactive feature through which students can mark their responses by just clicking the available answer on the screen. By this method integrated in the project, the mode of answering is simplified, the mental workload is reduced, and the accessibility of the system for students with special needs is promoted through a clear and direct interaction.
Differentiated Instruction	A teaching methodology that adapts and adjusts teaching approach, as well as materials and activities, to cater to the varying learning needs, skills, and preferences of the students. In the context of this interactive learning aid, it ensures that the whole project is accessible and provides diverse content suited for the needs and abilities of students under Special Needs Education (SNED) at different competency levels, which are beginner, proficient, and advanced.
Drag-and-Drop	An interactive functionality where students may select and relocate items from one part of the interface to another. In this project, students can engage with activities of matching, ordering, and sorting of numeric objects or values, and hence enhance the visual and real-life understanding of mathematical concepts.
Functional Mathematics	It is a branch of mathematics that applies mathematical concepts and principles in everyday life. It gives special emphasis on key competencies that include counting and recognizing numbers, basic operations, and recognizing money, enabling and empowering the students under Special Needs Education (SNED) facilitates and enables them to use mathematical abilities in their daily activities, problem-solving, and decision-making.

Information Awareness	An application of the Technology Acceptance Model (TAM) pertaining to students' knowledge and awareness regarding the purpose, nature, and advantage of the interactive learning tool in enhancing their skills in Functional Mathematics based on the Special Needs Education (SNED) program.
Interactive Learning Aid	It is a digital learning tool designed to ensure active participation of students by blending instructional content with interactive components. In this study, the interactive learning aid utilizes animation and motion graphics to enhance the understanding and engagement of the students in learning Functional Mathematics.
ISO 25010	An internationally accepted software quality standard that outlines the characteristics and evaluation criteria of relevance to a system, which includes functionality, reliability, usability, performance efficiency, maintainability, portability, security, and compatibility. In the context of this study, ISO 25010 was used as a framework or model of evaluation of the interactive learning aid to test its overall quality and effectiveness.
Perceived Cost	It is a part of the Technology Acceptance Model (TAM) that corresponds to the assessment of the user's costs associated with accessing and using the interactive learning aid in the areas of finance, time, and effort. It is a consideration of the cost of internet access or data usage for viewing animations and other visual elements, compatibility of devices with standard laptops/computers commonly employed in public schools, and other resources to ensure that students with special needs are given a smooth and sustainable learning experience.
Perceived Ease of Use	A key component of the Technology Acceptance Model (TAM), it refers to the degree up to which a student believes using the interactive learning instrument will be effortless and trouble-free, thus ensuring the ease of accessibility on the part of students with diverse learning needs.

Perceived Usefulness	An integral part of the Technology Acceptance Model (TAM), it is the extent up to which it is perceived by the learner that employing the interactive learning tool will enhance his/her performance, knowledge, and application of Functional Mathematics in practical situations and real-life scenarios.
Social Influence	It is one of the key components of the Technology Acceptance Model (TAM) that considers the impact of recommendations, encouragement, or social pressure from teachers, peers, or parents that can probably affect a student's decision in terms of adopting and interacting with the interactive learning aid.
Special Needs Education (SNED)	Special Needs Education (SNED) Program, formerly called the Special Education (SPED) Program, is an educational program of the Philippines under the Department of Education (DepEd) that is dedicated to delivering specialized instruction and assistance to learners with disabilities and differently structured learning needs. It serves the purpose of enabling students who have developmental, intellectual, behavioral, or physical challenges to have access to proper and inclusive learning opportunities suited to their individual and diverse needs.
Speech Recognition	An interactive feature incorporated in the interactive learning aid which allows the system to capture and convert verbal language into computer input. In this study, it enables students to orally or verbally provide answers to their activities, thus reducing the use of keyboard or mouse clicking, and increasing accessibility for learners with special needs.
Sublime Text Editor	A programming language application and text editor commonly known for its flexible and lightweight attributes and primarily used in developing code. In this study, it served as the primary instrument for developing the web application and embedding various interactive functionalities for the interactive learning aid, guaranteeing smooth and seamless integration of the overall content.

Technology Acceptance Model (TAM)	It is a theoretical framework utilized in the analysis of the extent to which individuals accept and use new technologies. It defines major elements which includes perceived usefulness, perceived ease of use, and social influence, which influence individuals' attitudes and behaviors toward technologies. In this project, TAM was used to analyze and evaluate the usage of the interactive learning aid developed for students under Special Needs Education (SNED), as well as its overall effectiveness and acceptance.
Web-Based Application	A type of software that is executed on a web browser and does not require installation. It serves as the platform of this study to ensure the functionality and accessibility across standard laptops, desktops, and mobile devices typically used in public schools.
Web Speech API	A browser-based technology and programming interface specifically created for the purpose of integrating voice data, which encompasses speech-to-text and text-to-speech abilities in web applications. In this project, it enables students to use the learning tool by speech or audio prompting and increases accessibility and encourages participation among students under Special Needs Education (SNED).

CHAPTER I

INTRODUCTION

PROJECT CONTEXT

Despite numerous government initiatives, access to quality education for students with special needs is considered a challenge in the Philippines. The Department of Education (DepEd) founded the Special Education (SPED) program to cater to learners with special needs. However, many students struggle with fundamental subjects, including Mathematics. The National Center for Education Statistics (NCES) found that 200,000 students are enrolled in SPED programs nationwide, but many have difficulties with their numeracy skills (NCES, 2022). Moreover, the United Nations Educational, Scientific, and Cultural Organization (UNESCO) stated that about 45% of students with disabilities complete their basic education, emphasizing the need for effective learning interventions to support them in achieving their educational potential (UNESCO, 2022).

Globally, numerous studies highlight that interactive and visual learning tools provide improvements for students with special needs in their educational outcomes. In multimedia learning, it is suggested that students, particularly those with cognitive challenges, digest information better when provided with engaging animated content (Mayer, 2021). Correspondingly, UNESCO advocates integrating digital technologies into special education for comprehension and retention improvement among students using interactive learning materials (UNESCO, 2023).

The SPED (presently called Special Needs Education or SNED) curriculum in the Philippines faces various challenges that contradict the positive effects of implementing the teaching and learning processes. Significant hindrances identified are low financial support, insufficient teacher training, and a lack of specialized instructional material tailored for Special Needs Education (SNED) students. The deficiencies of resources hamper teachers from supporting and teaching those with special needs, thereby degrading the quality of education (Espeño et al., 2024). Resolving these problems using creative and innovative resolutions, such as incorporating animation and motion graphics, can help facilitate students' understanding of core subjects, including Mathematics.

This capstone project, “Functional Mathematics: An Interactive Learning Aid for Students with Special Needs," aimed to develop a digital interactive educational tool that would be easy to experience and engaging for SNED students, mainly assessing their focus, comprehension, and retention. The project used animation and motion graphics to simplify and rationalize complex concepts so that the content will be interactive and visually stimulating. By integrating the principle of adaptive learning, the learning aid ensures that it will cater to the needs of the learners.

The interactive learning aid focused on essential Functional Mathematics areas. The module for Functional Mathematics will run through the lessons about whole number operations and concepts of money recognition and giving change. These topics are essential for SNED students to obtain efficiency in skills that can be practically applied in everyday life, providing both academic growth and independence.

PROJECT OBJECTIVES

Generally, the capstone project aimed to produce an interactive learning aid that effectively taught Functional Mathematics to Special Needs Education (SNED) students. Specifically, the project sought to:

1. Identify the core competencies in Functional Mathematics essential for SNED students, such as:
    a. Curriculum standards
    b. Learning challenges faced by the students
    c. Skills and knowledge areas that should be addressed
2. Develop an interactive learning aid aligned with the lessons and curriculum requirements of the SNED program for Functional Mathematics.
3. Test and evaluate the effectiveness, efficiency, and user acceptance of the interactive learning aid using the Technology Acceptance Model (TAM) and ISO 25010.

PROJECT PURPOSE

The primary purpose of this study was to develop an interactive learning aid that would allow students with special needs to aid students in learning Functional Mathematics. The project created engaging and interactive content appropriate for the students' levels to enhance understanding, retention, and use of core skills and promote practical use of critical skills for independence and confidence in overcoming real-life situations. This project resonates with the United Nations (UN) Sustainable Development Goal 4 (SDG 4- Quality

Education) of ensuring inclusive and equitable quality education and lifelong learning opportunities for all. It also correlates with the Philippine Development Plan (PDP) 2023-2028 in emphasizing the importance of inclusivity in education and developing innovative forms and instruments for learning provision to different learners. This project will be beneficial for the following:

Students with Special Needs. Instructional materials that assess learners' difficulties in understanding and utilizing the concepts of Functional Mathematics will benefit the learners under the Special Needs Education (SNED) program.

Special Education Teachers. Educators can access innovative teaching tools specifically designed to support differentiated instruction, allowing them to meet the diverse needs of the learners effectively.

Parents and Guardians. Family members can observe and support the enhanced learning experiences and the increased level of confidence of their children in terms of handling everyday tasks that require functional academic skills.

Educational Institutions. Schools and special education centers can implement interactive learning aids in their curriculum to promote inclusive education practices, enhancing the overall quality of learning through the adoption of technologies in teaching students with special needs.

Future Researchers. Practitioners in the field of special education can use the findings and methodologies of the project as a basis for developing educational tools. Moreover, this project can serve as a foundation or reference for future studies related to the effectiveness of interactive learning aids.

SCOPE AND LIMITATION OF THE STUDY

The project development included the creation of an interactive learning aid designed to enhance the learning experiences of students with special needs in the areas of Functional Mathematics. The content areas for Functional Mathematics include counting and recognizing numbers, basic operations on whole numbers, and recognizing money, as well as giving change. This learning aid aimed to provide learners with fun, engaging, and interactive content that can improve their comprehension, retention, and practical application of essential skills.

This project is focused on Special Needs Education (SNED) students of Los Baños National High School-Poblacion (LBNHS-Poblacion), with the participation of 40 (forty) learners under the supervision of their committed teachers and guardians. In addition, Mrs. Rhea P. Barboza, an SNED Teacher at LBNHS-Poblacion, served as the consultant for the creation of the project, which will ensure that the content and instructional composition align with the needs of the students, as well as provide guidance in the implementation of the project and the students' adaptability.

Despite the potential of the project, significant technological infrastructure gaps at LBNHS-Poblacion presented obstacles to full-scale implementation. The limited hardware of the school, which comprised only a single smart television and a sparse collection of desktop computers and laptops, created a bottleneck for accessibility because the entire student body must share these resources. This scarcity restricted the opportunity for frequent and hands-on interaction with the learning aid.

Furthermore, the digital divide was evident in the limited ownership of personal mobile devices among the students. This lack of hardware precluded the possibility of supplemental learning at home or individualized instruction during class hours. These constraints meant that the deployment of the interactive aid could not rely on modern or high-performance hardware. This necessitated a strategic focus on cross-platform compatibility and resource-efficient software design to ensure the project remained viable within a resource-constrained educational setting.

To address these hindrances, the project team optimized the interactive learning aid based on the available technological tools to ensure compatibility and functionality across the existing devices of the school. The project is designed to function efficiently on low-specification devices to counter the technological limitations. Furthermore, this will involve training the teachers to maximize the utilization of the learning aid under these limitations to still enhance the learning experiences of the students despite the difficulties.

CHAPTER 2

THEORETICAL FRAMEWORK OF THE PROJECT

Review of Related Literature, Studies, and Systems

This chapter reviews various existing studies and literature that provide the theoretical foundation for the project. The researchers have gathered relevant concepts and ideas regarding the difficulties faced by students with special needs in learning Functional Mathematics, as well as the potential of interactive learning aids to alleviate comprehension and engagement. The reviewed studies are categorized under the following headings:

1. Teaching Philosophy for Special Needs Education (SNED)
2. Classification of Students with Special Needs
3. Teaching Mathematics to Students with Special Needs
4. Numerical Comprehension of Students with Special Needs
5. Implementing Interactive Learning Aids in Special Education
6. Impact of Interactive Learning Aids on Cognitive and Behavioral Growth

Teaching Philosophy for Special Needs Education (SNED)

In handling students with special needs, differentiated instruction is a fundamental principle to establish quality education by ensuring that the teaching methods and materials are adjusted based on the individual learning profile of the students (Ugalde et al., 2021). Moreover, it is insisted that collaborative learning spaces, when combined with differentiated approaches, are most likely to be effective in terms of addressing the needs and different capabilities of the students. These strategies also allow inclusive teaching

practices and encourage flexible engagement by adapting to various learning styles and cognitive abilities (Rasnawati et al., 2019).

The most important improvement of differentiated instruction in special education is through flexible and adaptable learning methods that are provided by digital and technology resources. With such technologies, teachers can change activities, learning materials, and content to meet the different needs of students with disabilities. Examples of devices that are useful for students with cognitive or sensory disabilities include speech-to-text programs, screen readers, and adaptive learning interfaces. In addition, text-to-speech software and applications are crucial in the aspect of independent learning and comprehension, as students with reading impairments can access the content at their own pace. For instance, adaptive learning technologies, which employ modern technology to change class difficulty based on the success of students, ensure personalized learning experiences (Benedict et al., 2020).

By providing curriculum and instruction that allow inclusivity and equity for students with special needs, differentiated instruction, the potential impact of educational technology continues to grow. The emergence of innovations and possibilities in implementing technology in teaching can further accelerate this approach in future personalized student-centered learning in special education (Stingo, 2024).

Classification of Students with Special Needs

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition that has deficits in communication, social interaction, and behavior as its indicated characteristics. It is termed a spectrum disorder because the symptoms and

severity are so varied that these people may experience anything along an emotional scale. This condition is mainly characterized by problems involving communication, both verbal and nonverbal, stereotypies, extreme interests, and diminished skills in social interactions. However, all these conditions are highly variable in their expression on an individual basis (National Institute of Mental Health, 2024).

In line with this, the intervention and education strategies are critically important to the growth of individuals with ASD both developmentally and educationally; indeed, research evidence has established that teaching methods, visual aids, and support combinations such as individualized education programs bring students with ASD better learning outcomes. In addition, individualized support, including sensory adaptations, social skills instruction, and behavioral management techniques, highly augments the learning experience of students with autism spectrum disorder (McDougal et al., 2023).

Moreover, it has been substantiated that interactive and inclusive learning environments can improve the performance outcomes of students with autism. For instance, interventions using structured peer interaction and customized multimedia materials show a discrete improvement in the language and social communication skills of children with autism (Kirk, 2020; Ugalde et al., 2021). Their findings call for specific pedagogical modalities that will take into consideration the unique social and cognitive profiles of students with autism spectrum disorders.

Attention Deficit Hyperactivity Disorder (ADHD) is a developmental disorder characterized by regularities in behavior in terms of impulsivity, hyperactivity, and inattention. Individuals with this condition tend to have difficulty focusing on the task at hand or organizing their actions. Also, they might talk excessively, having continuous movements, and cannot stay still, even in situations that dictate such behaviors are socially unacceptable, and people affected by it often have a hard time waiting for their turn because they tend to interrupt others during conversations or intrude on activities (Cleveland Clinic, 2022).

There are current trends within special education supporting the fact that students with ADHD equally need interactive and participatory instruction approaches. According to emerging findings, if teacher instruction is adjusted to incorporate the use of visual stimuli, bodily activity-based work, and the use of instantaneous feedback, individuals with ADHD perform better in concentrating and learning outcomes, which implies that there is a need to adapt these interactive methods specifically for ADHD students. (Ugalde et al., 2021).

Teaching Mathematics to Students with Special Needs

Visual-spatial processing is a frequent problem for most students with special needs, especially in mathematics, where ideas usually need to be represented abstractly. Recent research has shown that the incorporation of interactive multimedia into mathematics teaching can narrow this gap by offering visual and kinesthetic scaffolds. The use of various software indeed helps students vividly imagine mathematical concepts as a way to assess their

spatial thinking and problem-solving skills (Lei et al., 2020; Lambert et al., 2020). These observations indicate the benefits of using motion graphics and dynamic visuals in enhancing mathematics understanding.

The integration of individually suited mathematics learning media, which includes adapting to learning methods for challenged learners, indicates that mathematical contents, multisensory teaching strategies, and using visual aids can facilitate understanding and engagement for the students (Chitiyo, 2019).

On the other hand, the case for autistic students is different as it does not raise concern because many of them have excellent mathematical skills, which causes teachers to develop effective strategies for teaching them. However, teaching mathematics to learners with autism is different from regular education programs because the teachers must adjust to the psychological condition of the students. Also, it was observed that students with autism had focusing problems, so it is significant to provide well-designed learning materials that are appropriate for them to learn and behave properly (Sabaruddin et al., 2020).

Numerical Comprehension of Students with Special Needs

The majority of special needs learners are impaired by numeracy comprehension, which will make it difficult for them to improve in the Mathematics subject, particularly since the lessons revolve around numbers. The implementation of interactive learning devices in countering these problems with the inclusion of audio-visual synchronization, kinetic typography, and multimodal teaching has been found to enhance the numeracy skills,

leading to improved Mathematics academic achievement (Chen et al., 2020; Ugalde et al., 2021).

Furthermore, customized lesson plans can also be developed by the teachers in various ways to suit varying learner-specific requirements that are created to cater to the needs of students who struggle to understand the mathematical language. For all students, including those with special needs, educators can establish more inclusive and accessible language learning approaches by using interactive engagement techniques, simplified texts, and visual aids (Upa & Mbato, 2020).

Implementing Interactive Learning Aids in Special Education

The use of interactive learning media in special education has continued to advance in recent years, demonstrating that multimedia equipment—from interactive simulations to motion graphics—has been introduced in classrooms to create more interactive and effective learning spaces (Fajriani et al., 2023). These technologies provide individualized learning experiences such that students can manage the pace and order of learning, which is highly advantageous for students with special needs (Ugalde et al., 2021).

In line with this, the utilization of assistive technology, which includes speech-to-text in teaching, along with visual instructional aids, demonstrates a positive effect on students in special education (Erdem, 2017).

In addition, when learning aids are implemented, they can increase the engagement, motivation, and understanding of learners with cognitive and language impairments (Smith & Jones, 2020). It was also observed that visually

interactive displays help students with special needs focus more effectively and increase retention (Garcia et al., 2021).

Impact of Interactive Learning Aids on Cognitive and Behavioral Growth

Interactive learning aids help children with a few specific disabilities with language acquisition, especially children with speech and language problems. In addition, interactive learning experiences can also help students with speech impairment and greatly enhance their abilities in expressive and receptive language. Along with cognitive benefits, this type of interactive learning source helps special needs children improve in social skills, emotional control, and levels of engagement (Lee et al., 2022).

Additionally, visual learning aids also benefit students with various learning disabilities by simplifying complex topics and maintaining their attention. In addition to this, visual learning tools such as animated video and interactive graphics have proved to enhance academic performance and development of students with special needs (O'Neill & Topping, 2019).

Furthermore, animated learning tools are highly significant, as these tools break down difficult concepts into more manageable pieces and provide opportunities for interactive engagement (Kucirkova & Cremin, 2020).

SYNTHESIS AND RELEVANCE OF THE STUDY

The literature review emphasizes the different nature of challenges that students with special needs encounter, particularly in understanding Functional Mathematics. In most cases, traditional teaching methods fail to address the various learning styles of the students, imposing the need for more engaging and adaptive kinds of learning aids. Moreover, studies related to Special Needs

Education (SNED) stated that integrating interactive learning aids can improve the students' comprehension by incorporating visual aspects into the learning process, making it more efficient and effective.

Furthermore, the use of animations and interactive elements in teaching mathematical concepts can simplify complex concepts to make them more understandable for learners under SNED, which leads to enhancing the mathematical skills and numerical comprehension of the learners by integrating interactivities, speech recognition, and contextual visual aids.

Along with these findings, this study proposed the development of an interactive learning aid, which can efficiently educate Functional Mathematics to students with special needs. It used animation, clickable elements, speech recognition, and audio-visual synchronization features in order to provide an engaging experience for learners. Moreover, the learning aid synthesized animation, clickable elements, speech recognition, and audio-visual synchronization to provide an engaging and immersive learning approach for the students. This project will address the needs of SNED students and will serve to bridge the gap between traditional instructional materials and modern educational technology toward effective teaching and creating a more inclusive, diverse environment for learning.

LITERATURE MATRIX

Table 1 summarizes the significant reviews of related literature and studies cited in this investigation. Relevant publications support incorporating animation and interactive elements in developing a learning aid for students with special needs in teaching Functional Mathematics.

Table 1. Matrix of Related Literature

Author(s)/Year	Title	Findings
NCES (2018)	Students with Disabilities: Condition of Education	200,000 students are enrolled in SNED programs nationwide, but many have difficulties with numeracy skills.
UNESCO (2022)	Disability Inclusive Education and Learning	45% of students with disabilities complete their basic education, emphasizing the need for effective learning interventions to support them in achieving their educational potential.
Mayer (2021)	Introduction to Multimedia Learning	In multimedia learning, it is suggested that students, particularly those with cognitive challenges, digest information better when provided with engaging animated content.
Ugalde et al. (2021)	Impact of Interactive Learning Environments on Learning and Cognitive Development of Children with Special Educational Needs: A Literature Review	Differentiated instruction is a fundamental principle to establish quality education by ensuring that the teaching methods and materials are adjusted based on the individual learning profiles of the students.
Rachmawati et al. (2019)	Inclusive Education Practices: Fostering Diversity and Equity in the Classroom	Collaborative learning spaces, with differentiated approaches, can address the needs and different capabilities of the students, allowing inclusive teaching practices and encouraging flexible engagement by adapting to various learning styles and cognitive abilities.
NIMH (2024)	Autism Spectrum Disorder	Autism Spectrum Disorder (ASD) is a neurodevelopmental condition that has deficits in communication, social interaction, and behavior as its indicated characteristics.
McDougal et al. (2023)	Understanding and Supporting ADHD in the Primary School Classroom: Perspectives of Children with ADHD and their Teachers	The intervention and education strategies are critically important to the growth of individuals with attention deficit hyperactivity disorder, in terms of their developmental and educational aspect.
Kirk (2020)	Practical Teaching Strategies for Students with Autism Spectrum Disorder: A Review of the Literature	Interactive and inclusive learning environments, wherein interventions that use structured peer interaction and customized multimedia materials improve the language and social communication skills of children that are diagnosed with ASD.

Cleveland Clinic (2022)	Attention Deficit Hyperactivity Disorder	Attention Deficit Hyperactivity Disorder (ADHD) is a developmental disorder characterized by regularities in behavior in terms of impulsivity, hyperactivity, and inattention, in which individuals with this condition tend to have difficulty focusing on the task at hand or organizing their actions.
Ugalde et al. (2021)	Impact of Interactive Learning Environments on Learning and Cognitive Development of Children with Special Educational Needs: A Literature Review	Incorporating the use of visual stimuli, and the use of instantaneous feedback, individuals with ADHD perform better in concentrating and learning outcomes, which implies that there is a need to adapt these interactive methods specifically for ADHD students.
Chitiyo (2019)	Exploring Teachers' Special and Inclusive Education Professional Development Needs	The integration of individually suited mathematics learning media can facilitate understanding and engagement for the students.
Chen et al. (2020)	Peer Language Resources and Development in Inclusive Classrooms	The implementation of interactive educational devices can enhance the learning experiences of students with special needs, which leads to improved Mathematics academic achievement.
Erdem (2017)	Students with Special Educational Needs and Assistive Technologies: A Literature Review	The utilization of speech-to-text in teaching, along with visual instructional aids, demonstrates a positive effect on students in special education.
O'Neill & Topping (2019)	Using Visual Aids to Improve Academic Performance for Students with ADHD: A Literature Review	Visual learning tools and interactive graphics have proven to enhance academic performance and the development of students with special needs.
Kucirkova & Cremin (2020)	Improving Engagement in Learning through Animated Tools: A Case Study in Primary Education	Animated learning tools are highly significant as they can break down difficult concepts into more manageable pieces and provide opportunities for interactive engagement.

CONCEPTUAL FRAMEWORK

Figure 1 illustrates the conceptual framework utilized by the researchers in developing the interactive learning aid for students with special needs. This framework served as a structured roadmap that guided the entire development

process, ensuring that the final output was both technically functional and pedagogically appropriate for learners with diverse learning needs. It presents a systematic sequence of phases, namely data gathering and analysis, planning of materials, content creation, implementation of interactivity, deployment and testing, and evaluation, all of which were aligned with the objectives of enhancing the teaching and learning of Functional Mathematics.

The process began with data gathering and analysis, which focused on requirements identification through a review of related literature, existing studies, and series of interviews with Special Needs Education (SNED) teachers. This phase allowed the researchers to identify key challenges in teaching Functional Mathematics, particularly the difficulty of translating abstract mathematical concepts into practical daily life skills. The insights gathered from real classroom experiences served as the foundation for designing a learning aid that addresses the actual needs of both learners and educators in SNED settings.

Following the initial analysis, the framework progressed to the planning of materials and content creation phases. During these stages, the researchers mapped out Functional Mathematics topics such as currency recognition, time-telling, and basic arithmetic while integrating differentiated instruction strategies. The integrated media learning assets, including high-contrast visuals, simplified text, clear audio cues, animations, and interactive activities, were developed to accommodate varying learning styles and sensory processing abilities of SNED students. These design considerations aimed to improve engagement, comprehension, and retention of mathematical concepts.

The development culminated in the implementation of interactivity, where the instructional materials were transformed into an engaging learning tool through interactive elements such as drag-and-drop activities, speech recognition, and immediate feedback mechanisms. The project then underwent deployment and testing to identify and resolve technical issues and ensure usability and stability. Finally, the framework concluded with evaluation using the Technology Acceptance Model) that was applied to assess learners' perceived usefulness and ease of use of the interactive learning aid, while the ISO 25010:2023 standard was utilized to evaluate the functional suitability, usability, and accessibility of the learning aid based on the perspectives of SNED teachers. This evaluation ensured that the learning aid was both educationally effective for students and technically reliable for classroom implementation.

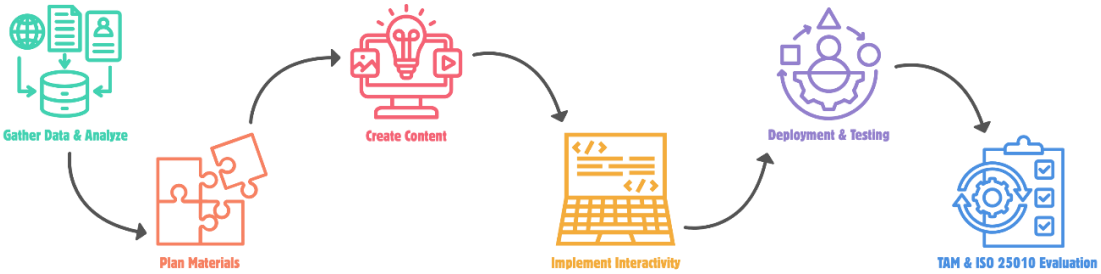


Figure 1. Conceptual Framework

CHAPTER 3

METHODOLOGY

This chapter presents the research design to be used in the project, the animation development model, the study population, data collection instruments, and the statistical treatment used throughout the research process.

PROJECT DESIGN

The learning aid for Functional Mathematics was designed to be interactive and specifically designed to cater to the different learning styles of Special Needs Education (SNED) students. The project will incorporate animated visual elements and interactive features into the learning concept of SNED students and aims to improve engagement, retention, and comprehension in Mathematics.

SOFTWARE DEVELOPMENT LIFE CYCLE

The researchers adopted the Software Development Life Cycle (SDLC) adapted from the sequential model approach (Priya Sr., 2018) to guide the development of the interactive educational tool from the initial concept to the final output. This model was selected for its disciplined and systematic structure, which is well-suited for producing high-quality animated content and integrating interactive features through a clearly defined and linear development process.

The sequential model was particularly appropriate for the specialized requirements of Functional Mathematics for Special Needs Education (SNED).

Its linear progression allowed the researchers to complete the Planning, Defining, and Designing phases before proceeding to technical development, ensuring that learning objectives, curriculum alignment, and learner needs were fully established. This approach minimized the risk of costly revisions to animation assets and interactive components, as all features were grounded in predefined instructional standards identified during the analysis phase.

Moreover, the structured transition between development stages provided a strong foundation for system stability and evaluation. By moving methodically from development to testing, the researchers ensured that the learning aid was fully functional prior to assessment. This was essential for the study's dual-layered evaluation using the Technology Acceptance Model (TAM) and ISO/IEC 25010, enabling accurate measurement of both user acceptance and technical quality for effective implementation within the SNED learning environment.

Figure 2. Software Development Life Cycle

Planning

In the planning phase of developing the interactive learning aid, an interview was conducted with Mrs. Rhea Barboza, a Special Needs Education (SNED) Teacher at Los Baños National High School-Poblacion (LBNHS-Poblacion), to take her input regarding the needs, capabilities, and learning styles of SNED students. In addition, the researchers studied existing literature on SNED and educational technology tools to conceptualize the structure and

features of the learning aid. To ensure accessibility, engagement, and appropriateness of learning materials for students with special needs, recent references regarding animation practices and visual tools were also studied.

Defining

After gathering information, the researchers focused on the significant findings that can support the development of the interactive learning aid. The insights obtained from the interview, along with existing studies on special needs education and visual educational tools, were incorporated to serve as the foundation for the project. In addition, based on the consultation with the SNED teacher, it is recommended to concentrate on developing a learning aid that can teach Functional Mathematics to students with special needs.

Designing

The designing phase converts the defined concepts into visual and interactive components. At this stage, character designs, script, storyboard, animations, audio elements, and interactive features are developed to ensure that each individual component of the project is aligned with the needs and capabilities of SNED students.

Character Design

In this phase, the researchers aimed for the characters in the interactive learning aid to be visually engaging, age-appropriate, and relevant to students with special needs. The character designs were digitally completed using iBis Paint, Adobe Illustrator, and Adobe Photoshop to ensure visual consistency and clarity.

The idea behind the character design phase was to build familiarity and establish an emotional bond between the learners and the project. Thus, the learning aid is intended to create a more supportive and inclusive learning environment for helping students with special needs feel comfortable, motivated, and involved throughout the interaction by having visually consistent and emotionally positive characters.

Figure 3. Sample Character Design

Scriptwriting

This phase focused on developing clear and well-structured scripts to guide the visual, auditory, and interactive elements of the learning aid. The scripts were based directly on the content needs identified during the data gathering phase and were designed specifically for students with special needs using simple language, professional yet friendly tones, and step-by-step instructions to ensure comprehension and engagement.

A main script for Functional Mathematics was created and divided into smaller lessons, each including an introductory snippet or animated tutorial that

clearly explained tasks. The scripts incorporated interactive components such as clickable elements, prompts, voice recognition instructions, and feedback to engage students at every stage. This structured approach ensured that all lessons, activities, quizzes, transitions, and short skits aligned with the visual assets and met the learning objectives, supporting both academic and user development effectively.

Storyboard and Interface

In this phase, the researchers translated the written script and instructional flow into a visual storyboard and user interface to guide the animation and interaction sequence of the learning aid. The storyboard provided a scene-by-scene layout for each module in Functional Mathematics, while the visual elements, including characters, backgrounds, objects, and interface components, were sketched based on the content, quizzes, and activities of each lesson. Individual panels included scene sketches, dialogue captions, text cues, and indicators of interaction to visually represent how the project would appear on-screen.

Interactive features, such as clickable buttons, drag-and-drop activities, and speech recognition prompts, were also depicted to ensure correct placement during development. Each lesson’s storyboard illustrated the learning process from instruction to activity and feedback, ensuring smooth transitions and content that are appropriate for the diverse needs of Special Needs Education (SNED) students. This approach ensured that all elements were interconnected, dynamic, and aligned with the educational goals of the project.

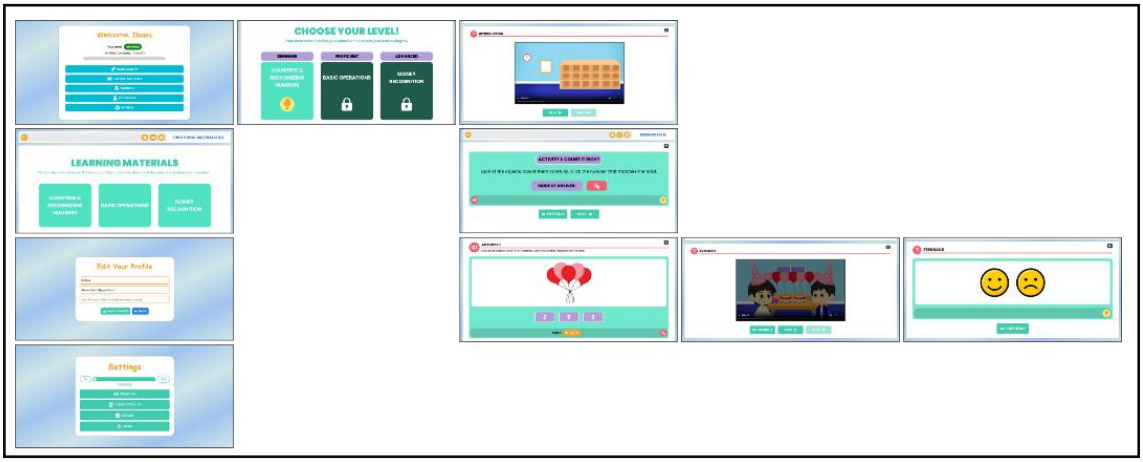


Figure 4. Sample Interface

Figure 5. Sample Storyboard

Illustration and Animations

The researchers started creating the final visual assets and motion sequences based on the storyboard and character designs. With the use of software applications, such as iBis Paint, Adobe Illustrator, and Adobe Photoshop, along with the use of a pen tablet, the visual elements that were styled to be simple, clear, and expressive were completed.

When the illustrations were completed, the work proceeded to the animation stage, wherein Adobe Animate, Adobe Photoshop, and Adobe After Effects were used to bring the different scenes from the lessons and skits to life. The key animated action included facial expressions, pointing gestures, guiding visual cues to highlight the correct answers, and transitions between the learning modules. In addition, short, animated skits were incorporated as

an introduction and ending for each subject and lesson to support context-based education and emotional engagement.

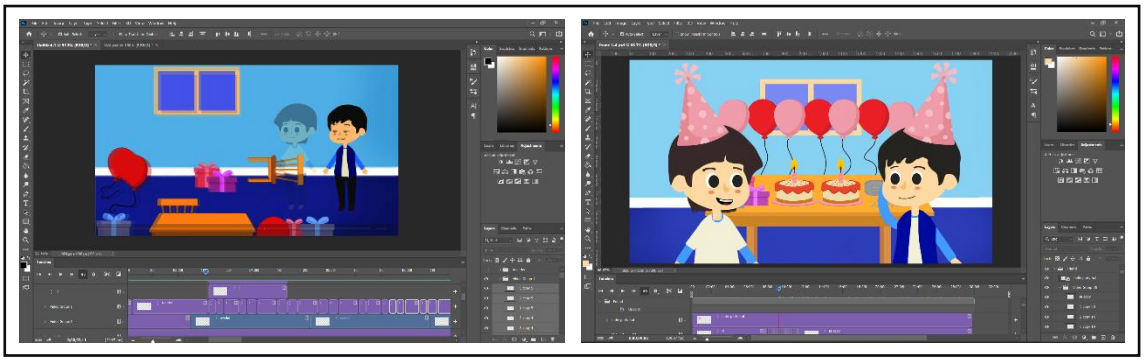


Figure 6. Sample Animation Process

Voiceover and Sound Design

The voiceover and sound design phase concentrated on improving the auditory engagement of the interactive learning aid by attempting to facilitate understanding and learning retention among Special Needs Education (SNED) students. The voiceover referred to clear and age-appropriate voices for each instructional content, animated skit dialogue, and quiz prompts. Furthermore, voiceover scripts were recorded in simple, yet highly encouraging language based on storyboards and scripts to ensure that lessons were consistent and equally easy to follow.

Apart from the voiceover, sound design comprised contextual music, paired with sound effects that match the pacing and mood of the quiz or enactment of a skit. A well-optimized volume and sound, along with further enhancements, made sure that learners do not feel overstimulated but will remain focused and engaged in the content. The whole phase ensured that the project was delivered as a diverse environment where visuals and sound were combined to make an inclusive and immersive learning platform for students.

Interactive Features

In this phase, the researchers sought to furnish the learning aid with functional, accessible, engaging elements that would accommodate the special engagement patterns and diverse learning styles of Special Needs Education (SNED) students. The HTML5 Canvas feature in Adobe Animate and the incorporation of JavaScript and Web Speech API, which were composed using Sublime Text Editor, were utilized to support the real-time interactivity of the lessons and activities.

The key interactive features included click-to-answer buttons, which allow students to visually select their answers in learning activities. Every clickable option was accompanied by sound feedback and visual highlighting to reinforce learning outcomes. Also, embedded was the function of converting speech into text, which allowed learners' verbal answers to be accepted, especially for vocabulary recognition and counting activities. This became an alternative input for students who may have problems with fine motor skills or reading comprehension.

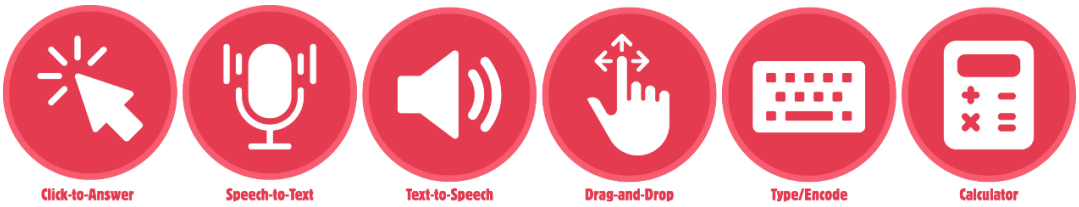


Figure 7. Interactive Features

Building

During the building phase, all major elements of the interactive learning aid were integrated to create a unified and functional educational platform for special needs education (SNED). This phase involved design, animation,

scripting, voiceover, interactivity, and testing to ensure that each of the components reinforced educational objectives and user-centered composition.

The project consisted of a module for Functional Mathematics. It has lessons arranged according to learning levels - Beginner, Proficient, and Advanced - with animated content, interactive quizzes, and speech-based tasks corresponding to the particular levels. It also incorporates animated skits that introduce the lesson in real-world scenarios to familiarize the learners and short tutorial components that will teach them how to deal with the features of the learning aid.

All the visual assets, which are the characters, icons, backgrounds, and animations, were produced using iBis Paint, Adobe Illustrator, and Adobe Photoshop, and animated in Adobe Animate and Adobe After Effects. For audio layering and synchronization, CapCut was employed. For example, interactive elements like click-to-answer, speech-to-text, and text-to-speech were integrated via the HTML5 Canvas environment using JavaScript and Web Speech API, and other code enhancements were in Sublime Text.

Testing

The Testing Phase secured the appropriate functioning of an interactive learning aid in terms of effectiveness, efficiency, and user acceptability. In line with that, it should be ensured that the project operates properly, even on standard laptops or average desktop computers, since classrooms that offer Special Needs Education (SNED) programs lack digital and technological resources.

Functionality testing encompassed the overall project composition, which includes animation, interactivity, and audio narration. Further testing was supported by the Technology Acceptance Model (TAM) and ISO 25010 Software Quality Model. The TAM was applied in a series of surveys and the collection of qualitative feedback to assess the perceived usefulness, ease of use, and user acceptance of the learning aid from SNED students and teachers. In addition, ISO 25010 was used to frame the assessment of the functional suitability, usability, performance efficiency, reliability, and maintainability of the project.

Deployment

The deployment phase represented the distribution and real-world implementation of the interactive learning aid for its intended users, who are the students enrolled in the Special Needs Education (SNED) program at Los Baños National High School-Poblacion (LBNHS-Poblacion). These students are classified as beginner, proficient, and advanced levels, wherein content delivery matches their individual learning needs and cognitive levels.

The use of desktop web-based applications in deploying the interactive learning aid imitates real technological conditions in classrooms where SNED students are commonly found, providing access through a web browser without the need for installation or high storage space. This ensures that the project will be compatible even with standard laptops, desktop computers, and Android mobile devices via the desktop site on a web browser, commonly used by the learners, which addresses the limitations of devices and insufficient capacity.

The web application format of the project allows it to be sustainable and inclusive, which lessens the dependency on device specifications.

In addition to student use, the deployment also saw participation of three SNED teachers from LBNHS-Poblacion who followed up students' interactions with the tool while providing feedback on the appropriateness of content, pace management, and feature usability for diverse learning profiles. Their involvement ensured that the final output suited both pedagogical motivations and actual classroom use.

MATERIALS

The researchers employed a combination of hardware and software to develop the interactive learning aid. Drawing tablets were used to digitize hand-drawn concepts, while laptops served as the main platform for project development. For designing visual and graphic elements, iBis Paint, Adobe Illustrator, and Adobe Photoshop were utilized, followed by Adobe Animate, Adobe After Effects, and CapCut to create dynamic sequences and finalize the animations. To implement interactive features such as clickable elements and speech-to-text functions, HTML5 Canvas, JavaScript, and the Web Speech API were applied. Additionally, Sublime Text was used to develop the user interface with HTML and CSS and to manage other coding and integration tasks.

Figure 8. Materials Used

Hardware Specification

In this project, two laptops were used to develop an interactive learning aid with more multitasking ability. The laptops are equipped with mid-range specifications, quad-core processors, entry-level dedicated graphics cards, and sufficient Random Access Memory (RAM) suited for the smooth running of multimedia editing and coding tasks.

In line with this, laptops were used to demonstrate the reality of most public schools that offer Special Needs Education (SNED) teaching, wherein standard laptops and/or average desktop computers are the only available devices. The researchers ensure that the final output will run efficiently in implementation, guaranteeing that the interactive learning aid is accessible, usable, and sustainable even if the technological resources are limited.

Table 2. Hardware Specification of Laptop 1

Laptop 1: **HP 15s**	
Hardware	**Specification**
Processor	AMD Ryzen 5 5500U 2.10 GHz
Memory	8GB
Graphics Processing Unit	AMD Radeon ™ Graphics
SSD	120GB
HDD	1TB
OS Version	Windows 11
System Type	64-bit operating system, x64-based processor

Table 3. Hardware Specification of Laptop 2

Laptop 2: **Asus Vivobook 16**	
Hardware	**Specification**
Processor	12th Gen Intel ® Core ™ i5-1235U 1.30 GHz
Memory	16GB
Graphics Processing Unit	Intel ® Iris ® Xe Graphics
SSD	120GB
HDD	1TB
OS Version	Windows 11
System Type	64-bit operating system, x64-based processor

In addition to these devices, the pen tablet was also used to create visual design and streamline the animation process. It allows accurate and extensive use of digital drawing and animation techniques to ensure that the visual assets produced are of high quality and compatible with standard devices.

Table 4. Hardware Specification of Pen

XPPen Artist 12 Pen Display (2nd Gen)	
Hardware	**Specification**
Wok Area	263.23 x 148.07 mm
Display Resolution	1920 x 1080
Color Gamut (typical)	90% NTSC, 127% sRGB, 94% Adobe RGB
Stylus	X3 Elite
Resolution	5080 LPI
Accuracy	±0.5 mm (center), ±1 mm (corner)

Software Specification

The software specifications served as a foundational framework for the technical requirements. This process established a comprehensive definition of the functional requirements, as well as the non-functional requirements including usability and cross-platform stability. By formalizing these criteria, the researchers created a technical roadmap that ensured every visual and interactive element remained aligned with the intended outcomes of the study.

The specification ensured project flow clarity and served as a guide for the development throughout the process to meet users' needs and expectations. Specifically, it will serve as a foundation for streamlining the visual elements, animation-based learning sequences, and interactive features, including clickable content and speech recognition, to create an interactive learning aid that is effective and user-friendly, intended for school environments.

Table 5. Software Specification

Specification	Description
Project Name	Functional Mathematics: An Interactive Learning Aid for Students with Special Needs
Platform	Web-based application, compatible with standard laptops and desktop computers, and mobile devices
Development Tools	iBis Paint, Adobe Illustrator, and Adobe Photoshop: These are the tools used for creating the visual elements, illustrations, and other graphics assets for the project. (Graphic design and enhancement) Adobe Animate, Adobe After Effects, and CapCut: These are used to apply the dynamic sequences and edit the overall animation to make the educational content engaging. (Animation development and video editing) Sublime Text: It was used to build the user interface by incorporating HyperText Markup Language (HTML) and Cascading Style Sheet (CSS). It was also used for implementing the interactive features using JavaScript and Web Speech API. (Interactivity and platform design)
Programming Language	JavaScript (interactive features and speech recognition)

EVALUATION PROCEDURE

Sampling Design

In evaluating the user acceptability of the interactive learning aid, forty students with special needs and three Special Needs Education (SNED) teachers were given a survey questionnaire. Aligning with this survey, purposive sampling was used to ensure that the evaluation would be derived from a focused group whose experiences, difficulties, and needs are aligned with the project's objective. This method will ensure that the feedback will come from individuals who are most suited to assessing the effectiveness and usability of the project.

Data Gathering Instrument and Validation

Figure 9 illustrates the Technology Acceptance Model (TAM) that was utilized to gather data regarding the acceptability of the project. It is utilized to systematically assess the acceptability and adoption of the interactive learning aid among the target users. This theoretical framework allowed the researchers to evaluate the project through seven core components: perceived usefulness, perceived ease of use, attitude, behavior, perceived cost, social influence, and information awareness. Specifically, the model assisted in quantifying user perceptions regarding how effectively the tool aided in learning Functional Mathematics and how user-friendly the interface remained for the students. By analyzing these perceptions, the researchers identified critical areas for refinement, ensuring the final output was optimized for inclusive educational environments.

To ensure the integrity of the data gathered, a rigorous validation process was conducted on the survey instrument prior to its distribution. The questionnaire was first subjected to content and face validity by experts, which are the Special Needs Education (SNED) teachers. This step ensured that the questions were appropriate for the students' cognitive levels and technically relevant to the features of the software. Following the experts' recommendations, the researchers refined the language and layout of the instrument to accommodate the specific communication needs of the SNED students at Los Baños National High School-Poblacion (LBNHS-Poblacion).

Once validated, the survey was distributed to a sample of 40 SNED students. The results of this survey served as the primary foundation for

analyzing the effectiveness of the learning aid and determining how well the students engaged with its interactive features. This systematic evaluation not only measured the current success of the project but also provided a data-driven roadmap for future enhancements in specialized educational technology.

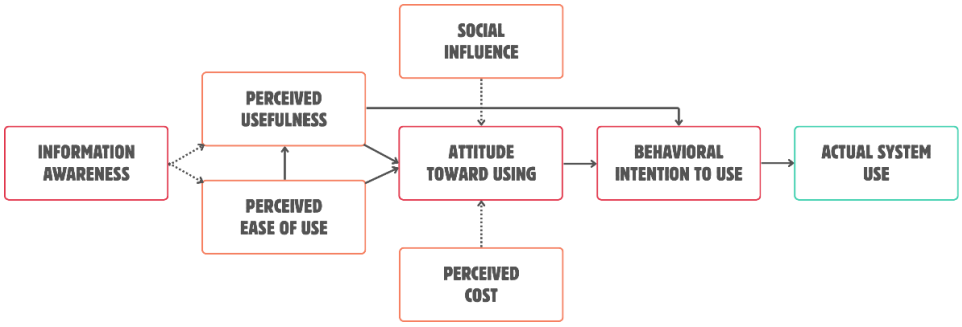


Figure 9. Technology Acceptance Model (TAM)

Figure 10 shows the ISO 25010:2023 Software Quality Model, it was utilized to establish a rigorous, international framework for assessing the efficacy, usability, and sustainability of the interactive learning aid. Several quality characteristics were specifically selected to evaluate the system's fitness for use within the context of Special Needs Education (SNED). The evaluation concentrated on six main attributes: Functional Suitability, which ensured the aid met the specific pedagogical needs of SNED learners; Usability, which assessed the accessibility and ease of navigation of the interface; Performance Efficiency, which guaranteed the software ran smoothly on standard devices found in public schools; and Reliability, which confirmed consistent performance across all modules. Additionally, Maintainability was assessed to ensure the tool could be readily upgraded, while Compatibility was verified to ensure seamless operation across laptops, desktops, and smart TVs.

To gather data for this evaluation, the researchers developed a structured survey instrument. To ensure the accuracy and credibility of the findings, the validation process of the survey instrument was conducted through

a two-step approach. First, the questionnaire underwent content validation by a panel of experts, consisting of seasoned SNED educators and software engineers. These experts reviewed the items to ensure the technical terms were translated into understandable criteria and that the questions accurately measured the specific ISO 25010 sub-characteristics.

Following the expert review, the instrument was subjected to face validation to check for clarity, formatting, and readability. This ensured that the teachers participating in the evaluation could interpret the questions consistently and without ambiguity. These validated attributes were then assessed through thorough system tests, teacher validations, and documented feedback. This comprehensive validation of both the software and the measurement tool guaranteed that the learning aid met both the specialized educational requirements of the students and internationally recognized standards of software quality.

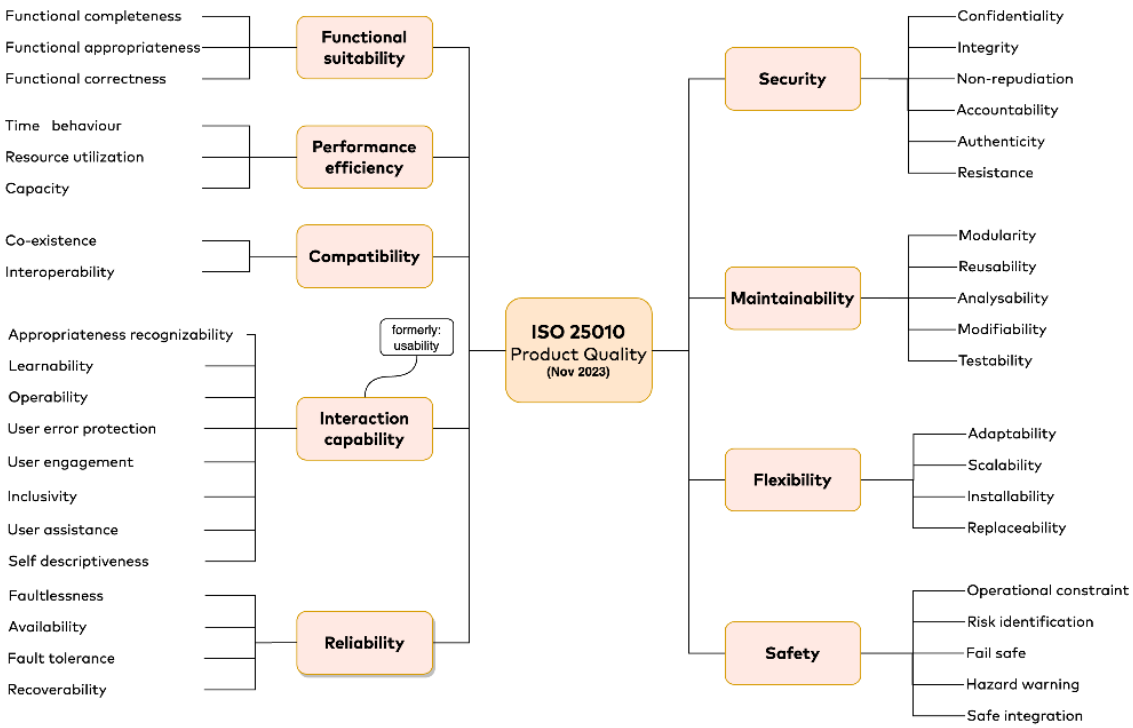


Figure 10. ISO 25010:2023 Software Quality Model

Population of the Study

This project involves students enrolled in the Special Needs Education (SNED) program at Los Baños National High School-Poblacion (LBNHS-Poblacion). They are classified into beginner, proficient, and advanced categories according to their skill levels, allowing proper alignment of the interactive learning aid according to their competency levels. Furthermore, three SNED teachers were consulted as additional sources to give their insights and guidance on the learning aid, ensuring that it is suited for the learners.

STATISTICAL TREATMENT OF DATA

The statistical tool utilized in this project is descriptive statistics, in which the data were analyzed and summarized to assess the collected data for the application of the overall acceptability of the interactive learning aid. In line with this, a four-point Likert-type scale was used as a response to ensure a focused and accessible method for students with special needs to report their feedback.

Table 6. Likert Scale

Rating	Verbal Interpretation
4	Highly Acceptable
3	Moderately Acceptable
2	Slightly Acceptable
1	Not Acceptable

The data obtained was collated and encoded in Microsoft Excel, then analyzed using IBM SPSS software to generate the findings. Median (M) was computed for each item to summarize the central tendencies. To establish the consistency and reliability of the responses, Cronbach's Alpha Reliability Test was also utilized to determine whether the items in the questionnaire under

each Technology Acceptance Model (TAM) factor and ISO 25010 were consistent. The results then would be used in interpreting how effective, engaging, and user-friendly interactive learning aids are from these perspectives of Special Needs Education (SNED) students and teachers.

Table 7. Reliability Test Results for Technology

Categories	Cronbach's Alpha	Verbal Interpretation
Perceived Usefulness	0.832	Good
Perceived Ease of Use	0.827	Good
Attitude toward Using	0.882	Good
Behavioral Intention to Use	0.711	Acceptable
Perceived Cost	0.749	Acceptable
Information Awareness	0.709	Acceptable
Social Influence	0.831	Good

Legend: *above .90 = Excellent, 0.80-0.89 = Good, 0.70-0.79 = Acceptable, 0.60-0.69 = Questionable, 0.50-0.59 = Poor, below 0.50 = Unacceptable*

Table 8. Reliability Test Results for ISO 25010 Software Quality Model

Categories	Cronbach's Alpha	Verbal Interpretation
Functional Suitability	0.750	Acceptable
Performance Efficiency	0.750	Acceptable
Compatibility	0.923	Excellent
Interaction Capability	0.857	Good
Reliability	0.857	Good
Security	0.857	Good
Maintainability	0.857	Good
Flexibility	0.808	Good
Safety	0.750	Acceptable

Legend: *above .90 = Excellent, 0.80-0.89 = Good, 0.70-0.79 = Acceptable, 0.60-0.69 = Questionable, 0.50-0.59 = Poor, below 0.50 = Unacceptable*

ETHICAL CONSIDERATION

The researchers secured all requisite approvals and permissions to ensure that the project meets ethical standards. A letter of request was sent to the school principal and the Special Needs Education (SNED) teacher to carry

out the study within the school. In alignment, parental consent forms were administered to, and signed by, the respective guardians of students who permitted the participation of SNED students. This way, all individuals involved, which includes the school administration, teachers, parents, and students, were adequately informed, and their voluntary consent was sought for participation in the research.

It was emphasized that taking part in the research is voluntary, allowing the respondents, particularly the SNED students, to decline or withdraw their participation at any stage. On the other hand, the aim of the study, possible benefits, and procedures were thoroughly explained to the people involved. Respect, understanding, and fairness to the participants were general principles adhered to by the researchers at all times during the research process, and these principles secured a safe environment in which students felt supported and valued.

Integrity and transparency were ensured so that all the data collected would be handled in a responsible, confidential way by the researchers. The data were strictly for research purposes only, and truthful reports were made from them without any manipulation or distortion. Participants were informed about their rights, the confidentiality of their responses, and the need for honest feedback. Furthermore, the researchers guarantee to pursue ethical standards while ensuring to build trust and maintain the dignity and welfare of all respondents throughout the research.

CHAPTER 4

RESULTS AND DISCUSSION

This chapter presents the findings and analysis obtained from the development and evaluation of Functional Mathematics: An Interactive Learning Aid for Students with Special Needs. This section aims to assess the project's effectiveness, usability, and quality using user feedback and a systematic software evaluation. The results and discussions are concluded based on the project objectives, software development process, and software evaluation results. Key findings are interpreted within the framework of the Technology Acceptance Model (TAM) to analyze the factors that influence the engagement of learners with the animated contents and other visual elements. In addition, ISO 25010:2023 Software Quality Model is utilized to have a complete understanding of the quality, effectiveness, and potential impact of the learning aid towards accessible and inclusive mathematics education.

Results by Objectives of the Study

The capstone project mainly focuses on creating an interactive learning aid that will effectively teach Functional Mathematics to Special Needs Education (SNED) students by integrating animation, motion graphics, visual elements, and interactive components to support their various learning needs and participation. Overall, the objectives were attained through planning, alignment to the curriculum, software creation, and testing in alignment with both the Technology Acceptance Model (TAM) and the ISO 25010 Software Quality Model.

Identification of the core competencies in Functional Mathematics essential for SNED students

To achieve this objective, the researchers interviewed one of the SNED teachers in Los Baños National High School-Poblacion and gathered information regarding the Functional Mathematics curriculum appropriate to special education learners to identify key learning competencies. Amongst these are primary skills such as the ability to count and recognize numbers, perform basic calculations, and handle money, which are important to develop mathematical competency and daily living skills of the students.

Curriculum-based lesson plans served as the direct evidence of these competencies, as they outlined instructional goals such as identifying numbers, solving simple math problems, and using money in real-life transactions. These competencies were validated through teacher interviews, validating that these foundational skills are the primary focus of Functional Mathematics instruction for students with special needs.

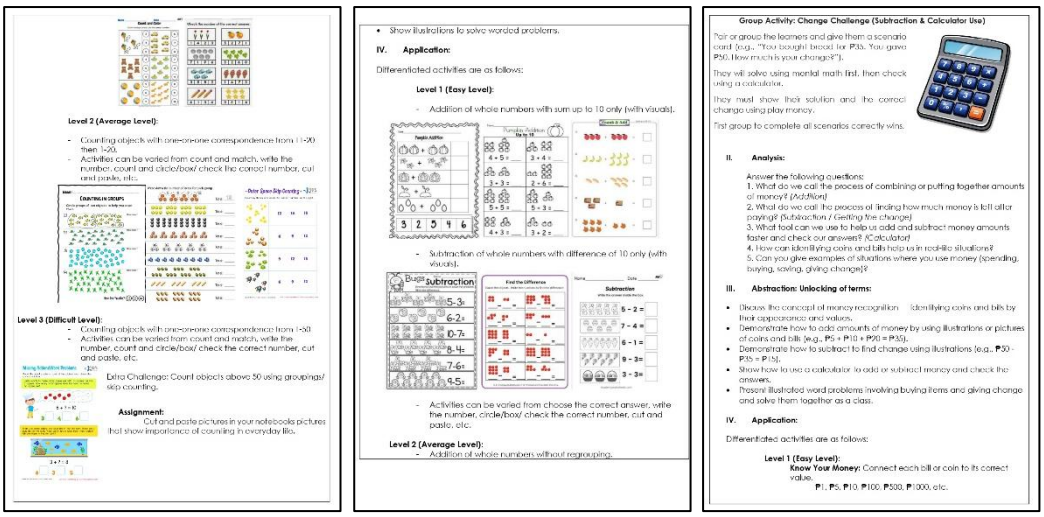


Figure 11. Sample Activities from Curriculum-Based Lesson Plans

These core competencies served as the primary basis in conceptualizing and structuring the activity order and animated skits, thereby ensuring that the

interactive learning aid demonstrates actual classroom practices. This ensures that lessons and activities are aligned with the approach used by SNED teachers in teaching and also certifies that the contents are appropriate, accurate, and sensitive to the learners' needs. Furthermore, general learning challenges, which include issues in recognizing symbols, memorization, and applying mathematical concepts to real-life situations, were observed and aimed to be solved through answering the activities.

Development of the interactive learning aid aligned with the lessons and curriculum requirements of the SNED program

Based on the identified competencies, researchers conceptualized and developed an interactive learning aid that utilized animation and motion graphics as the primary instruction medium. The project integrated HTML5 Canvas, JavaScript, and the Web Speech API with the use of Sublime Text Editor to implement interactive features, which included click-to-answer, drag-and-drop, speech-to-text, text-to-speech, type or encode, and calculator options. Each lesson has animated video discussions, along with downloadable review notes in PDF format, and interactive activities that provide various modalities for answering to address different learning styles. Storyline, user interface, and activity flow have been carefully planned in order to maintain engagement and interest, while ensuring that lessons are aligned with curriculum standards. Furthermore, functionality testing using standard laptops and desktop computers , as well as Android mobile devices has proven the effective functioning of the application without requiring high-powered devices, thus making it feasible for public school settings.

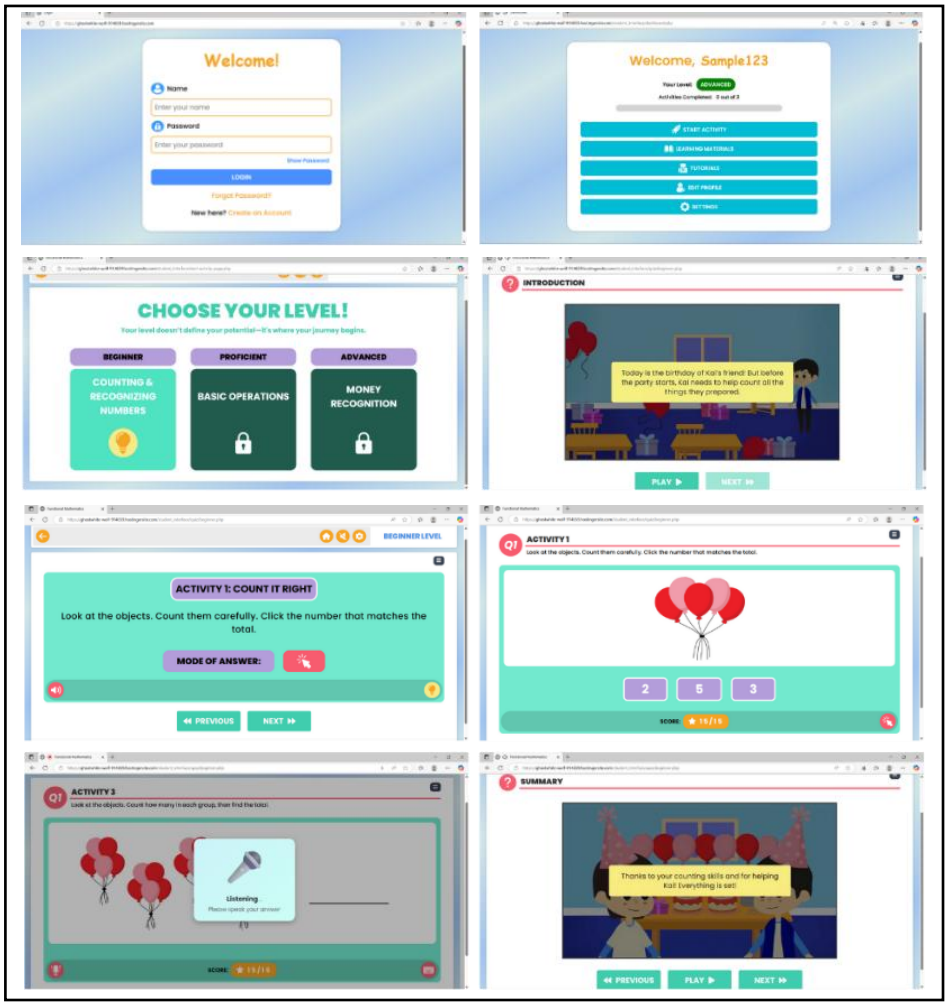


Figure 12. Sample Interface of Interactive Learning Aid

Result of testing and evaluation of the effectiveness, efficiency, and user acceptance of the interactive learning aid using the Technology Acceptance Model (TAM) and ISO 25010

The researchers used two frameworks, which are the Technology Acceptance Model (TAM) and ISO 25010:2023, in evaluating the overall quality, usability, and user acceptance of the interactive learning aid. Surveys and testing sessions were conducted with students and teachers, wherein the project gained high levels of acceptance, which indicates that the animations and interactive features effectively support the various learning needs of the learners. In addition, the system met the required software quality standards, which demonstrate proper execution and ease of use on standard devices.

Animated Content Effectiveness Evaluation

Table 9 shows the perceptions of the respondents regarding the usefulness of animations and visuals in learning mathematics. The statements “The animations make learning more meaningful and engaging for students” and “The visuals help simplify abstract or difficult topics for every learner” obtained the highest mean score which 4.00, corresponded as “highly acceptable,” which implies that the animations and visuals incorporated were highly effective in terms of improving the engagement and comprehension of the students, particularly in understanding complex mathematical concepts.

On the other hand, the lowest mean score of 3.95, which is still interpreted as "highly acceptable," is from the statement "Overall, animations and visuals are useful for achieving the learning objectives of each lesson." This hints that the interactive learning aid is consistently effective in achieving the educational objectives, although minor differences in perception may vary due to individual differences in learning preferences. This variation suggests that while the multimedia elements are highly functional, some students may prioritize different sensory inputs depending on their specific cognitive profiles. Consequently, the high rating confirms that the visual design remains a potent instructional tool even when accounting for the diverse processing needs of SNED learners.

The composite mean of 3.98 substantiates the claim that the respondents highly accepted the usefulness of the animations and visuals. This result overall suggests that the visual and animated parts of the learning aid did play an important role in the attainment of lesson objectives and the imparting of knowledge to the learners.

Table 9. Perceived Usefulness based on Respondents' Response

	STATEMENTS	MEAN	SD	VERBAL INTERPRETATION
1.	The animations and visuals help students better understand mathematical concepts.	3.98	0.16	Highly Acceptable
2.	The visual elements support the lessons and make explanations clearer.	3.98	0.16	Highly Acceptable
3.	The animations make learning more meaningful and engaging for students.	4	0	Highly Acceptable
4.	The visuals help simplify abstract or difficult topics for every learner.	4	0	Highly Acceptable
5.	Overall, animations and visuals are useful for achieving the learning objectives of each lesson.	3.95	0.22	Highly Acceptable
	Composite Mean	**3.98**	**0.10**	**Highly Acceptable**

Legend: *1.0-1.75: Not Acceptable; 1.76-2.50: Slightly Acceptable; 2.51-3.25: Moderately Acceptable; 3.26-4.0: Highly Acceptable*

Table 10 presents the respondents' perceptions of the ease of use of animation and visuals. The highest mean scores of 3.98, interpreted as "highly acceptable" were from the statements "The pacing, transitions, and movements are appropriate for every learner", "Text, symbols, and visuals are presented clearly and consistently", and "Students can easily focus on and interact with the animations without difficulty." demonstrating that the overall animation flow and presentation were very smooth, clear, and engaging, indicating that the system was very accessible and inclusive for the different types of learners.

On the other hand, the statement "The animations and visual illustrations are easy to follow and understand" received the lowest mean score of 3.90, which was still interpreted as "highly acceptable." This still indicates that the animations were very effective in helping students understand the content and that the lessons were clear all throughout.

The composite mean of 3.95, which was also equivalent to "highly acceptable," indicates that the respondents mostly considered the interactive

learning aid to be easy to navigate and user-friendly. The results suggest that the interface and visual presentation of the project were well-suited to facilitate the learning experience of special needs students, allowing them to concentrate, interact, and comprehend the lessons with proper assistance.

Table 10. Perceived Ease of Use based on Respondents' Response

	STATEMENTS	MEAN	SD	VERBAL INTERPRETATION
1.	The animations and visual illustrations are easy to follow and understand.	3.90	0.30	Highly Acceptable
2.	The pacing, transitions, and movements are appropriate for every learner.	3.98	0.16	Highly Acceptable
3.	Text, symbols, and visuals are presented clearly and consistently.	3.98	0.16	Highly Acceptable
4.	The visuals support comprehension without causing confusion or overload.	3.93	0.27	Highly Acceptable
5.	Students can easily focus on and interact with the animations without difficulty.	3.98	0.16	Highly Acceptable
	Composite Mean	**3.95**	**0.07**	**Highly Acceptable**

Legend: *1.0-1.75: Not Acceptable; 1.76-2.50: Slightly Acceptable; 2.51-3.25: Moderately Acceptable; 3.26-4.0: Highly Acceptable*

Table 11 shows respondents' perceptions of their attitudes towards using the interactive learning aid. The statement "I have a positive impression of the animations and visual presentations used in the learning aid" obtained the highest mean score of 4.00, which was rated as “highly acceptable.” This implies that all the participants had a similar positive attitude towards the use of animations and visuals, and the design and display of the animations played a major part in the learners' satisfaction and positive interaction with the material.

In contrast, the statement "I believe using animations is an effective way to teach Functional Mathematics" got the lowest mean of 3.95, which is still considered “highly acceptable.” This implies that the respondents consistently

viewed the animated approach as an effective medium for teaching and learning mathematical concepts.

In general, the composite mean of 3.98 reflects a high acceptability level in terms of user attitude, which denotes the fact that the presence of animations and visuals in the interactive learning aid has a great impact on learners' motivation, enjoyment, and perception of Functional Mathematics and their positive acceptance.

Table 11. Attitude toward Using based on Respondents' Response

	STATEMENTS	MEAN	SD	VERBAL INTERPRETATION
1.	The animations and visuals make the lessons more enjoyable to watch and learn from.	3.98	0.16	Highly Acceptable
2.	I believe using animations is an effective way to teach functional mathematics.	3.95	0.22	Highly Acceptable
3.	I have a positive impression of the animations and visual presentations used in the learning aid.	4	0	Highly Acceptable
4.	I would recommend this type of animated and visual learning approach for other lessons.	3.98	0.16	Highly Acceptable
5.	The animations make the learning experience feel more dynamic and engaging.	3.98	0.16	Highly Acceptable
	Composite Mean	**3.98**	**0.08**	**Highly Acceptable**

Legend: *1.0-1.75: Not Acceptable; 1.76-2.50: Slightly Acceptable; 2.51-3.25: Moderately Acceptable; 3.26-4.0: Highly Acceptable*

Table 12 indicates the degree of behavioral intention among the participants to use animations and visuals in teaching and learning mathematics. The highest mean score of 4.00, which means "highly acceptable," was obtained by the statement "I will integrate similar animation-based materials in my teaching/learning strategies." This positively indicates the effectiveness of animated visuals in understanding and engagement with mathematical concepts.

Meanwhile, the statement “I intend to continue using animated and visual learning materials for teaching/learning mathematics” gained the lowest mean of 3.95, but still remained as “highly acceptable,” thus indicating the users' positive perception and continued interest in the application of animation-supported instruction.

The findings present an overall mean of 3.98, which is considered a high composite rating. It is an indicator that the interactive learning aid has been successful in promoting the users' positive behavioral intention, which in turn leads to their ongoing use, recommendation, and integration of similar digital and animation-based resources in educational settings. Thus, it is confirmed that the project has played a significant role in technology acceptance and long-term adoption in the SNED learning environment.

Table 12. Behavioral Intention to Use based on Respondents’ Response

	STATEMENTS	MEAN	SD	VERBAL INTERPRETATION
1.	I intend to continue using animated and visual learning materials for teaching/learning mathematics.	3.95	0.22	Highly Acceptable
2.	I plan to recommend the use of these animations to other teachers or students.	3.98	0.16	Highly Acceptable
3.	I am likely to use these kinds of visual materials in future lessons.	3.98	0.16	Highly Acceptable
4.	I will integrate similar animation-based materials in my teaching/learning strategies.	4	0	Highly Acceptable
5.	I support the continued use of these visuals in classrooms.	3.98	0.16	Highly Acceptable
	Composite Mean	**3.98**	**0.08**	**Highly Acceptable**

Legend: *1.0-1.75: Not Acceptable; 1.76-2.50: Slightly Acceptable; 2.51-3.25: Moderately Acceptable; 3.26-4.0: Highly Acceptable*

Table 13 presents the respondents' perceptions regarding the perceived cost of using animations and visuals in the teaching and learning process. The obtained results revealed that the statements "The benefits of the animations outweigh any cost of production or access," and "Schools can adopt these materials without heavy financial requirements," are equally rated by researchers to a maximum average of 4.00, which is then verbally interpreted as "highly acceptable." Thus, the respondents are in strong agreement that the learning aid is cost-efficient and feasible for classroom use without demanding a high amount of expenses or requiring extensive resources.

Conversely, the statement regarding the cost-effectiveness of the animations and visuals as educational tools obtained the lowest mean score of 3.95, which remains within the "highly acceptable" range. This suggests that while respondents recognize the significant investment of time, specialized software, and technical expertise required for high-quality multimedia production, they perceive these expenditures as justifiable, indicating that the pedagogical benefits, such as increased student engagement and improved retention of mathematical concepts, provide a substantial return on investment. Ultimately, the stakeholders view the enhanced learning experience and the tailored accessibility of the interactive platform as a superior alternative to traditional, less resource-intensive teaching methods.

Overall, the composite mean of 3.98, interpreted as "highly acceptable," further suggests that the interactive learning aid is considered a practical and economical teaching/learning aid and thus, it could be extensively used in the Special Needs Education (SNED) classrooms and other similar learning setups.

Table 13. Perceived Cost based on Respondents' Response

	STATEMENTS	MEAN	SD	VERBAL INTERPRETATION
1.	The animations can be used without requiring costly resources.	3.98	0.16	Highly Acceptable
2.	The time needed to use or integrate the visuals is manageable.	3.98	0.16	Highly Acceptable
3.	The benefits of the animations outweigh any cost of production or access.	4	0	Highly Acceptable
4.	Schools can adopt these materials without heavy financial requirements.	4	0	Highly Acceptable
5.	Overall, the animations and visuals are cost-effective educational tools.	3.95	0.22	Highly Acceptable
	Composite Mean	**3.98**	**0.11**	**Highly Acceptable**

Legend: *1.0-1.75: Not Acceptable; 1.76-2.50: Slightly Acceptable; 2.51-3.25: Moderately Acceptable; 3.26-4.0: Highly Acceptable*

Table 14 presents the perception of the respondents regarding information awareness in using the interactive learning aid. The statements "The content of the visuals clearly shows what the lesson is about," "The animations help clarify common misunderstandings in the topic," and "I gained a better understanding of the lesson through the visuals and animations" received the highest mean score of 3.98, which was interpreted as "highly acceptable." This means that the visual design and animated content were clear, effective, and directly linked to the learners' comprehension of the lesson topics.

On the other hand, the statement "The animations provide new insights or reinforce important mathematics concepts" obtained the lowest mean score of 3.90, but it is still equivalent to "highly acceptable." The results imply that although every feature of the animated visuals was positively received, the areas of real-life application and conceptual reinforcement could be refined further to increase learner engagement and strengthen the overall content.

In summary, the mean score of 3.95 suggests that the respondents considered the animations and visual materials to be “highly acceptable.” This guarantees that the animated parts of the learning material contributed significantly to the process of students' understanding and awareness of the mathematical concepts, attaining the objective of the learning to provide accessible, engaging, and meaningful instruction for learners with special needs.

Table 14. Information Awareness based on Respondents’ Response

	STATEMENTS	MEAN	SD	VERBAL INTERPRETATION
1.	The animations provide new insights or reinforce important mathematics concepts.	3.90	0.30	Highly Acceptable
2.	The content of the visuals clearly shows what the lesson is about.	3.98	0.16	Highly Acceptable
3.	The visual materials increased awareness and understanding of mathematical skills used in daily life.	3.93	0.27	Highly Acceptable
4.	The animations help clarify common misunderstandings in the topic.	3.98	0.16	Highly Acceptable
5.	I gained a better understanding of the lesson through the visuals and animations.	3.98	0.16	Highly Acceptable
	Composite Mean	**3.95**	**0.21**	**Highly Acceptable**

Legend: *1.0-1.75: Not Acceptable; 1.76-2.50: Slightly Acceptable; 2.51-3.25: Moderately Acceptable; 3.26-4.0: Highly Acceptable*

Table 15 shows the results of the evaluation of social influence when assessing the animations integrated into the interactive learning aid. The statement with the highest mean score of 4.00, which is “Positive feedback from teachers or peers influences my willingness to use the animations,” indicates that the students consistently agreed that they consider the learning aid beneficial or relevant, particularly if their teachers and classmates greatly enjoy using the interactive learning aid.

Meanwhile, the statement “Endorsement by teachers or schools increases my trust in the animations” gained a slightly lower mean, which is 3.93, but still indicated as “highly acceptable,” confirming the significant role of the support from teachers and educational institutions in the level of student engagement and trust toward the learning materials and the project itself.

The results reveal a composite mean of 3.97, which is verbally interpreted as “highly acceptable.” This suggests that social influence plays a strong and positive role in the learners’ acceptance and engagement with the interactive animations, indicating that the involvement of peers and teachers can further promote the use of the learning aid in the education setting.

Table 15. Social Influence based on Respondents’ Response

	STATEMENTS	MEAN	SD	VERBAL INTERPRETATION
1.	Positive feedback from teachers or peers influences my willingness to use the animations.	4	0	Highly Acceptable
2.	I am more inclined to use these materials if others recommend them.	3.95	0.22	Highly Acceptable
3.	Endorsement by teachers or schools increases my trust in the animations.	3.93	0.27	Highly Acceptable
4.	Seeing other students engaged with the visuals motivates me to use them too.	3.98	0.16	Highly Acceptable
5.	I am more likely to participate in learning activities that use these kinds of visuals.	3.98	0.16	Highly Acceptable
	Composite Mean	**3.97**	**0.16**	**Highly Acceptable**

Legend: *1.0-1.75: Not Acceptable; 1.76-2.50: Slightly Acceptable; 2.51-3.25: Moderately Acceptable; 3.26-4.0: Highly Acceptable*

Table 16 presents the overall summarized results of the Technology Acceptance Model (TAM) evaluation. The highest mean of 3.98 was obtained by Perceived Usefulness, Attitude toward Using, Behavioral Intention to use, and Perceived Cost, among all the assessed components, which were

interpreted as "highly acceptable." This serves as a leading signal of the practicality, cost-effectiveness, efficiency, and sustainability of the interactive learning aid for educational purposes.

On the other hand, the Perceived Ease of Use and Information Awareness got recorded with a mean of 3.95, which translates still as "highly acceptable." This implies that although a few users might have needed some initial assistance in exploring particular features, the whole system would still be identified as simple and friendly for the users in general. The rest of the dimension, which is Information Awareness, was rated with a mean of 3.97, showing that it was equally accepted with a high score.

In general, with an overall composite mean of 3.97, verbally interpreted as “highly acceptable,” the interactive learning aid was well-received by the users, thus underscoring its effectiveness, ease of use, and relevance for an engaging and accessible learning experience.

Table 16. Summary Mean Table of TAM

	MEAN	SD	VERBAL INTERPRETATION
Perceived Usefulness	3.98	0.10	Highly Acceptable
Perceived Ease of Use	3.95	0.07	Highly Acceptable
Attitude toward Using	3.98	0.08	Highly Acceptable
Behavioral Intention to Use	3.98	0.08	Highly Acceptable
Perceived Cost	3.98	0.11	Highly Acceptable
Information Awareness	3.95	0.21	Highly Acceptable
Social Influence	3.97	0.16	Highly Acceptable
Composite Mean	**3.97**	**0.12**	**Highly Acceptable**

Legend: *1.0-1.75: Not Acceptable; 1.76-2.50: Slightly Acceptable; 2.51-3.25: Moderately Acceptable; 3.26-4.0: Highly Acceptable*

Software Quality Evaluation

Table 17 presents the assessment results in terms of functional suitability for the interactive learning aid. The "highly acceptable" mean of 4.00 was

gained from the statement “The learning aid provides appropriate and accurate lessons," which, upon interpretation, means that the lessons are suitably aligned with the intended learning objectives of the topics it represents and relevant to the learners' educational needs. This signifies that the overall content is properly organized and is in accordance with the SNED curriculum to easily support the learners' comprehension and engagement, meeting their diverse needs.

The statement “All interactive features (e.g., click-to-answer, speech-to-text, animations) work as intended.” gathered the lowest mean of 2.67, but was still interpreted verbally as "moderately acceptable." This indicates that there are still some minor areas left for improvement and optimization to provide seamless functioning of the interactive components.

The composite mean of 3.44, verbally interpreted as "highly acceptable," indicates that the project has satisfactorily performed its functions, and serves the purpose of providing lessons accurately to meet educational objectives and is functionally suitable as a digital educational platform.

Table 17. Functional Suitability based on Respondents’ Response

	STATEMENTS	MEAN	SD	VERBAL INTERPRETATION
1.	The learning aid provides accurate and appropriate lessons.	4	0	Highly Acceptable
2.	All interactive features *(e.g., click-to-answer, speech-to-text, animations)* work as intended.	2.67	0.58	Moderately Acceptable
3.	The learning aid meets the educational objectives of the students.	3.67	0.58	Highly Acceptable
	Composite Mean	**3.44**	**0.33**	**Highly Acceptable**

Legend: *1.0-1.75: Not Acceptable; 1.76-2.50: Slightly Acceptable; 2.51-3.25: Moderately Acceptable; 3.26-4.0: Highly Acceptable*

Table 18 shows the results from the assessment of the performance efficiency of the interactive learning aid. The "highly acceptable" mean of 4.00 was recorded in the statement “Activities run smoothly on standard school laptops, desktops, and similar devices,” showing that the system operates efficiently and performs well within standard school equipment. This result highlights the learning aid's ability to maintain responsiveness in typical classroom technology settings.

The statement "The interactive learning aid performs consistently even with extended use" displayed the lowest mean of 2.67. Although slightly lower compared to other indicators, it still lies within the "moderately acceptable" range and still reflects an acceptable level of system stability, which suggests that the learning aid remains functional during prolonged sessions while maintaining an overall reliable performance.

Overall, the composite mean of 3.44, which corresponds to a "highly acceptable" verbal interpretation, demonstrates that the interactive learning aid operates efficiently, responsively, and reliably behaves in a manner that is consistent with expectations for performance efficiency derived from observation and user experience.

Table 18. Performance Efficiency based on Respondents’ Response

	STATEMENTS	MEAN	SD	VERBAL INTERPRETATION
1.	The learning aid loads quickly and responds without noticeable delay.	3.67	0.58	Highly Acceptable
2.	Activities run smoothly on standard school laptops, desktops, and similar devices.	4	0	Highly Acceptable
3.	The interactive learning aid performs consistently, even with extended use.	2.67	0.58	Moderately Acceptable
	Composite Mean	**3.44**	**0.33**	**Highly Acceptable**

Legend: *1.0-1.75: Not Acceptable; 1.76-2.50: Slightly Acceptable; 2.51-3.25: Moderately Acceptable; 3.26-4.0: Highly Acceptable*

Table 19 shows the assessment results of the compatibility of the interactive learning aid. The highest mean score of 3.67, which is equal to a "highly acceptable" rating, was obtained from the statements “The learning aid works properly across different devices (e.g., laptops, desktops, smart TVs)” and “The learning aid can be accessed by multiple users simultaneously without issues.” These results show that the system effectively works on various devices and accommodates multiple users, making it accessible with the same functionality in various settings.

The lowest average, 2.00, was recorded in the statement “The interface and functions remain consistent on different browsers.” While this result is somewhat low, lying in the range of "slightly acceptable," it still shows good performance that can be improved by optimizing the browsers so that the system behaves uniformly and properly on every platform.

On average, the compatibility attribute reached a composite mean of 3.11, interpreted verbally as "moderately acceptable." This means that the learning aid is still considered compatible across devices and user access, though there are minor areas for enhancement to further raise consistency in performance across different browsers and devices, if necessary.

Table 19. Compatibility based on Respondents’ Response

	STATEMENTS	MEAN	SD	VERBAL INTERPRETATION
1.	The learning aid works properly across different devices (laptops, desktops, etc.).	3.67	0.58	Highly Acceptable
2.	The interface and functions remain consistent on different browsers.	2.00	1.00	Slightly Acceptable
3.	The learning aid can be accessed by multiple users simultaneously without issues.	3.67	0.58	Highly Acceptable
	Composite Mean	**3.11**	**0.24**	**Moderately Acceptable**

Legend: *1.0-1.75: Not Acceptable; 1.76-2.50: Slightly Acceptable; 2.51-3.25: Moderately Acceptable; 3.26-4.0: Highly Acceptable*

Table 20 shows respondents' assessment of the interaction capability of the learning aid developed. The highest mean of 3.67 corresponds to "highly acceptable" and was recorded in the statement “The interface is clear, easy to understand, and accessible for every student.” This shows that the design and layout of the interface are visually organized and appropriate for its users, allowing easy access and comprehension during use. The proper structure of the interface contributes significantly to the overall usability of the learning aid.

Meanwhile, the lowest mean was 3.33, shared by the statements “Instructions and navigation help learners use the web application independently” and “Interactive elements are responsive and support student engagement.” Though these were the lowest, these results still fell into the "highly acceptable" range, demonstrating that the instructional and interactive features have been successful in encouraging user participation and navigation within the system.

Overall, a composite mean of 3.44 for "highly acceptable" confirms that the system demonstrates a strong level of interaction capability, as well as promoting accessibility and engagement throughout the learning experience.

Table 20. Interaction Capability based on Respondents’ Response

	STATEMENTS	MEAN	SD	VERBAL INTERPRETATION
1.	The interface is clear, easy to understand, and accessible for every student.	3.67	0.58	Highly Acceptable
2.	Instructions and navigation help learners use the web application independently.	3.33	0.58	Highly Acceptable
3.	Interactive elements are responsive and support student engagement.	3.33	0.58	Highly Acceptable
	Composite Mean	**3.44**	**0.58**	**Highly Acceptable**

Legend: *1.0-1.75: Not Acceptable; 1.76-2.50: Slightly Acceptable; 2.51-3.25: Moderately Acceptable; 3.26-4.0: Highly Acceptable*

Table 21 shows how reliable the interactive learning aid is according to the responses of the respondents. The indicative statement "The learning aid operates without crashes or major errors during use" obtained an average of 3.67, equivalent to "highly acceptable." This clearly shows that the learning aid works well all the time it is used and is still reliable when used on a regular basis.

The other statements, "Lessons and activities can be completed without data loss or interruptions" and "The learning aid remains stable and dependable during extended periods of use," each received a mean rating of 3.33, which is still considered "highly acceptable." This indicates that the interactive learning aid is considered both stable and responsive even after long periods of use.

In general, the average score of 3.44 indicates that the project reliability is rated as "highly acceptable." Thus, the interactive learning aid continuously provides the user with a reliable performance, and this enables the users to fully benefit from its features without experiencing significant issues.

Table 21. Reliability based on Respondents' Response

	STATEMENTS	MEAN	SD	VERBAL INTERPRETATION
1.	The learning aid operates without crashes or major errors during use.	3.67	0.58	Highly Acceptable
2.	Lessons and activities can be completed without data loss or interruptions.	3.33	0.58	Highly Acceptable
3.	The learning aid remains stable and dependable during extended periods of use.	3.33	0.58	Highly Acceptable
	Composite Mean	**3.44**	**0.58**	**Highly Acceptable**

Legend: *1.0-1.75: Not Acceptable; 1.76-2.50: Slightly Acceptable; 2.51-3.25: Moderately Acceptable; 3.26-4.0: Highly Acceptable*

Table 22 shows the evaluation result of the security aspect of the project. For the statement “The web app ensures that personal data is not exposed to

unauthorized users," the highest mean of 3.67, which corresponds to a "highly acceptable" rating, was reported. This shows that the respondents had great confidence in the data protection measures of the system and in keeping up with privacy standards. It also indicates that the web application was being perceived as a trustworthy instrument for protecting user data and barring unauthorized access.

The lowest mean values of 3.33 were noted for the statements "User accounts and information (e.g., name, email) are protected within the web application" and "Only authorized users can access their profiles and activities." The following scores are comparatively lower but still remain within the "highly acceptable" range, indicating that the web application is providing effective, secure access and data protection facilities for all users.

The composite mean of 3.44, in this case, conveys that the web application's security features are "highly acceptable." The implications of these results are that the project is successful in keeping user information confidential and in making the education tool inherently acceptable for a safe and secure learning environment, as it prevents the leakage of sensitive information.

Table 22. Security based on Respondents' Response

	STATEMENTS	MEAN	SD	VERBAL INTERPRETATION
1.	User accounts and information (e.g., name, email) are protected within the web application.	3.33	0.58	Highly Acceptable
2.	Only authorized users can access their profiles and activities.	3.33	0.58	Highly Acceptable
3.	The web app ensures that personal data is not exposed to unauthorized users.	3.67	0.58	Highly Acceptable
	Composite Mean	**3.44**	**0.58**	**Highly Acceptable**

Legend: *1.0-1.75: Not Acceptable; 1.76-2.50: Slightly Acceptable; 2.51-3.25: Moderately Acceptable; 3.26-4.0: Highly Acceptable*

Table 23 presents how the respondents rated the maintainability of the learning aid. The highest mean score of 3.67 was given to the statement "Bugs and errors can be fixed without affecting other parts of the learning aid or causing confusion for student users." This indicates that the learning aid's structure was recognized as quite stable and controllable, hence facilitating issues being solved quickly without interfering with other system functions.

The lowest means, both of 3.33, were attributed to the statements "The interactive learning aid's structure allows for easy updates or improvements without altering the familiar interface used by students" and "The project can be maintained smoothly over time, ensuring stable performance and accessibility for learners." Although these scores are the lowest, they are still in the "highly acceptable" range, which means that the respondents still saw maintaining the system and giving it consistent performance as a possible challenge, but not a hard one to overcome.

Generally, the maintainability feature received a mean score of 3.44, which was interpreted as "highly acceptable." This implies that the system is not only well constructed but also can be maintained for a long time without losing its quality or accessibility to the learners.

Table 23. Maintainability based on Respondents' Response

	STATEMENTS	MEAN	SD	VERBAL INTERPRETATION
1.	The interactive learning aid's structure allows for easy updates or improvements without altering the familiar interface used by students.	3.33	0.58	Highly Acceptable
2.	Bugs and errors can be fixed without affecting other parts of the learning aid or causing confusion for student users.	3.67	0.58	Highly Acceptable
3.	The project can be maintained smoothly over time, ensuring stable performance and accessibility for learners.	3.33	0.58	Highly Acceptable
	Composite Mean	**3.44**	**0.58**	**Highly Acceptable**

Legend: *1.0-1.75: Not Acceptable; 1.76-2.50: Slightly Acceptable; 2.51-3.25: Moderately Acceptable; 3.26-4.0: Highly Acceptable*

Table 24 presents the assessment results of the flexibility aspect of the learning aid. The mean of 3.67, verbally interpreted as "highly acceptable," was recorded for the statement "The learning aid remains usable and effective across different environments and teaching contexts without requiring major web application changes," among the indicators. This result indicates that the system is adaptable to different educational settings and that it remains consistent in terms of usability and performance regardless of the environment it is used in.

On the other hand, the lowest but still "highly acceptable" mean of 3.33 was achieved in the statements "The learning aid can adapt to different learning setups (e.g., classroom, computer lab) and supports individual pacing and varied teaching strategies" and "The web app allows adjustments to meet varying student needs (e.g., audio instructions, or alternative interaction methods) or school requirement." These findings signify that the learning aid effectively supports different teaching setups and accessibility options, which cater to the needs of the diverse classroom.

The project achieved a composite mean of 3.44 for flexibility, resulting in a "highly acceptable" interpretation. This score suggests that the system allows for seamless transitions between various instructional modes, such as individualized self-paced study and teacher-led demonstrations. Such flexibility is critical in a Special Needs Education (SNED) setting, where delivery must be modified to suit the unique cognitive and behavioral profiles of each learner. By accommodating these diverse requirements, the system ensures that learning objectives remain reachable despite varying hardware constraints or environmental settings.

Table 24. Flexibility based on Respondents' Response

	STATEMENTS	MEAN	SD	VERBAL INTERPRETATION
1.	The learning aid can adapt to different learning setups *(e.g., classroom, computer lab*) and supports individual pacing and varied teaching strategies.	3.33	1.15	Highly Acceptable
2.	The web app allows adjustments to meet varying student needs *(e.g., audio instructions or alternative interaction methods)* or school requirements.	3.33	0.58	Highly Acceptable
3.	The learning aid remains usable and effective across different environments and teaching contexts without requiring major web application changes.	3.67	0.58	Highly Acceptable
	Composite Mean	**3.44**	**0.33**	**Highly Acceptable**

Legend: *1.0-1.75: Not Acceptable; 1.76-2.50: Slightly Acceptable; 2.51-3.25: Moderately Acceptable; 3.26-4.0: Highly Acceptable*

Table 25 shows the results of the evaluation for the safety attribute of the interactive learning aid. The highest mean of 4.00 was obtained by the statement "The web-app operates in a way that does not expose learners or devices to harm or inappropriate content," which is verbally interpreted as "highly acceptable."

The statements— "The learning aid functions properly under defined conditions without compromising data or device security" and "The platform promotes a safe and controlled digital learning environment"—all had a mean score of 3.67, still interpreted as "highly acceptable," which means that the system provides a secure and well-regulated experience for the users.

A composite mean of 3.78 indicates that the interactive learning aid has been successful in adhering to safety standards in both content and operation. This proves that the interactive learning aid can generate a secure and safe digital learning space by taking proper care of the security of user data and

device functionality, as well as providing a suitable environment for the students.

Table 25. Safety based on Respondents' Response

	STATEMENTS	MEAN	SD	VERBAL INTERPRETATION
1.	The web-app operates in a way that does not expose learners or devices to harm or inappropriate content.	4	0	Highly Acceptable
2.	The learning aid functions properly under defined conditions without compromising data or device security.	3.67	0.58	Highly Acceptable
3.	The platform promotes a safe and controlled digital learning environment.	3.67	0.58	Highly Acceptable
	Composite Mean	**3.78**	**0.33**	**Highly Acceptable**

Legend: *1.0-1.75: Not Acceptable; 1.76-2.50: Slightly Acceptable; 2.51-3.25: Moderately Acceptable; 3.26-4.0: Highly Acceptable*

Table 26 summarizes the software quality evaluation of the interactive learning aid using the ISO 25010 model. Among all the parameters, the Safety attribute was rated the highest with an average of 3.78, corresponding to the verbal interpretation of "highly acceptable." This implies that the digital learning environment provided by the system is secure and controlled, and protection is offered to both the data and the functioning of the devices.

Conversely, the Compatibility attribute was given the lowest score of 3.11, which corresponds to the verbal interpretation of "moderately acceptable." Nevertheless, it can still be concluded from the result that the learning aid provides a similar performance across different devices and platforms, together with no barriers in terms of access and usability.

In summary, the study shows that most of the quality characteristics, including Functional Suitability, Performance Efficiency, Interaction Capability, Reliability, Security, Maintainability, and Flexibility, also had means of 3.44,

which were indicated as high ratings of “highly acceptable.” With an overall composite mean of 3.44, verbally interpreted as “highly acceptable,” this indicates that the interactive learning aid is capable of meeting the required software quality standards as per ISO 25010, thus confirming its reliability, efficiency, and adaptability for prolonged use in the education sector.

Table 26. Summary Mean Table of ISO 25010

	MEAN	SD	VERBAL INTERPRETATION
Functional Suitability	3.44	0.33	Highly Acceptable
Performance Efficiency	3.44	0.33	Highly Acceptable
Compatibility	3.11	0.24	Moderately Acceptable
Interaction Capability	3.44	0.58	Highly Acceptable
Reliability	3.44	0.58	Highly Acceptable
Security	3.44	0.58	Highly Acceptable
Maintainability	3.44	0.58	Highly Acceptable
Flexibility	3.44	0.33	Highly Acceptable
Safety	3.78	0.33	Highly Acceptable
Composite Mean	**3.44**	**0.43**	**Highly Acceptable**

Legend: *1.0-1.75: Not Acceptable; 1.76-2.50: Slightly Acceptable; 2.51-3.25: Moderately Acceptable; 3.26-4.0: Highly Acceptable*

CHAPTER 5

SUMMARY, CONCLUSION, AND RECOMMENDATIONS

This chapter outlines the main plans, approaches, development processes, testing procedures, and implementation plans carried out during the study. It also highlights the key findings drawn from the assessment and the feedback from the users about the project entitled "Functional Mathematics: An Interactive Learning Aid for Students with Special Needs." Furthermore, this section discusses the conclusions drawn from the outcomes and presents recommendations to enhance the project's effectiveness, usability, and sustainability for potential application in learning settings.

Summary of Findings

This project, “Functional Mathematics: An Interactive Learning Aid for Students with Special Needs,” was created to enhance and support the teaching and learning of Functional Mathematics among Special Needs Education (SNED) students. The project provides the need for accessible, engaging, and curriculum-aligned learning materials that are suited to the diverse learning needs of SNED learners.

The project development was aligned with a systematic approach, which started by determining the core competencies in Functional Mathematics, including curriculum standards, learning challenges faced by the students, and skills and knowledge areas that should be addressed. In addition, teacher feedback and curriculum-based lesson plans were used to conceptualize and build the structure of the animated lessons, activities, and interactive aspects

of the learning aid to guarantee that the overall content will be appropriate and applicable to actual classroom practices.

The interactive learning aid was developed as a desktop web application consisting of three main topics, namely, counting and recognizing numbers, basic operations, and money recognition. Each includes downloadable video discussions and Portable Document Format (PDF) reviewer notes. Activities were also aligned according to the students' level, which are beginner, proficient, and advanced. In line with this, varied mode of answering was also included, such as click-to-answer, drag-and-drop, type/encode, speech-to-text, and calculator, as well as text-to-speech for instructions and questions, to support the different learning approaches of the learners. Short animated skits that serve as an opening and ending scene for each level were embedded to provide visual context and enhance understanding. Lastly, as an overall reflection for the whole learning aid, an animated learning wrap-up will be shown after all the activities have been answered and completed.

The project went through software development and evaluation phases for the purpose of ensuring the suitability, efficiency, and learning effectiveness. The ISO 25010:2023 Software Quality Model helped in determining the software quality in terms of functional suitability, performance efficiency, compatibility, interaction capability, reliability, security, maintainability, flexibility, and safety. On the other hand, the Technology Acceptance Model (TAM) helped in the measurement of the perceived usefulness, ease of use, attitude, behavior, perceived cost, information awareness, and social influence of the visual aspects and animations.

Testing and evaluations with teachers and SNED students were also conducted to assess the performance and overall content of the interactive learning aid. High acceptance rates were registered for the interactive system and the learning material animations. The learning aid was observed to be operational, user-friendly, and efficient in operating with regular school hardware, and the animations were efficient in promoting student interaction and understanding.

Overall, this study was successful in terms of developing an interactive learning aid, which also demonstrates alignment with SNED learning outcomes, inclusive learning methodologies, and meets learning software quality conditions. The design facilitates sustainable classroom implementations and generates a useful prototype with future potential in the incorporation of interactive, visualized, and animated materials into specialized learning programs.

Conclusion

The project executed the integration of interactive features, 2D animation, and desktop-based web technology to develop an effective teaching and learning instrument for Functional Mathematics, specialized for students with special needs. Throughout the process of project development, all activities, animated components, and the interactive features were mapped to the curriculum standards of the Special Needs Education (SNED) program to ensure that the overall content remained appropriate, structured, and aligned with the learners' educational goals. The project attained its objectives as summarized:

1. **Identify the core competencies in Functional Mathematics.**

Curriculum-based lesson plans served as the foundation for structuring the overall content so that the lessons remained focused on relevant skills and knowledge areas outlined in the SNED program and curriculum. This process facilitated the planning of animated and visual content, instructional materials, and interactive activities, making the learning process structured and aligned with the students' actual classroom practices.

2. **Develop the interactive learning aid.**

The researchers properly designed and developed an interactive learning aid via a desktop web application that incorporates 2D-animated skits and interactive features. The interface of the learning platform was created to be user-friendly, ensuring its ease of use, accessibility, and engagement, to empower the students' independence to learn at their own pace.

3. **Evaluate the effectiveness and acceptability of the project using the Technology Acceptance Model (TAM) and ISO 25010.**

The respondents in this study included 40 students with special needs and 3 teachers of SNED at Los Baños National High School-Poblacion (LBNHS-Pob.). Results showed high acceptability at all TAM aspects, and this implies that animations and the visual aspects benefited understanding and interaction. Furthermore, the ISO 25010 software quality model established that the system satisfied fundamental software quality attributes in terms of functional suitability, performance

efficiency, compatibility, interaction capability, reliability, security, maintainability, flexibility, and safety.

Recommendations

Guided by the findings and results of the study, the following recommendations have been gathered and presented to further enhance the development, usability, and learning effectiveness of the project entitled “Functional Mathematics: An Interactive Learning Aid for Students with Special Needs.” The goal of these suggestions is to widen the operational efficiency and accessibility of the learning aid, while ensuring its long-term effectiveness as a learning material. By incorporating these recommendations, educators, developers, and stakeholders can provide an inclusive and sustainable learning experience to students with special needs:

1. **Expand accessibility features for different disabilities:** Future project enhancements may incorporate other accessibility features to accommodate students with other special learning needs, including those who are deaf or hard-of-hearing, through the inclusion of sign language interpretation and other visual cues.
2. **Develop a version for mobile devices or a standalone application:** To make it more accessible and flexible, it is recommended to convert the learning aid into a mobile or standalone application. The said conversion provides the learners with the benefit of being able to access the learning platform even when there is no network or internet connection, or when they are in an environment that is outside the

classroom. This way, it can promote learning independence by allowing the learners to continuously learn at all times and in all places.

3. **Regular content review and technical maintenance:** Periodically, the web application, along with its content, needs to be checked and updated to confirm its quality, reliability, and good performance. This will include fixing the bugs, making the app work on other devices or browsers, and going through the lesson contents to validate that they are still aligned with the current curriculum standards.
4. **Provide training for teachers:** Training or workshop sessions could be held to upgrade their knowledge and skills to use visual elements, animations, and interactive components in their teaching methods. Consequently, the learning material will be part of the classroom practices and will be efficiently used to cater to various learning needs.
5. **Explore integration with other learning areas**: Future studies and development may be conducted to expand the other learning areas aside from Functional Mathematics. By adapting the same framework and format, the interactive learning aid can support broader subject coverage while still providing accessibility and inclusivity.

REFERENCES

Alanazi, A. S., Almulla, A. A., & Khasawneh, M. A. S. (2023). Exploring the E-learning Supporting Systems for Students with Special Needs. Revista de Gestão Social e Ambiental, retrieved from proquest.com: https://www.proquest.com/openview/9bb51e6b5fea7c5bd147313fd1b18f46/1?cbl=2031968&pq-origsite=gscholar

Alcantara, A., Bangotb, S., Geronac, M., Jimenezd, M., Talone, J., Cagape, W., EdD, PhDf (2023). Lived experiences of Special Needs Education Professionals in Handling Learners with Intellectual Disability. Retrieved from https://www.researchgate.net/profile/Wenefredo-Cagape/publication/371285822_Lived_experiences_of_Special_Needs_Education_Professionals_in_Handling_Learners_with_Intellectual_Disability/links/654a0587ce88b87031d1a426/Lived-experiences-of-Special-Needs-Education-Professionals-in-Handling-Learners-with-Intellectual-Disability.pdf

Ballin, A., Davidson, E., Caron, J., & Drago, M. (2022). Making math add up for students receiving special education. International Journal of Whole Schooling, Retrieved from eric.ed,gov: https://eric.ed.gov/?id=EJ1339890

Benedict, J., Green, D., & Watson, A. (2020). Using digital tools to differentiate instruction for students with special needs: A review of current practices. Journal of Special Education Technology, 35(2), 124-135.

Chen, L., et al. (2020). Peer language resources and language development in inclusive classrooms. Journal of Inclusive Education, 22(4), 345–360.

Cleveland Clinic. (2025). Attention-Deficit/Hyperactivity Disorder. ADHD. Retrieved from https://my.clevelandclinic.org/health/diseases/4784-attention-deficithyperactivity-disorder-adhd

Erdem, R. (2017). Students with Special Educational Needs and Assistive Technologies: A Literature Review. TOJET: The Turkish Online Journal of Educational Technology, retrieved from eric.ed.gov: https://eric.ed.gov/?id=EJ1124910

Espeño, S.R., Babiano, E.H., Bucoy, M.R., et al. (2024). Issues and Challenges of Implementing Special Education (SPED) Curriculum in the Philippines: A Systematic Literature Review. Retrieved from https://edukhasi.org/index.php/jip/article/view/390?utm_source

Fajriani, A., Adnan, A., & Faisal, F. (2023). The effect of interactive multimedia on students' cognitive learning outcomes. Biodidaktika: Jurnal Biologi dan Pembelajarannya, 18(1), 9–23. https://doi.org/10.30870/biodidaktika.v18i1.17808

Ferguson, J., et al. (2020). Inclusive early intervention for children with ASD: Impact on language development. Journal of Autism Research, 14(2), 120–135.

Franco, J. (2023). Socioeconomic Issue on Spotlight: Persons with Disabilities. Philippine Institute for Development Studies. Retrieved from https://serp-p.pids.gov.ph/feature/public/index-view?feauredtype_id=1&slug=persons-with-disabilities#:~:text=Agbon%20and%20Mina%20(2017)%20revealed%20that%20the,to%20a%20rising%20dropout%20rate%20among%20them

Garcia, M., Patel, N., & Lin, C. (2021). Visual learning aids and engagement among students with autism spectrum disorder. Journal of Autism and Developmental Disorders, 51(4), 1987-2003.

Gernot Starke. (2023). Arc42 Quality Model: System and Product Quality Made Easy. Retrieved from https://quality.arc42.org/

Harahap, L. W., & Surya, E. (2017). Development of Learning Media in Mathematics for Students with Special Needs. International Journal of Sciences: Basic and Applied Research. Retrieved from https://core.ac.uk/download/pdf/249335803.pdf

International Organization for Standardization. (2023). Systems and Software Engineering — Systems and Software Quality Requirements and Evaluation (SQuaRE) — Product Quality Model. Retrieved from https://www.iso.org/standard/78176.html

James R. Lewis. (2021). Comparison of Four TAM Item Format: Response Option Labels and Order. Retrieved from https://uxpajournal.org/tam-formats-effect-response-labels-order/

Kassim, A., & Nordin, M. N. (2024). AN EFFECTIVE TEACHING AIDS USING VISUAL, AUDITORY AND KINESTHETIC LEARNING STYLES FOR STUDENTS WITH SPECIAL NEEDS. Special Education. Retrieved from: https://prademicpress.com/index.php/specialeducation/article/view/9

Kucirkova, N., & Cremin, T. (2020). Improving engagement in learning through animated tools: A case study in primary education. Journal of Computer-Assisted Learning, 36(4), 489-498. https://doi.org/10.1111/jcal.12402

Lambert, M., et al. (2020). Enhancing mathematical thinking in students with autism. Journal of Special Education Technology, 35(3), 215–229.

Lei, X., et al. (2020). Teacher scaffolding in mathematics for students with disabilities. Journal of Special Education Research, 15(2), 45–60.

Luo, Y., Weibman, D., Halperin, J. M., & Li, X. (2019). A Review of Heterogeneity in Attention Deficit/Hyperactivity Disorder (ADHD). Frontiers in Human Neuroscience, retrieved from: https://www.frontiersin.org/journals/human-neuroscience/articles/10.3389/fnhum.2019.00042/full

Mayer, R. (2024). The Past, Present, and Future of the Cognitive Theory of Multimedia Learning. Educational and Psychology Review. Retrieved from https://link.springer.com/article/10.1007/s10648-023-09842-1#:~:text=Abstract,based%20principles%20of%20multimedia%20design

McDougal, E., Tai, C., Stewart, T. M., Booth, J. N., & Rhodes, S. M. (2023). Understanding and Supporting Attention Deficit Hyperactivity Disorder (ADHD) in the Primary School Classroom: Perspectives of Children with ADHD and their Teachers. Journal of Autism and Developmental Disorders, retrieved from https://link.springer.com/article/10.1007/s10803-022-05639-3

NIMH. (2024). Autism Spectrum Disorder. National Institute of Mental Health. NIMH Information Resource Center. Retrieved from https://www.nimh.nih.gov/health/topics/autism-spectrum-disorders-asd

O'Neill, J., & Topping, K. (2019). Using visual aids to improve academic performance for students with ADHD: A review. Journal of Special Education Technology, 34(4), 273-285. https://doi.org/10.1177/0162643419849264

Pocaan, J.M. (2022). Exploring Teaching Strategies and Challenges towards a Holistic Context-Based Special Education Teaching Strategies Program. The Normal Lights. Retrieved from https://www.researchgate.net/publication/361487607_Exploring_teaching_strategies_and_challenges_towards_a_holistic_context-based_special_education_teaching_strategies_program_The_Normal_Lights

Priya, Sr. (2018). SDLC – Sequential Model. H2K Information Systems. Retrieved from https://www.h2kinfosys.com/blog/sdlc-sequential-model/?fbclid=IwY2xjawOEhtNleHRuA2FlbQIxMABicmlkETFYek9VVENOUjdTZXpVZnROc3J0YwZhcHBfaWQQMjIyMDM5MTc4ODIwMDg5MgABHp1FIQZwha2A2TpwT_5pHTYZQ_Q-dTLLGXsp951xGphLPjlADDXNMPr3s7Am_aem_qrhPgshFERJckVMJCDeaBw

Rasnawati, A., Rahmawati, W., Akbar, P., & Putra, H. D. (2019). Analisis kemampuan berfikir kreatif matematis siswa SMK pada materi sistem persamaan linier dua variabel (SPLDV) di kota Cimahi. Jurnal Cendekia:

Jurnal Pendidikan Matematika, 3(1), 164–177. Retrieved from https://d1wqtxts1xzle7.cloudfront.net/111997966/71-libre.pdf?1709329224=&response-content-disposition=inline%3B+filename%3DAnalisis_Kemampuan_Berfikir_Kreatif_Mate.pdf&Expires=1743482540&Signature=dnEvnAhJ5pJnZKeoU99Sm42Qr4McSgIXqtw-aUVxDjGQkOHN1P5RMEJdWb4g__&Key-Pair-Id=APKAJLOHF5GGSLRBV4ZA

Rebecca Gross. (2024). Canva Design Inspiration: Designing Website for Kids: Trends and Best Practices. Retrieved from https://www.canva.com/learn/kids-websites/

Rosamsi, S., Miarsyah, M., & Ristanto, R. H. (2019). Interactive multimedia effectiveness in improving cell concept mastery. Journal of Biology Education, 8(1), 56–61. Retrieved from https://journal.unnes.ac.id/sju/ujbe/article/view/28154

Sabaruddin, S., Mansor, R., Rusmar, I., & Husna, F. (2020). Student with Special Needs and Mathematics Learning: A Case Study of an Autistic Student. Journal of Research and Advances in Mathematics Education. Retrieved from https://files.eric.ed.gov/fulltext/EJ1280943.pdf

Sanchhaya Education Private Limited. (2023). Software Development Life Cycle (SDLC). Retrieved from https://www.geeksforgeeks.org/software-engineering/software-development-life-cycle-sdlc/

Santos, J. (2023). A Qualitative Study: Special Education Teachers' Perceptions of Instruction during COVID-19. American College of Education ProQuest Dissertations & Theses. Retrieved from https://www.proquest.com/openview/34d303f27deac55755313ec2f3542e1d/1?cbl=18750&diss=y&pq-origsite=gscholar

Smith, A., & Jones, R. (2020). Interactive learning tools for students with cognitive impairments: A longitudinal study. Journal of Learning Disabilities Research, 35(3), 287-305.

Stingo, J. (2024). The Effects of Differentiated Instruction and Individualized Instruction on Special Education Students. Retrieved from https://www.researchgate.net/publication/380392960_The_Effects_of_Differentiated_Instruction_and_Individualized_Instruction_on_Special_Education_Students

Subarna, M. T. N., Masud, N. A., Mensah, J., San, S. S. S., Hasan, M., & Tania, J. S. (2022). Teaching Strategies for Students with Disabilities in Regular Classes. Creative Education, retrieved from https://www.scirp.org/journal/paperinformation?paperid=117809

Ugalde, L., Santiago-Garabieta, M., Villarejo-Carballido, B., & Puigvert, L. (2021). Impact of interactive learning environments on learning and cognitive development of children with special educational needs: A literature review. Frontiers in Psychology, 12, 674033. https://doi.org/10.3389/fpsyg.2021.674033

United Nations Educational, Scientific, and Cultural Organization. UNESCO. (2023). Technology in Education. Good, Impartial Evidence Report. Retrieved from https://gem-report-2023.unesco.org/technology-in-education/

Upa, Y., & Mbato, C. L. (2020). Teacher Identity Construction: Indonesian Teachers' Motivation and Strategies in Teaching Mathematics for Special Needs Students. PROJECT (Professional Journal of Education), retrieved from https://core.ac.uk/download/pdf/323115397.pdf

Wilsa, A. W. (2022). The Effect of Interactive Multimedia-Based Learning on Students' Mathematical Critical Thinking Ability in the Course of Structure Algebra. Jurnal Didaktik Matematika. Retrieved from https://www.researchgate.net/publication/365796415_The_Effect_of_Interactive_Multimedia-Based_Learning_on_Students'_Mathematical_Critical_Thinking_Ability_in_the_Course_of_Structure_Algebra/fulltext/6384abe0c2cb154d2937ae2c/The-Effect-of-Interactive-Multimedia-Based-Learning-on-Students-Mathematical-Critical-Thinking-Ability-in-the-Course-of-Structure-Algebra.pdf?origin=journalDetail

Appendix A

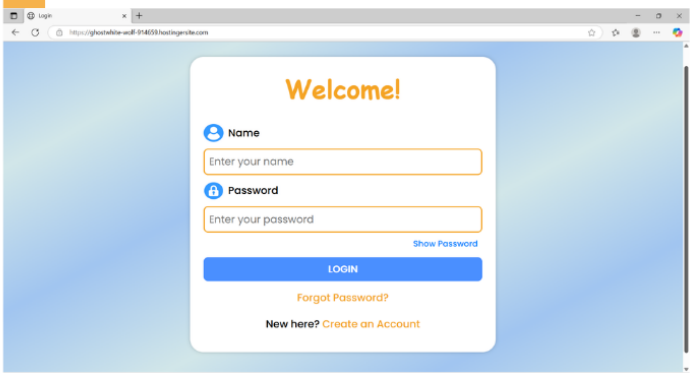
Interface and Storyboard

FUNCTIONAL MATHEMATICS: AN INTERACTIVE LEARNING AID FOR STUDENTS WITH SPECIAL NEEDS

FLOW AND INTERFACE OF THE ACTUAL DESKTOP WEB APPLICATION (INTERACTIVE LEARNING AID)

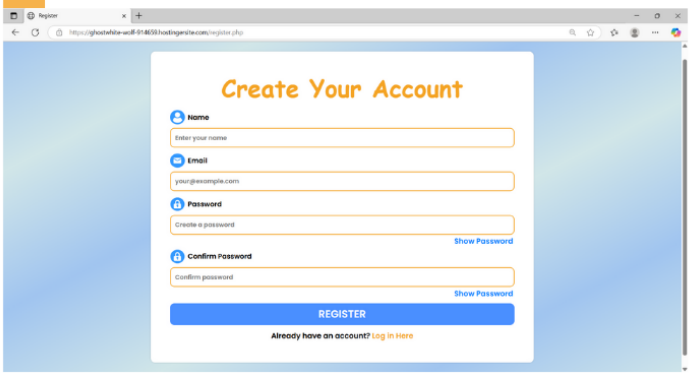
WEB BROWSER: **Microsoft Edge**

1



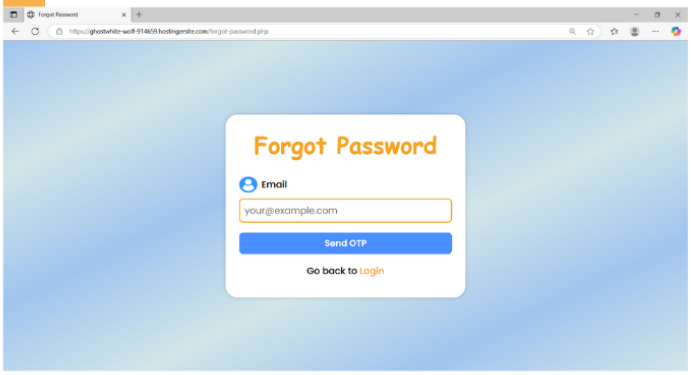
LOGIN PAGE

2



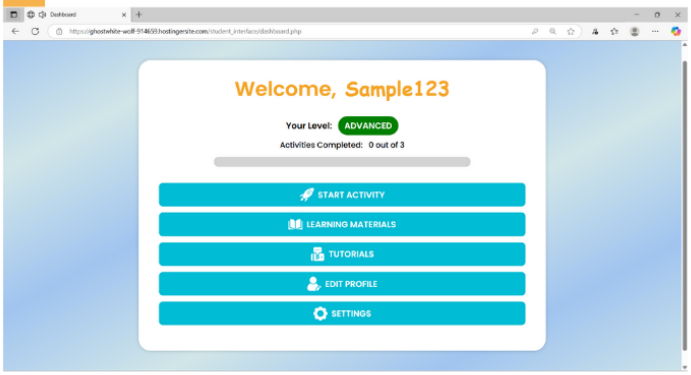
ACCOUNT CREATION PAGE

3



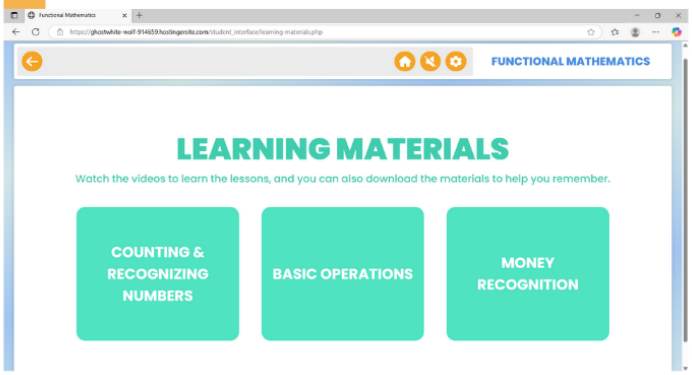
'FORGOT PASSWORD' PAGE

4



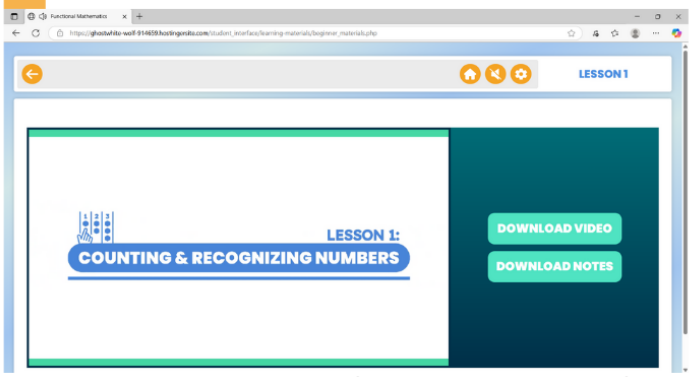
USER DASHBOARD

5



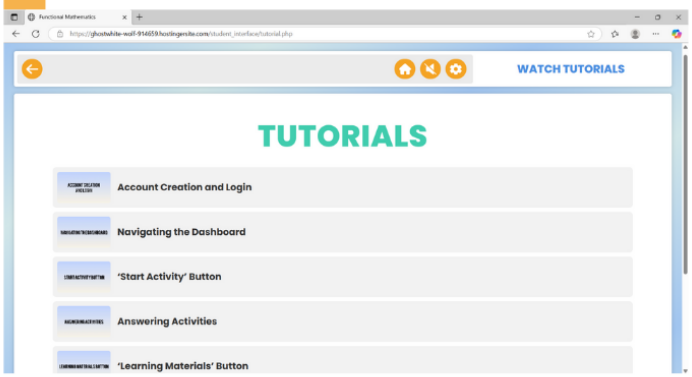
'LEARNING MATERIALS' PAGE

6



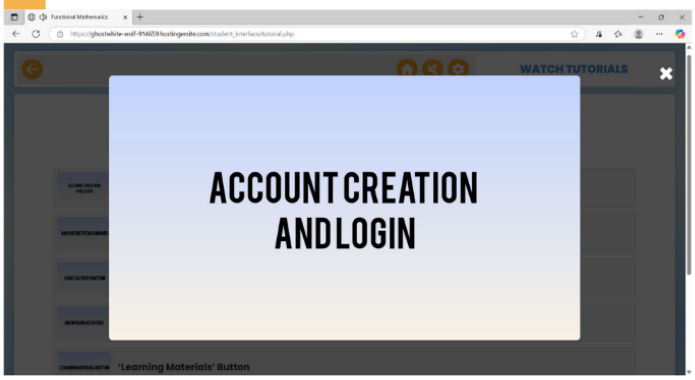
SAMPLE LEARNING MATERIAL (WITH DOWNLOAD OPTIONS)

7



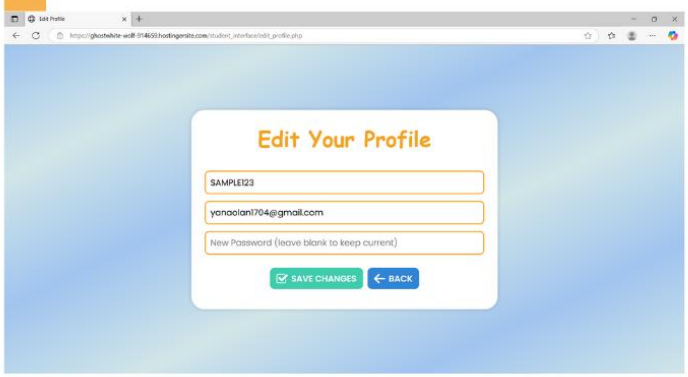
'TUTORIALS' PAGE

8



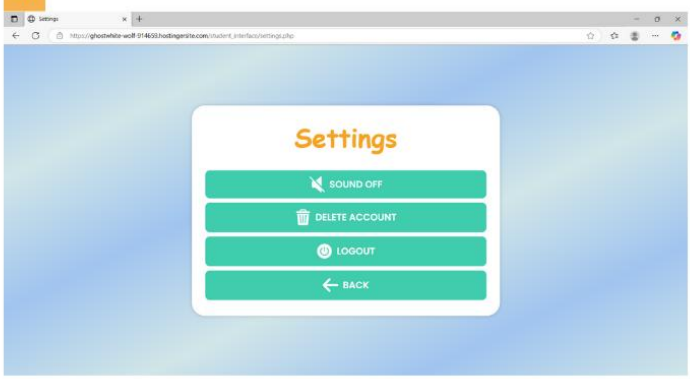
SAMPLE TUTORIAL VIDEO

9



'EDIT PROFILE' PAGE

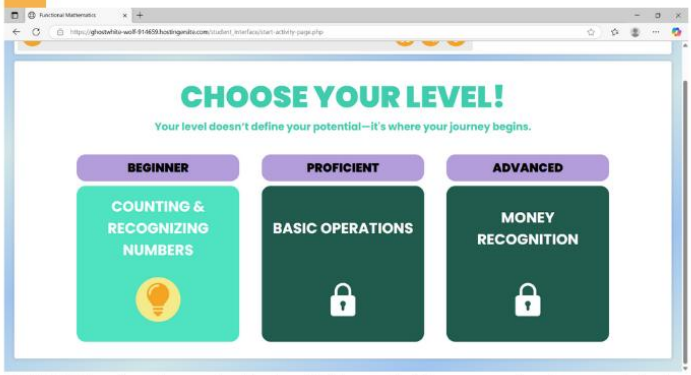
10



'SETTINGS' PAGE

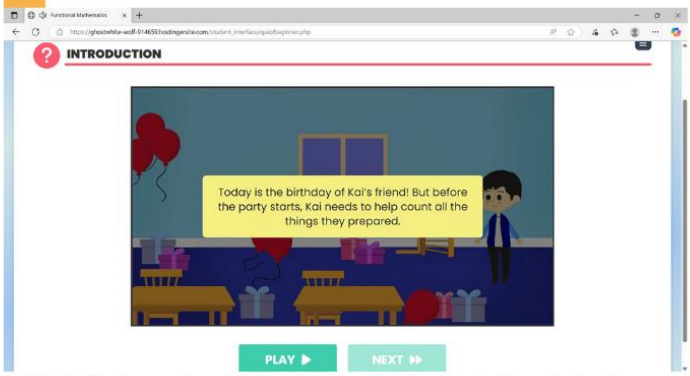
-START ACTIVITY : ***ANSWERING ACTIVITIES-***
BEGINNER LEVEL

11 *'START ACTIVITY' PAGE - BEGINNER LEVEL*



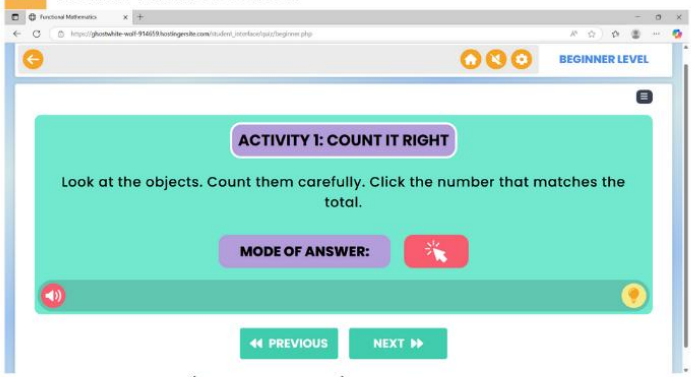
NOTE: Other levels are locked until the activities are not yet completed.

12 *INTRODUCTION / OPENING SKIT*



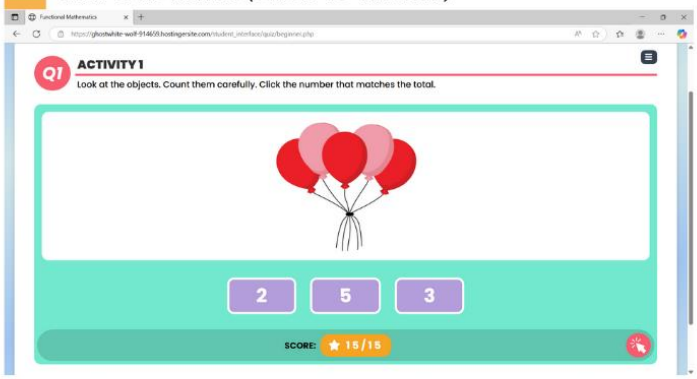
NOTE: A short opening skit to introduce the lesson to the students.

13 *ACTIVITY INTRODUCTION*



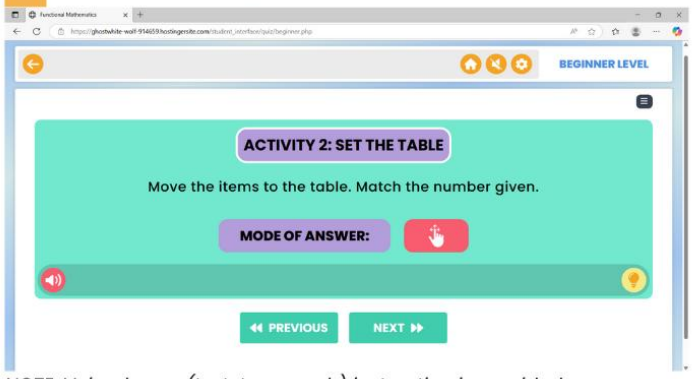
NOTE: Voiced-over (text-to-speech) instruction is provided.

14 *SAMPLE ACTIVITY 1 (CLICK-TO-ANSWER)*



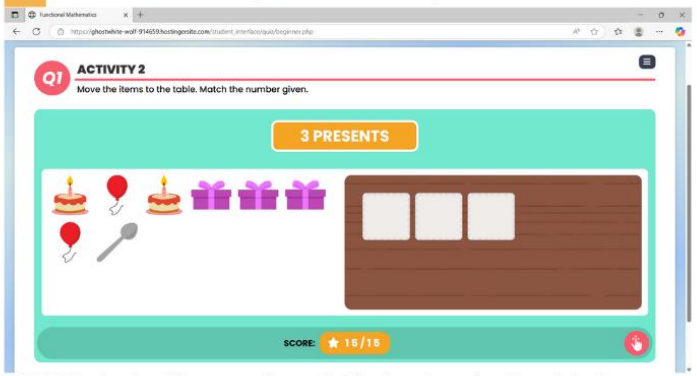
NOTE: Students will answer the activities by clicking the correct answer.

15 *ACTIVITY 2 INTRODUCTION*



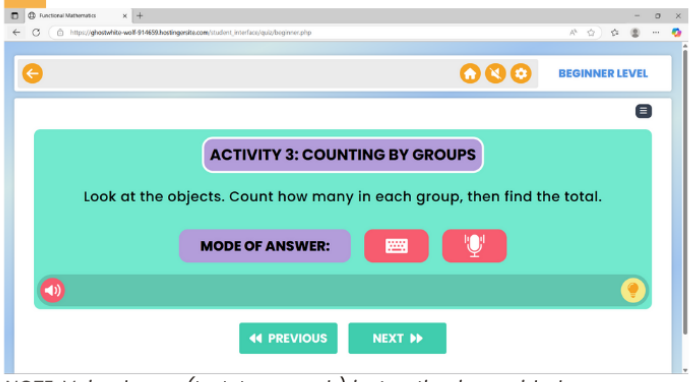
NOTE: Voiced-over (text-to-speech) instruction is provided.

16 *SAMPLE ACTIVITY 2 (DRAG-AND-DROP)*



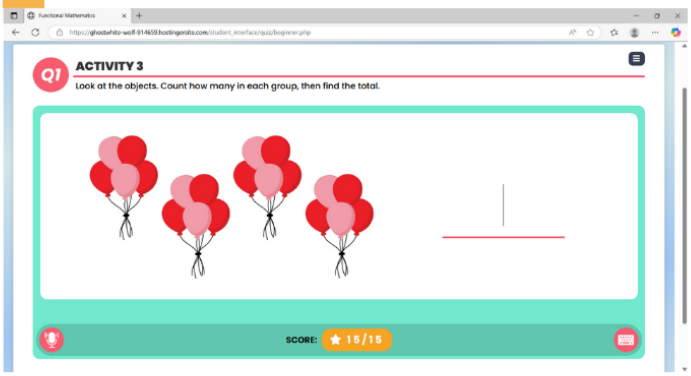
NOTE: Students will answer the activities by dragging the objects.

17 *ACTIVITY 3 INTRODUCTION*



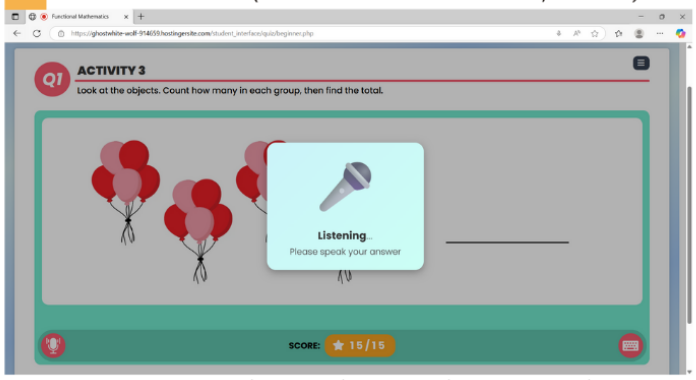
NOTE: Voiced-over (text-to-speech) instruction is provided.

18 *SAMPLE ACTIVITY 3 (SPEECH-TO-TEXT & **TYPE-IN/ENCODE**)*



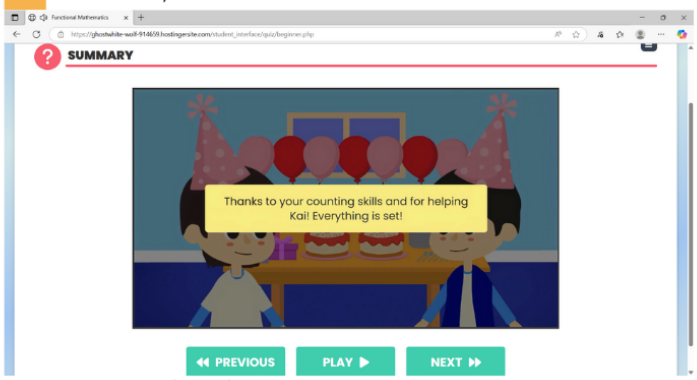
NOTE: The students can type-in their answer on the blank.

19 *SAMPLE ACTIVITY 3 (**SPEECH-TO-TEXT** & TYPE-IN/ENCODE)*



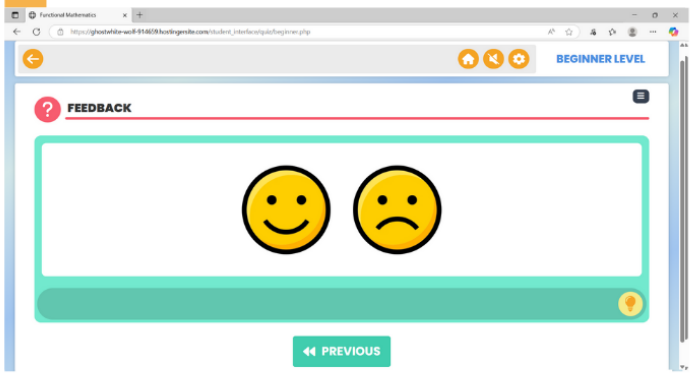
NOTE: The students can click the 'microphone' icon to say their answer.

20 *SUMMARY / ENDING SKIT*



NOTE: A short ending skit to conclude the level.

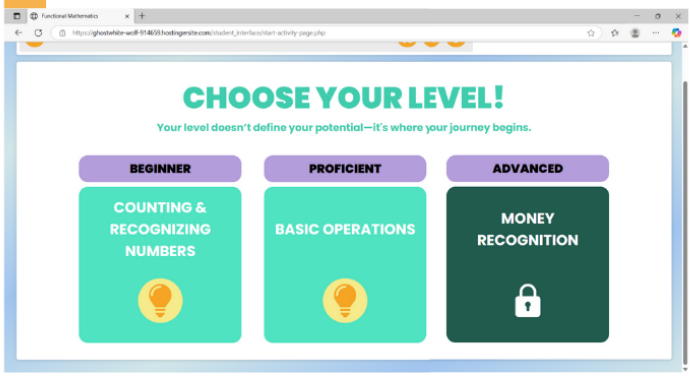
21 *'FEEDBACK' PAGE*



NOTE: The feedback will indicate that the student finished the level.

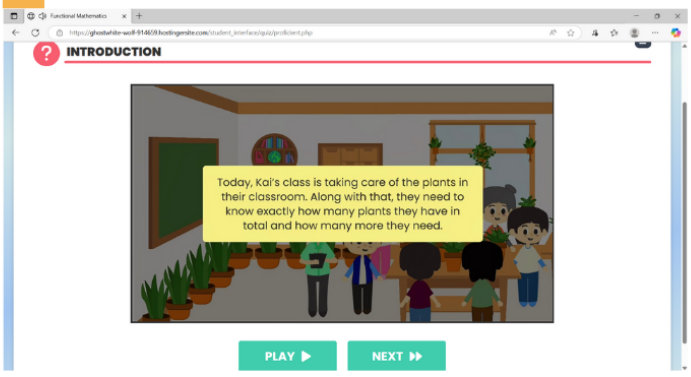
*-START ACTIVITY : **ANSWERING ACTIVITIES-PROFICIENT LEVEL***

22 *'START ACTIVITY' PAGE - PROFICIENT LEVEL*



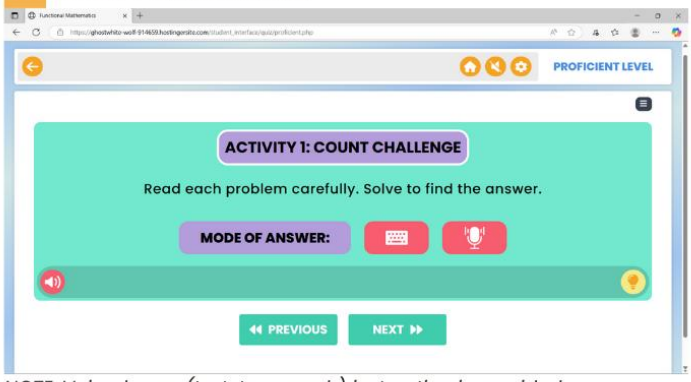
NOTE: Other levels are locked until the activities are not yet completed.

23 *INTRODUCTION / OPENING SKIT*



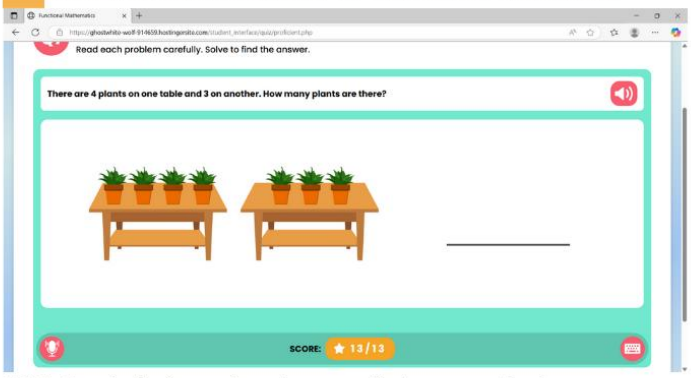
NOTE: A short opening skit to introduce the lesson to the students.

24 *ACTIVITY 1 INTRODUCTION*



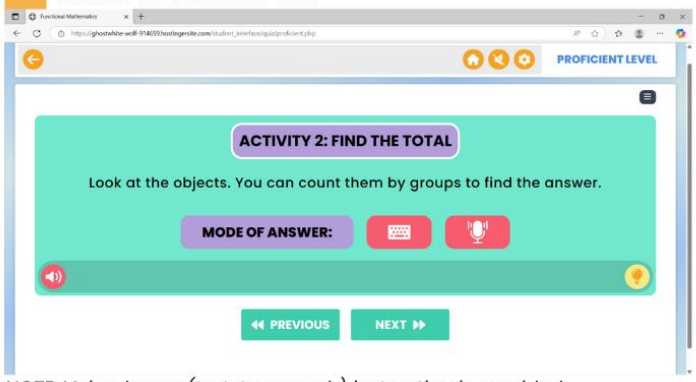
NOTE: Voiced-over (text-to-speech) instruction is provided.

25 *SAMPLE ACTIVITY 1 (SPEECH-TO-TEXT & TYPE-IN/ENCODE)*



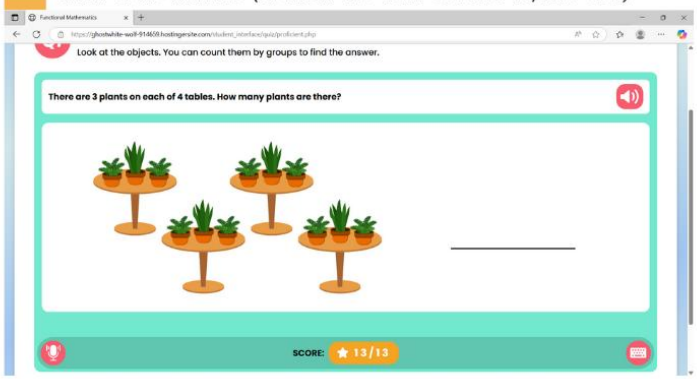
NOTE: The students can type-in or say their answer aloud.

26 *ACTIVITY 2 INTRODUCTION*



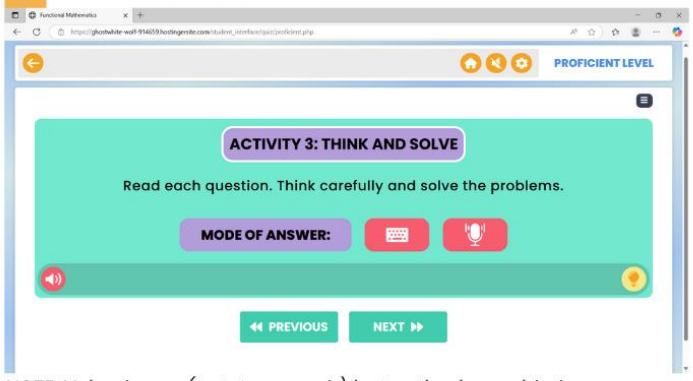
NOTE: Voiced-over (text-to-speech) instruction is provided.

27 *SAMPLE ACTIVITY 2 (SPEECH-TO-TEXT & TYPE-IN/ENCODE)*



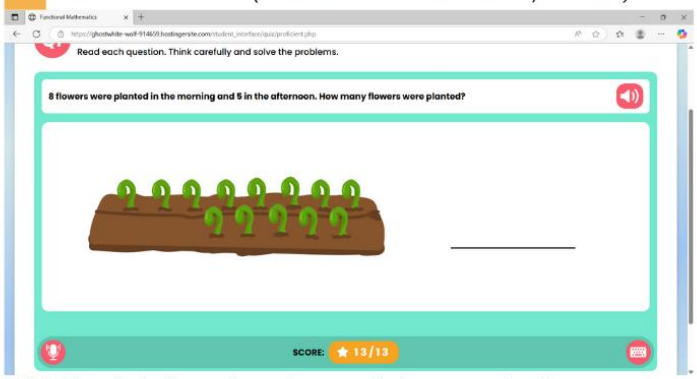
NOTE: The students can type-in or say their answer aloud.

28 *ACTIVITY 3 INTRODUCTION*



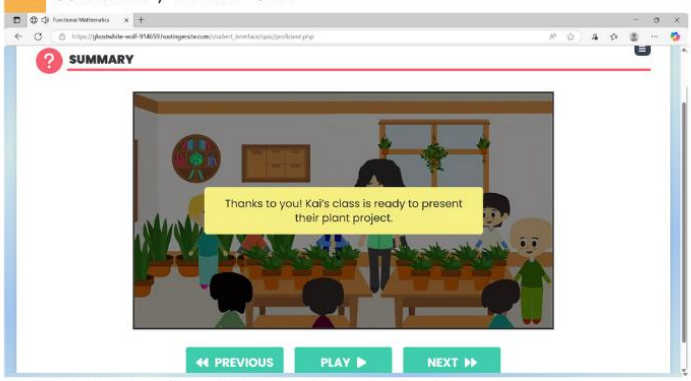
NOTE: Voiced-over (text-to-speech) instruction is provided.

29 *SAMPLE ACTIVITY 3 (SPEECH-TO-TEXT & TYPE-IN/ENCODE)*



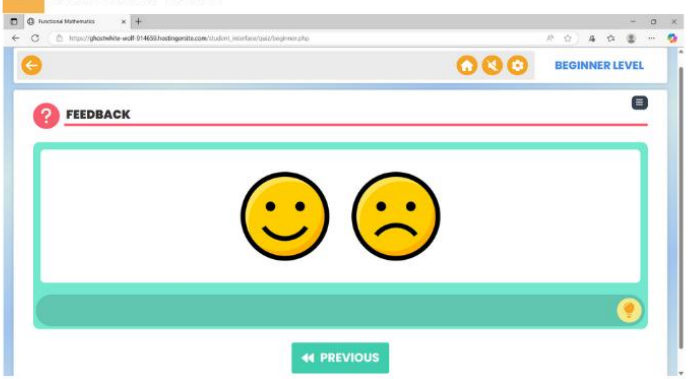
NOTE: The students can type-in or say their answer aloud.

30 *SUMMARY / ENDING SKIT*



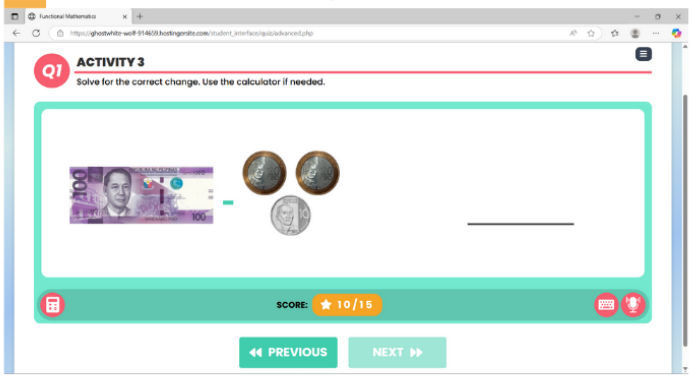
NOTE: A short ending skit to conclude the level.

31 *'FEEDBACK' PAGE*



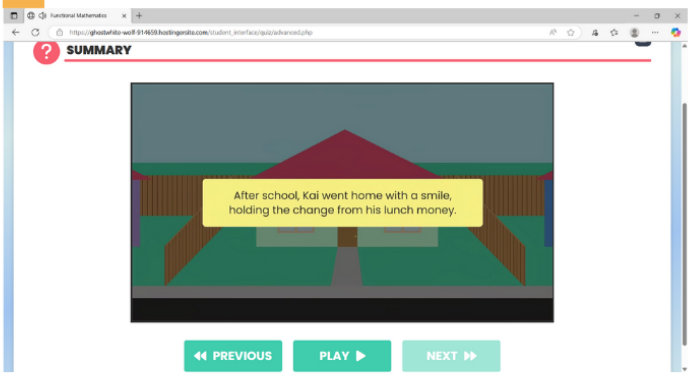
NOTE: The feedback will indicate that the student finished the level.

40 *SAMPLE ACTIVITY 3 (SPEECH-TO-TEXT & TYPE-IN/ENCODE & OPTIONAL CALCULATOR USE)*



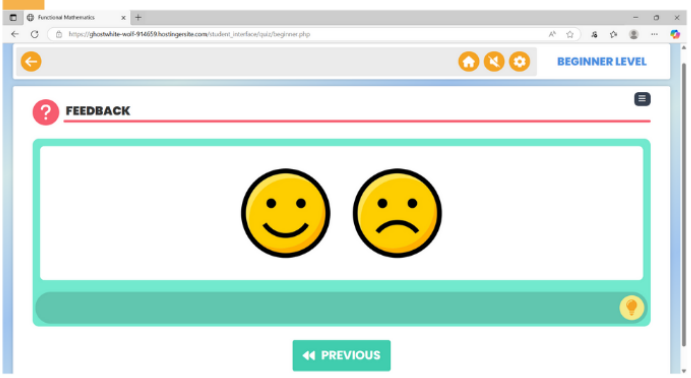
NOTE: The students can encode or say their answer aloud. Calculator use is also available.

41 *SUMMARY / ENDING SKIT*



NOTE: A short ending skit to conclude the level.

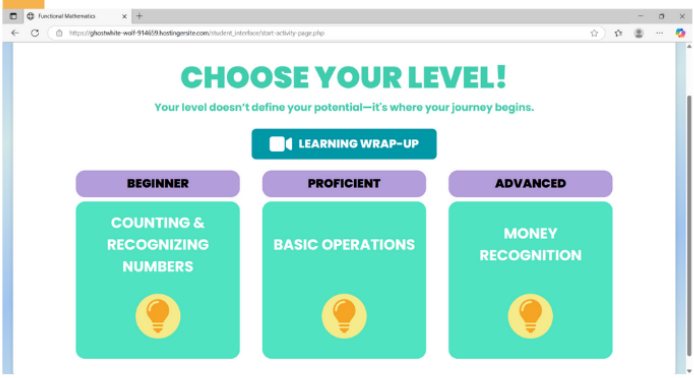
42 *'FEEDBACK' PAGE*



NOTE: The feedback will indicate that the student finished the level.

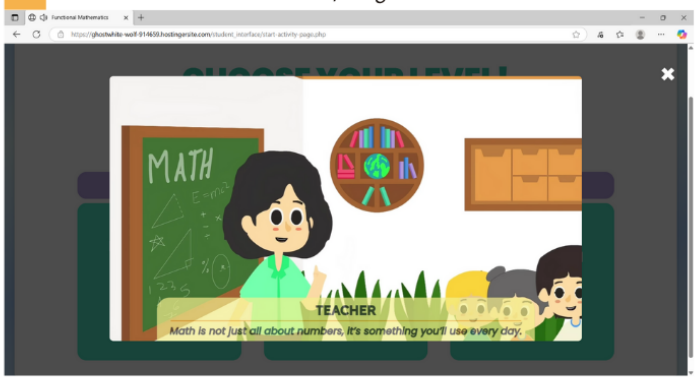
*-START ACTIVITY : **AFTER ANSWERING ACTIVITIES-LEARNING WRAP-UP***

43 *'START ACTIVITY' PAGE - ALL LEVELS COMPLETED*



NOTE: The 'LEARNING WRAP-UP' will appear after finishing all the levels.

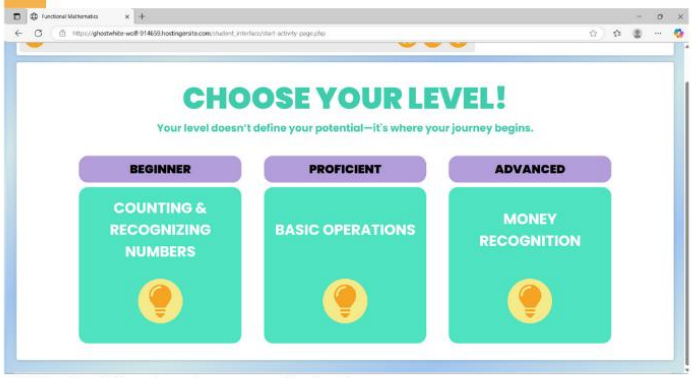
44 *'LEARNING WRAP-UP' Video / Page*



NOTE: A short epilogue skit to summarize the lessons. It will serve as a reflection of the overall content/activities of the learning aid, demonstrating how Functional Mathematics is always present in everyday life.

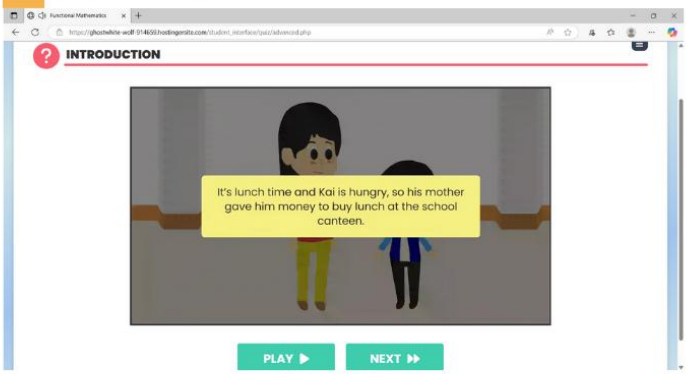
-START ACTIVITY : ***ANSWERING ACTIVITIES- ADVANCED LEVEL***

32 *'START ACTIVITY' PAGE - ADVANCED LEVEL*



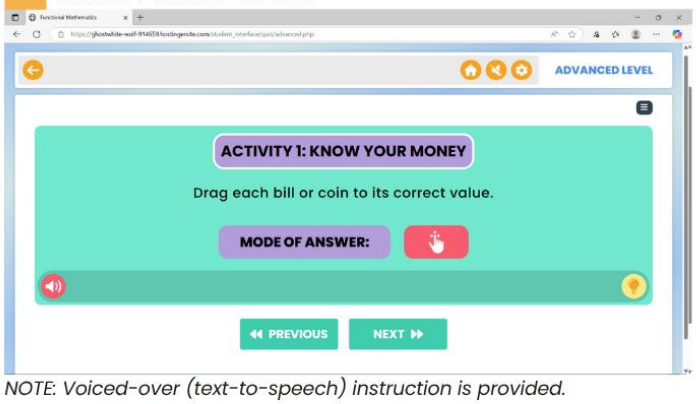
NOTE: All of the levels were unlocked.

33 *INTRODUCTION / OPENING SKIT*



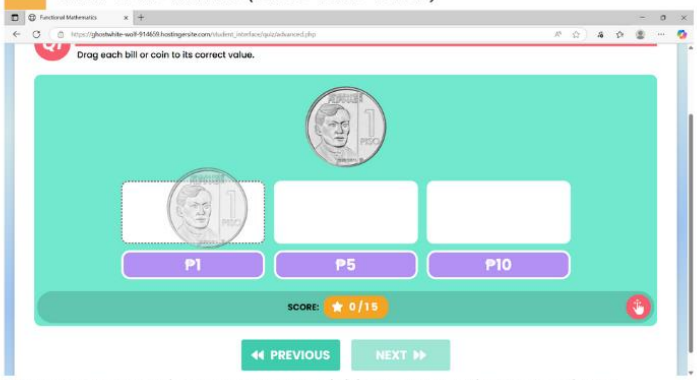
NOTE: A short opening skit to introduce the lesson to the students.

34 *ACTIVITY 1 INTRODUCTION*



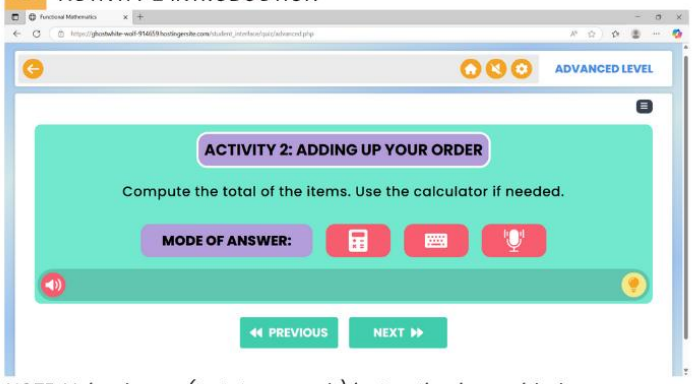
NOTE: Voiced-over (text-to-speech) instruction is provided.

35 *SAMPLE ACTIVITY 1 (DRAG-AND-DROP)*



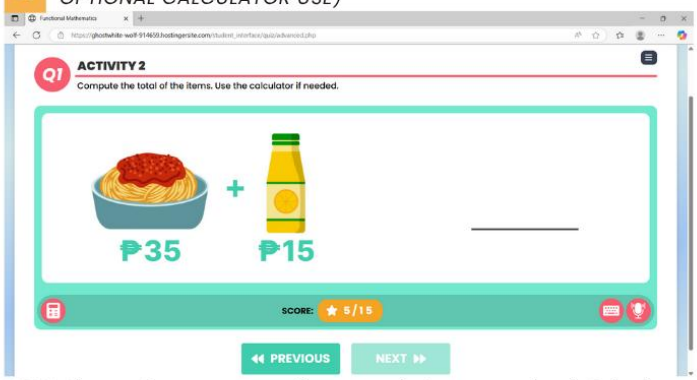
NOTE: Students will answer the activities by dragging the objects.

36 *ACTIVITY 2 INTRODUCTION*



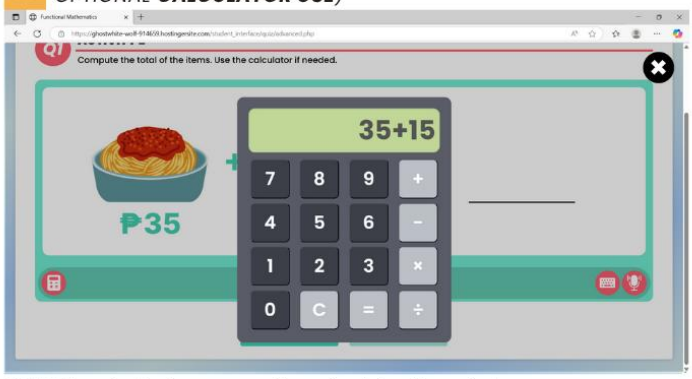
NOTE: Voiced-over (text-to-speech) instruction is provided.

37 *SAMPLE ACTIVITY 2 (SPEECH-TO-TEXT & TYPE-IN/ENCODE & OPTIONAL CALCULATOR USE)*



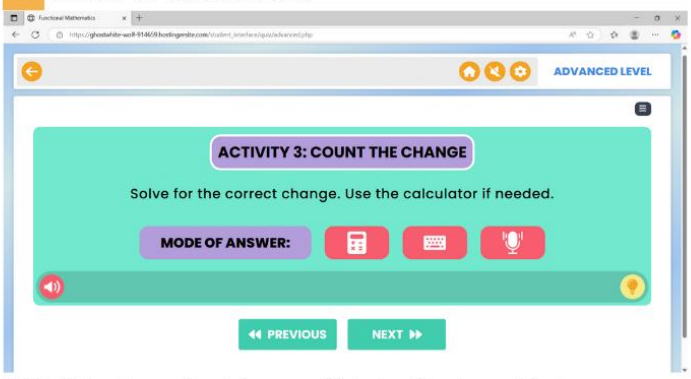
NOTE: The students can encode or say their answer aloud. Calculator use is also available.

38 *SAMPLE ACTIVITY 2 (SPEECH-TO-TEXT & TYPE-IN/ENCODE & OPTIONAL **CALCULATOR USE**)*



NOTE: The students can use the calculator if needed.

39 *ACTIVITY 3 INTRODUCTION*



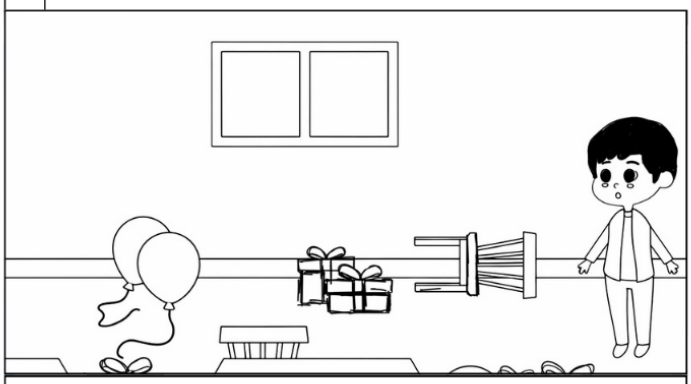
NOTE: Voiced-over (text-to-speech) instruction is provided.

FUNCTIONAL MATHEMATICS: AN INTERACTIVE LEARNING AID FOR STUDENTS WITH SPECIAL NEEDS

STORYBOARD OF THE SHORT SKITS IN THE INTERACTIVE LEARNING AID

BEGINNER LEVEL : COUNTING AND RECOGNIZING NUMBERS

1

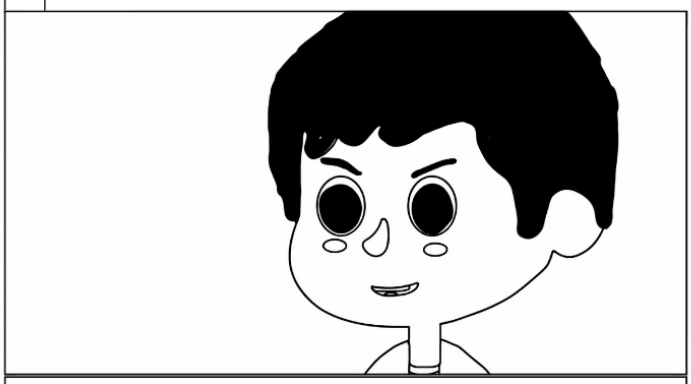


A room filled with balloons, chairs, and a table with cupcakes, as well as boxes scattered on the surroundings. Kai enters, wide-eyed (shocked/overwhelmed), looking at the things to be arranged.

2

A friend enters, pointing at the items and explaining to arrange them.

3



Zoom to Kai's determined/smiling face.

4

Kai who wears a party hat, talking to his friend who is also wearing a party hat.

5

Kai's friend proudly chaps for him as he successfully arranged all the party needs and setup.

6

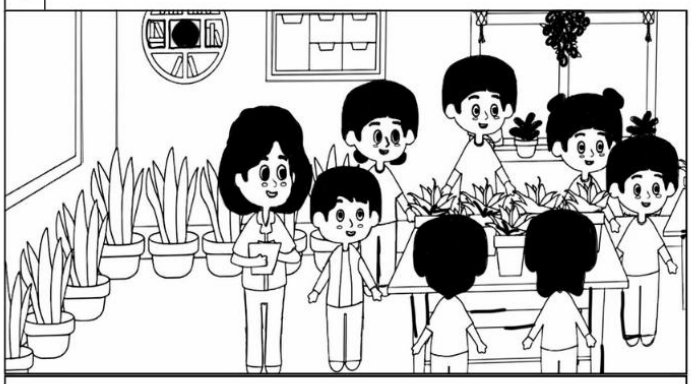
Kai is also proud of himself for helping to organize his friend's party.

FUNCTIONAL MATHEMATICS: AN INTERACTIVE LEARNING AID FOR STUDENTS WITH SPECIAL NEEDS

STORYBOARD OF THE SHORT SKITS IN THE INTERACTIVE LEARNING AID

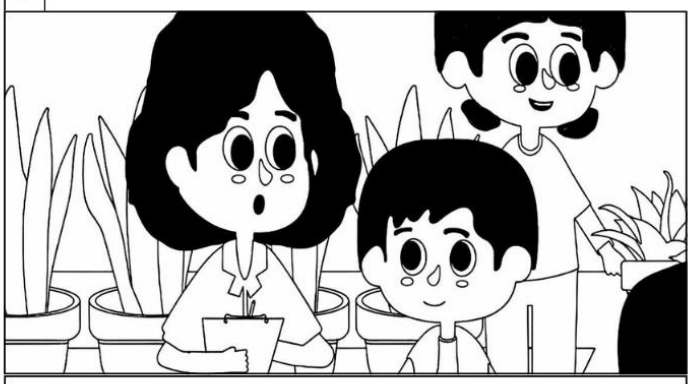
PROFICIENT LEVEL : BASIC OPERATIONS

1



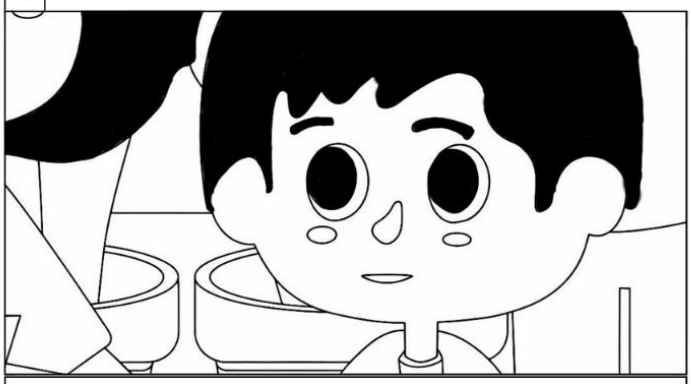
Wide shot of the classroom scenario with tables, potted plants, and students looking at them.

2



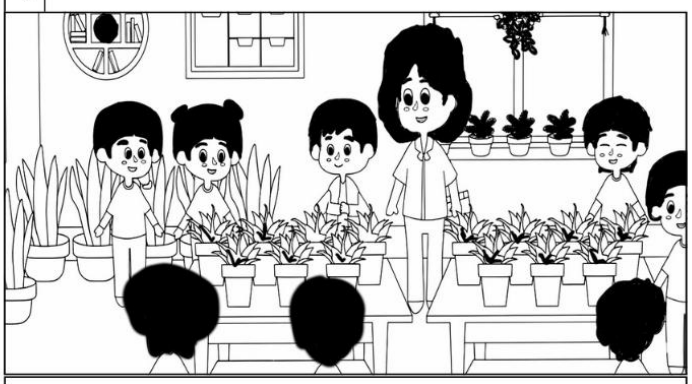
A teacher holding a clipboard with a plant checklist, talking to Kai.

3



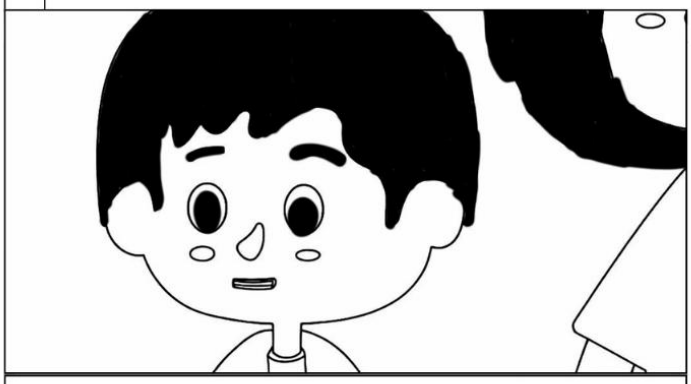
Zoom to Kai, looking nervous but smiling.

4



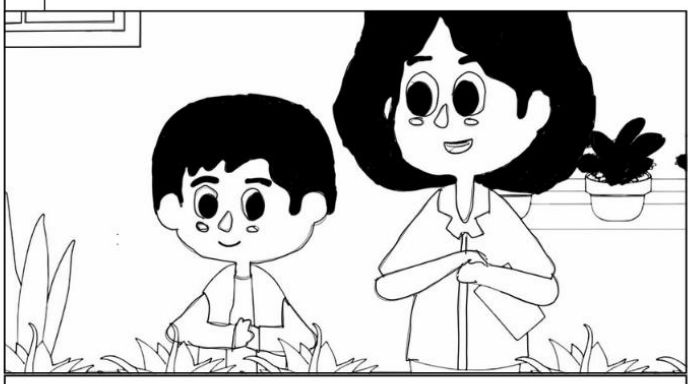
The class gathered in front of a table with neatly arranged plants, Kai standing with his classmates.

5



Zoom to Kai, smiling and looking proud.

6



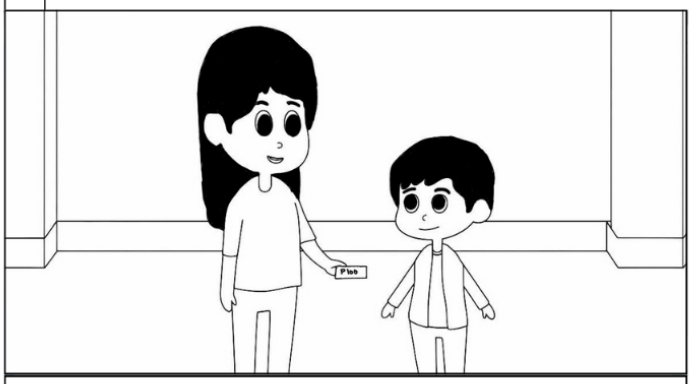
The teacher claps and nods approvingly.

FUNCTIONAL MATHEMATICS: AN INTERACTIVE LEARNING AID FOR STUDENTS WITH SPECIAL NEEDS

STORYBOARD OF THE SHORT SKITS IN THE INTERACTIVE LEARNING AID

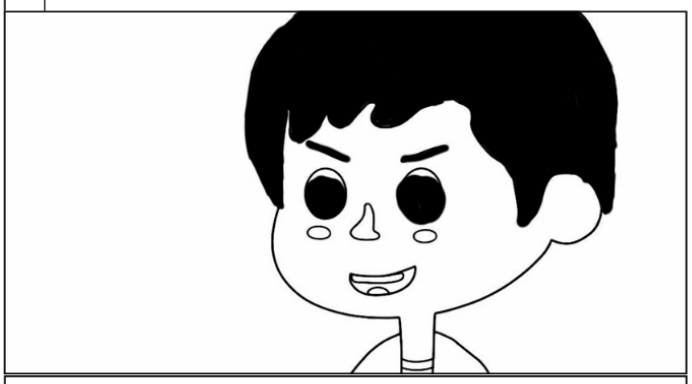
ADVANCED LEVEL : MONEY RECOGNITION

1



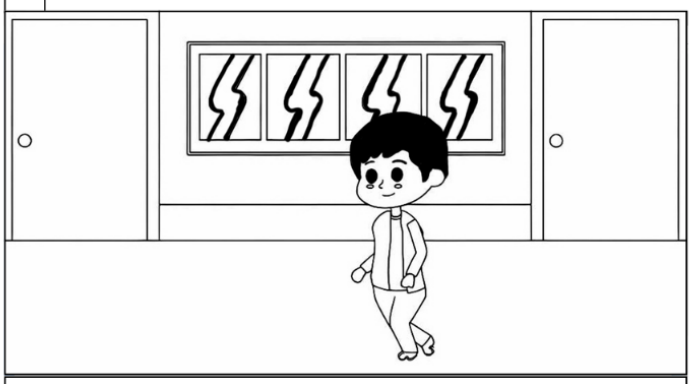
Outside the canteen, Kai's mother gave him one hundred pesos. Kai accepted the money.

2



Kai, determined face. Confident that he can handle his money himself.

3

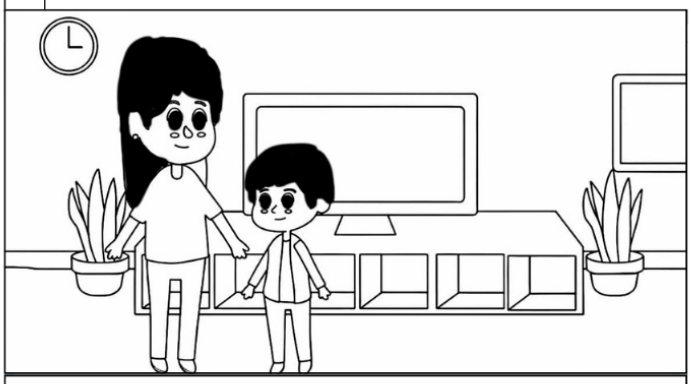


Kai walking towards the canteen while holding his money.

4

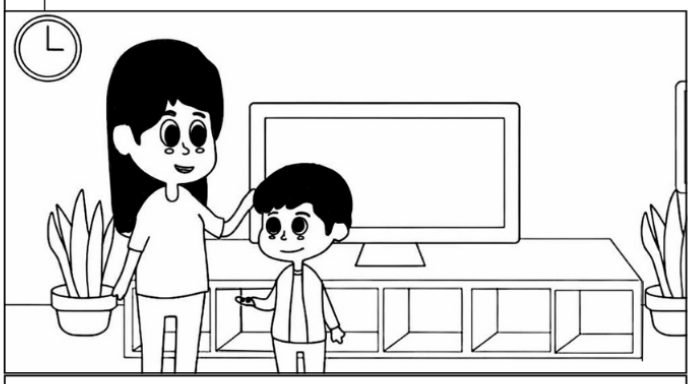
Kai looking at the food items, thinking what to buy for his lunch.

5



Kai arrives home after school, smiling and with coins in his pocket.

6



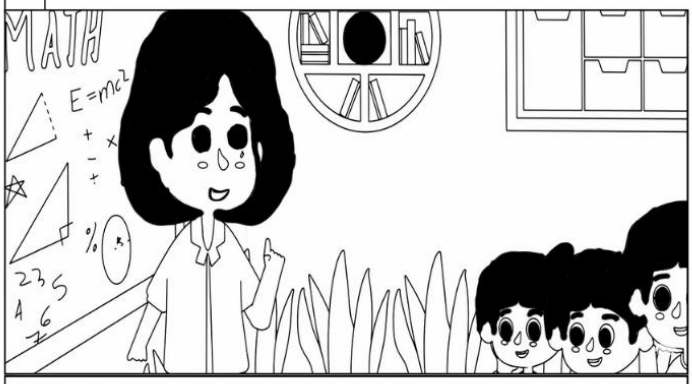
The mother pats Kai's head, smiling proudly.

FUNCTIONAL MATHEMATICS: AN INTERACTIVE LEARNING AID FOR STUDENTS WITH SPECIAL NEEDS

STORYBOARD OF THE SHORT SKITS IN THE INTERACTIVE LEARNING AID

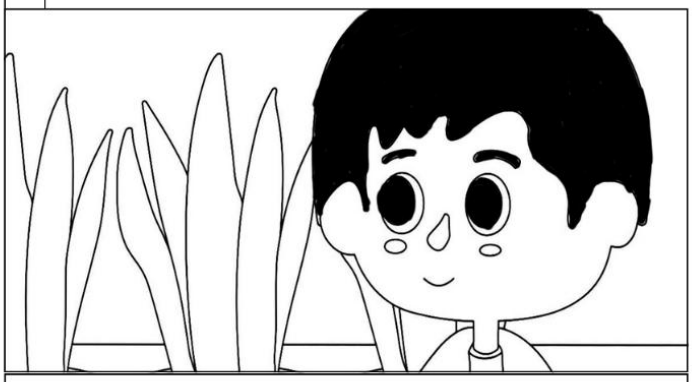
EPILOGUE : LEARNING WRAP-UP

1



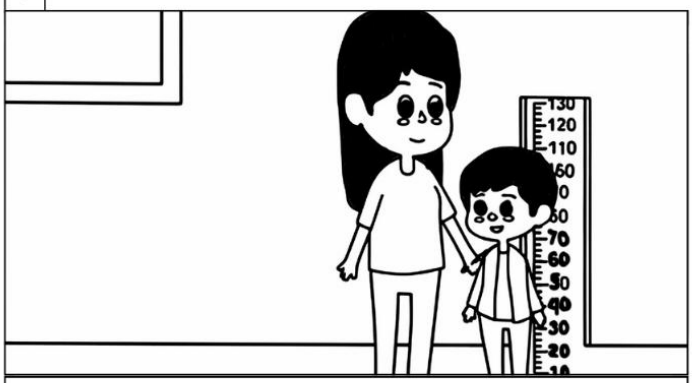
At the classroom, the mathematics class just ended. Kai and classmates are packing up their things.

2



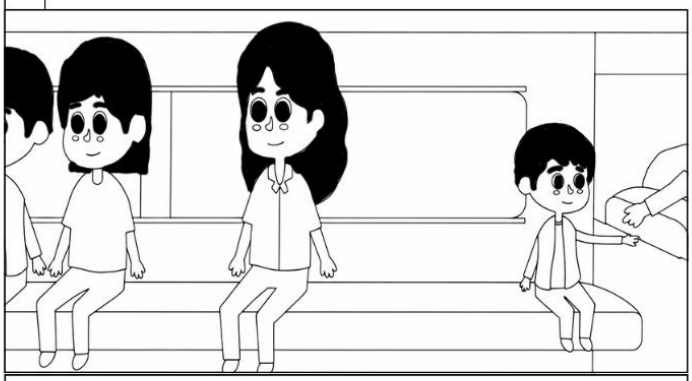
Kai will realize that math is everywhere (thinking of the happenings)

3



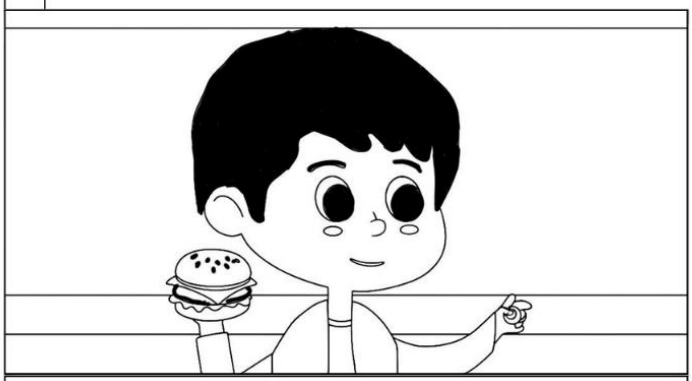
At home – Kai stands against a wall height chart while his mother marks his height

4



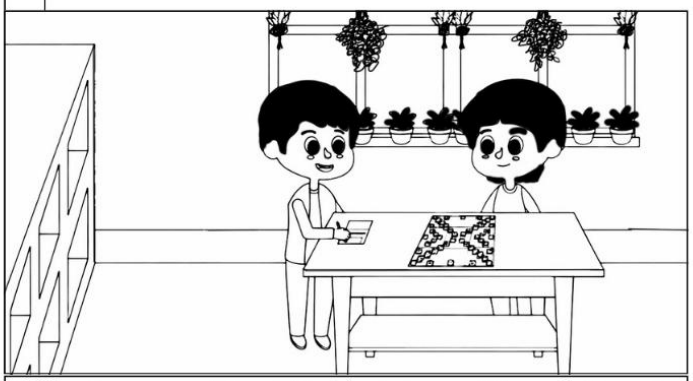
Inside a jeepney – Kai hands money and checks the fare change.

5



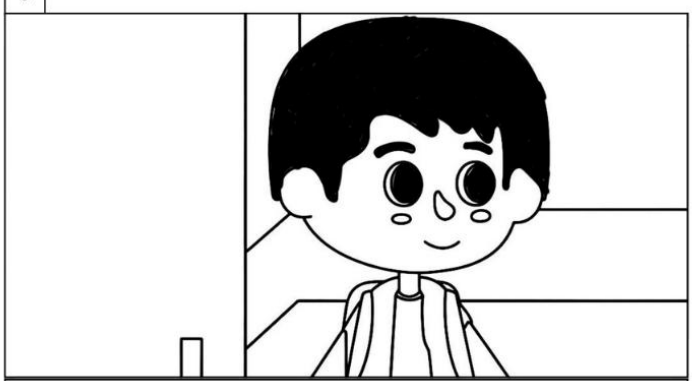
At the canteen – Kai buys snacks and counts change.

6



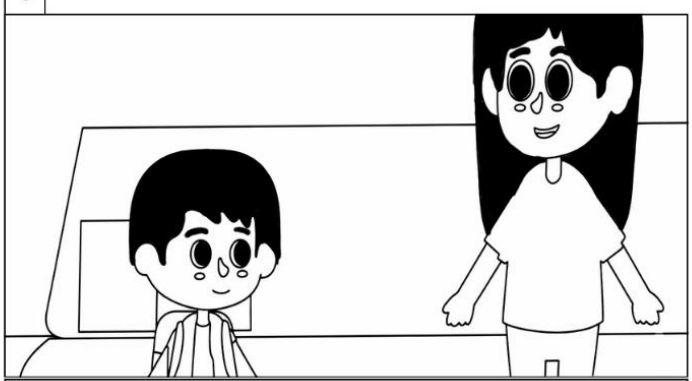
In the classroom – Kai and his friend play scrabble and add up the scores.

5



After class, Kai arrived home.

6



Near the door, Kai just arrives home. His mother was waiting near the couch.

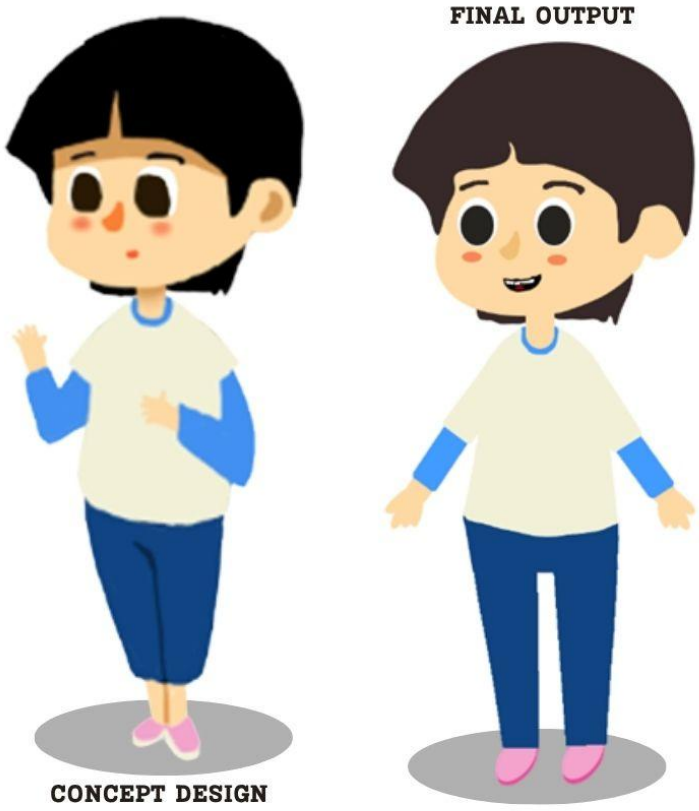
Appendix B

Character Design

KAI

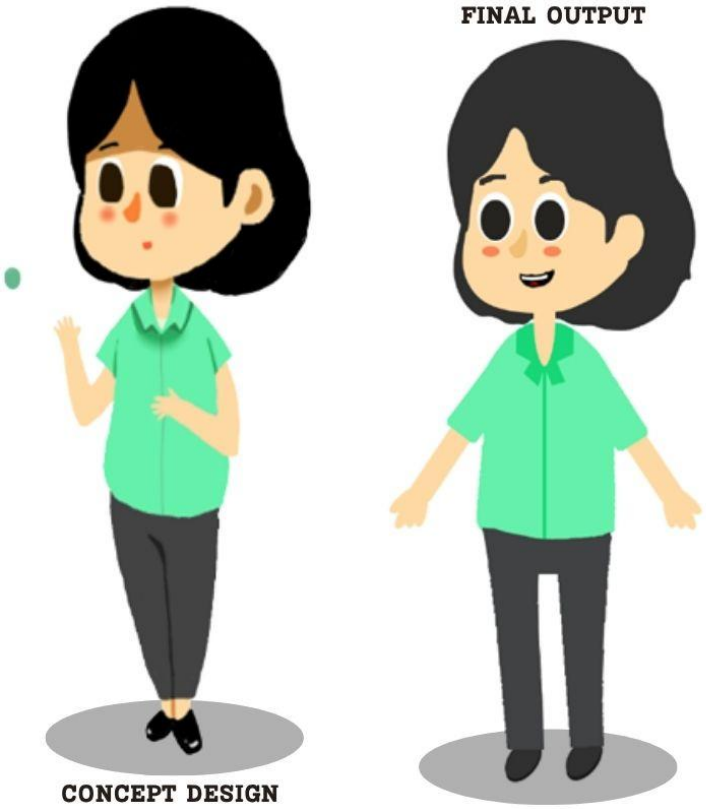
Kai, the main character in this interactive learning aid and skit animation, is intentionally designed to be both engaging and easy to familiarize for students with special needs. ***His light, dark hair and cheerful, open smile are meant to convey a sense of openness and approachability***, while his plain clothes—a white T-shirt, blue jacket, black trousers, and brown shoes—***strike a balance between informal neatness and lively, down-to-earth appearance.*** All these aspects of design are intended to render Kai as s***omeone who is non-intimidating and approachable***. Other than his physical appearance, the character of Kai is built on his inherent curiosity, worldliness, and being nice, which qualify him as an ideal facilitator of learning resources. His cultivated spirit, along with his upbeat demeanor and going-the-extra-mile in serving other people, establishes a culture of safety and care, which is an absolute necessity for students with special needs. By taking these reflective design decisions, Kai not only illustrates to students that they can believe in him as an individual, but he also indirectly aids learning, creating a caring and positive learning environment.

FRIEND



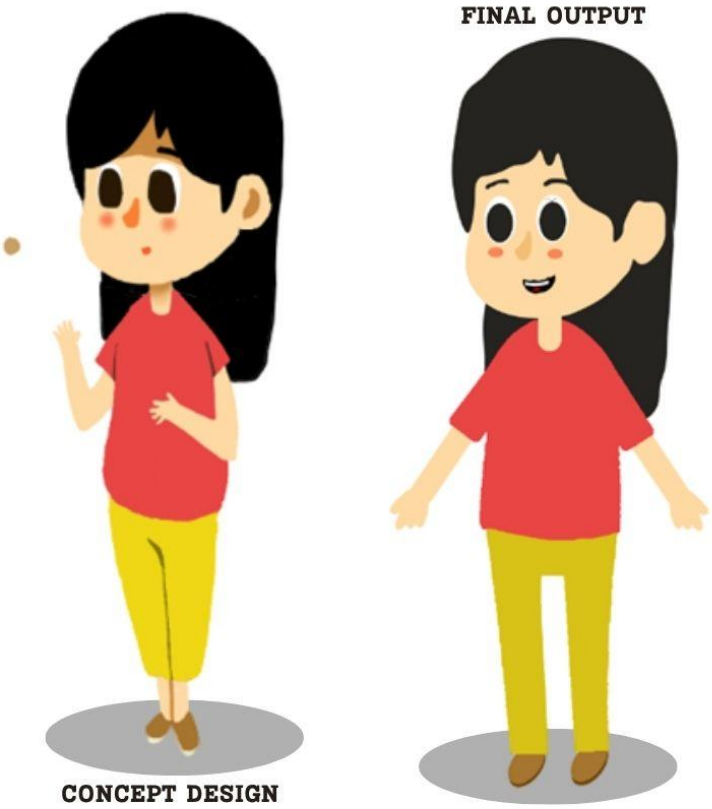
This character is ***one of Kai's friends;*** the character design certainly has a role in making the interactive learning experience, as it would be a very pleasant, relaxed, and inclusive environment. She ***represents the happiest and most sociable personality***. She has short dark brown hair, a warm smile that shows her ***friendly and easy-going character***, and that is how she is depicted. Her dress is made of a cream-colored top with blue cuffs, blue pants, and pink shoes, which together give her a ***very playful and lively look***. The use of simple but at the same time lively light colors strengthens the positive aura around her and thus, an inviting and light-hearted atmosphere is created for learners.

TEACHER



This is ***Kai's teacher*** in the short skit animation. The design choice brings comfort and motivation, which in turn makes it easier for students to participate actively and become emotionally involved. The teacher's personality is portrayed through the application of smooth contours, flat shades, and soft shadows in a ***friendly and warm*** manner. These visual methods are very successful in converting the teacher into a character that is not only visually attractive but also ***very approachable***, which is the main characteristic of an interactive learning device that is to be used for capturing the interest of students, especially those who are disabled. The teacher's ***happy face and very vivid facial expressions*** are important in creating the warm and friendly atmosphere of the learning environment.

MOTHER



The ***mother***'s personality, created as the ***parent of Kai***, plays a crucial role in the interactive learning aid and companion skit animation. She is created with ***warmth, emotional safety, and approachability in mind,*** factors necessary for building trust and a sense of comfort among the special needs students. Her dark hair and soft, warm smile are contributors to projecting a ***calming and maternal presence***. The red blouse, yellow shorts, and brown shoes color combination is also one that is visually stimulating but not overstimulating, which serves to project a ***sense of friendliness and approachability.*** The flash of radiance and elegance of her dress is designed to be effective but never disconcerting, and she is a type of ***comforting, reassuring presence*** in consequence. The overall tone of the mother figure is ***stability and affective care***, and it is something that can be used to ***generate a sense of safety and hospitality for learning.***

Appendix C

Scripts, Curriculum-Based Leson Plans and Learning Materials

Learning Area	Functional Academics – Math (Beginner)	**Grade Level**	Non-Graded
Quarter	First	**Date**	
I. LESSON TITLE		Counting objects from 1-50	
II. LEARNING COMPETENCY		Count objects with one-to-one correspondence up to quantities of 50 CODE: TP_MFP-FO-NG-1-6	
III. CONTENT STANDARD		The learners demonstrate understanding of the concept of whole numbers.	
IV. PERFORMANCE STANDARD		The learners recognize and organize numbers according to value up to 50	

OBJECTIVES:

a. Recognize numbers from 1-10, 11-20, 21-30, 31-40, 41-50.
b. Rote count numbers from 1-50.
c. Count objects with one-to-one correspondence up to quantities of 50.
d. Appreciate the importance of counting objects and its application in everyday living.

V. LEARNING PHASES AND LEARNING ACTIVITIES

PRELIMINARIES:

- Prayer
- Checking attendance
- Motivation

Game: Hide and Seek

The class will be instructed to look for objects in the classroom or in their belongings with numbers. (Examples: clock, calendar, pencil, etc.)

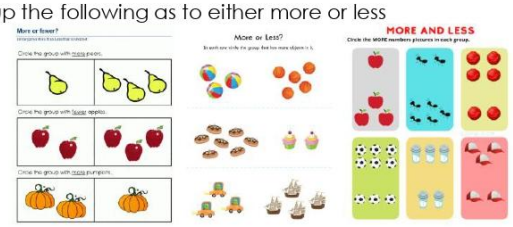
After the activity, the teacher will ask about the learners' experiences. Was it easy or difficult? Let the learners share their experiences in class.

LESSON PROPER:

I. Activity:

More or Less?

- Group the following as to either more or less



II. Analysis:

Answer the following questions:
1. How do we count objects?
2. What technique can we use to count larger groups of objects?
3. Why do we need to count objects around us?

III. Abstraction: Unlocking of terms:

- Discuss concept of counting whole numbers.
- Demonstrate how to count using one-to-one correspondence.
- Show videos of practical applications of counting objects

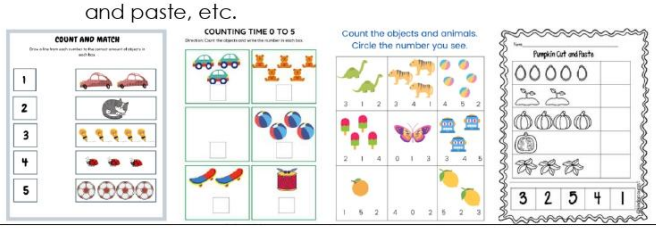
IV. Application:

Differentiated activities are as follows:

The class will again be divided into 3 groups. The groups and activities are as follows:

Level 1 (Easy Level):

- Counting objects with one-on-one correspondence from 1-5 then 1-10
- Activities can be varied from count and match, write the number, count and circle/box/ check the correct number, cut and paste, etc.



Reflection:

How do you feel after the lesson. Encircle it and explain. Learners can be guided in answering this part.

Complete the following:

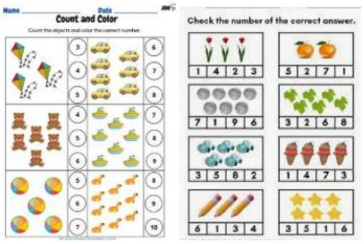
I am HAPPY/ SAD because _____
_____
_____
_____.

Remarks:

_____
_____
_____
_____
_____
_____.

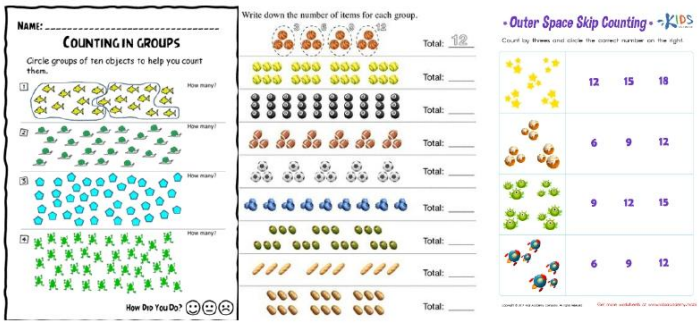
Prepared by:

RHEA P. BARBOZA
SPECIAL EDUCATION TEACHER I



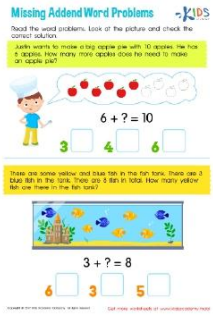
Level 2 (Average Level):

- Counting objects with one-on-one correspondence from 11-20 then 1-20.
- Activities can be varied from count and match, write the number, count and circle/box/ check the correct number, cut and paste, etc.



Level 3 (Difficult Level):

- Counting objects with one-on-one correspondence from 1-50
- Activities can be varied from count and match, write the number, count and circle/box/ check the correct number, cut and paste, etc.



Extra Challenge: Count objects above 50 using groupings/ skip counting.

Assignment:

Cut and paste pictures in your notebooks pictures that show importance of counting in everyday life.

Learning Area	Functional Academics – Math (Proficient)	**Grade Level**	Non-Graded
Quarter	Second	**Date**	
I. LESSON TITLE		Basic Operations	
II. LEARNING COMPETENCY		To teach fundamental number operations. CODE: TP_MFP-FO-NG-1-14	
III. CONTENT STANDARD		The learners demonstrate understanding of symbolic mathematics and real-life computation.	
IV. PERFORMANCE STANDARD		The learners perform addition and subtraction in context, mixed addition/subtraction for multiple numbers.	

OBJECTIVES:

Addition

a. Illustrate addition as putting together or combining/ joining sets.
b. Adds quantities with and without regrouping.
c. Solves simple worded problems using the concept of addition with illustrations.

Subtraction

a. Illustrate subtraction as taking away sets/ objects.
b. Subtracts quantities with and without regrouping.
c. Solves simple worded problems using the concept of subtraction with illustrations.

II. LEARNING PHASES AND LEARNING ACTIVITIES

PRELIMINARIES:

- Prayer
- Checking attendance
- Motivation

Game: The Highest Tower (Addition Lesson)

The class will be divided into 3 groups. At the teacher's signal, learners will get anything in their bags that they could stack up together to form a tower. At the end of 2 minutes, the group with the tallest tower winds. – teaches collaboration, communication and strategic planning.

instead of blocks, the learners may use pencil case, food containers, bags, shoes or anything they could find in their own belongings.

After the activity, the teacher will ask about the learners' experiences and ask Why did team __ win and what is their strategy? Lead the discussion to ADDITION since stacking up is like adding up.

Game: Bowling (Subtraction Lesson)

The learners will take turns in bowling. All will start with 10 pins, and they will remember how many pins fell after they hit it with the bowling ball.

After the activity, the teacher will ask about the learners' experiences and ask who among them made a strike (all pins fell)? Lead the discussion to SUBTRACTION since bowling starts with 10 pins then take away the number of pins that fell.

LESSON PROPER:

I. Activity:

Simon says group yourselves according to…

This activity allows the learners to add/ subtract members to get the correct number that Simon says. This is both for Addition and Subtraction lessons.

II. Analysis:

Answer the following questions:

1. What is the process of putting things/ objects together (Addition) and taking away (Subtraction)?
2. Give situations where you use ADDITION and SUBTRACTION in everyday living.

III. Abstraction: Unlocking of terms:

- Discuss concept of Addition/ Subtraction whole numbers with and without regrouping.
- Demonstrate how to add/ subtract numbers with and without regrouping.

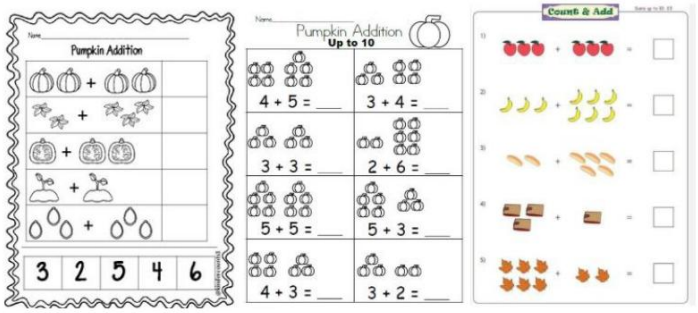
- Show illustrations to solve worded problems.

IV. Application:

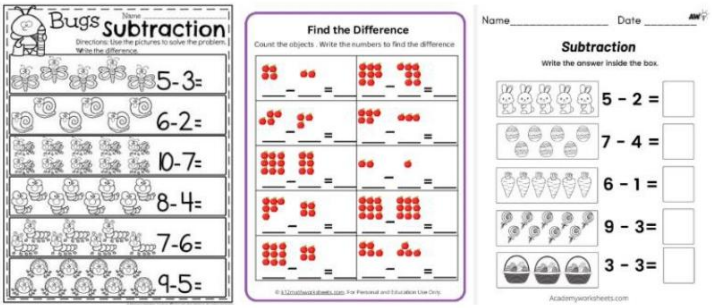
Differentiated activities are as follows:

Level 1 (Easy Level):

- Addition of whole numbers with sum up to 10 only (with visuals).



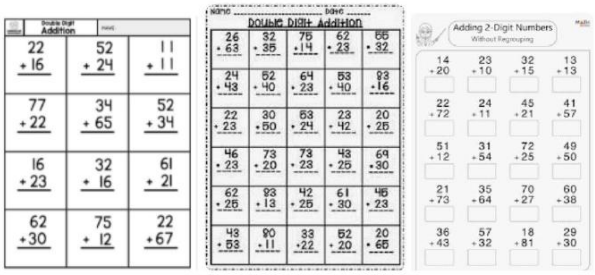
- Subtraction of whole numbers with difference of 10 only (with visuals).



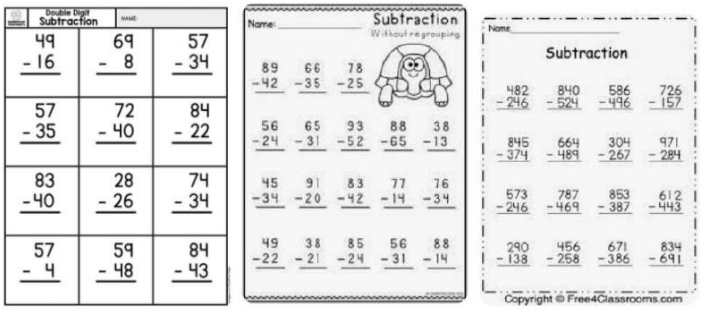
- Activities can be varied from choose the correct answer, write the number, circle/box/ check the correct number, cut and paste, etc.

Level 2 (Average Level):

- Addition of whole numbers without regrouping.

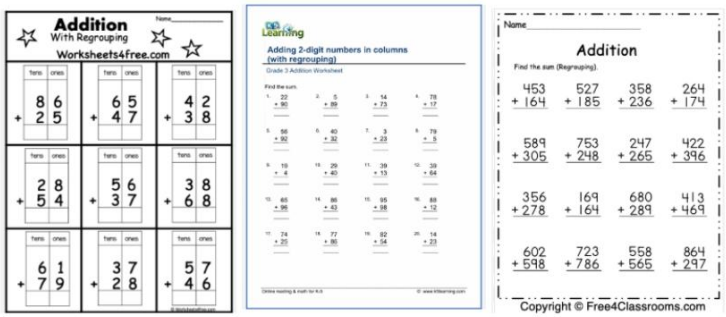


- Subtraction of whole numbers without regrouping.

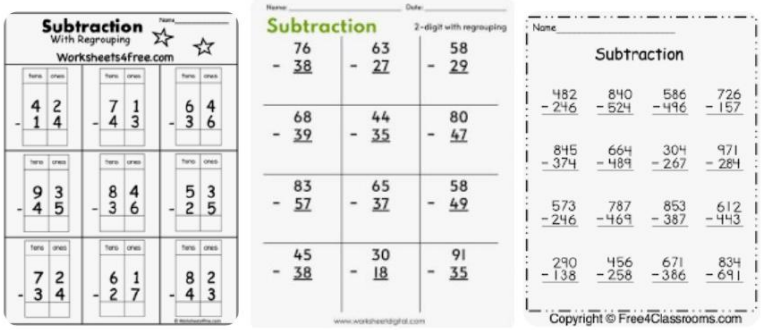


Level 3 (Difficult Level):

- Addition of whole numbers with regrouping.



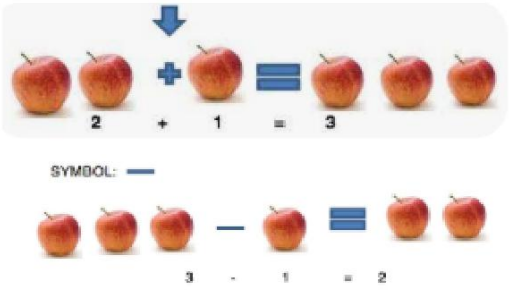
- Subtraction of whole numbers with regrouping.



Assignment:

Cut and paste pictures in your notebooks that illustrate addition/ subtraction.

Example:



Reflection:

How do you feel after the lesson. Encircle it and explain. Learners can be guided in answering this part.

Complete the following:

I am HAPPY/ SAD because _____
_____
_____
_____.

Remarks:

_____
_____
_____
_____
_____
_____.

Prepared by:

RHEA P. BARBOZA
SPECIAL EDUCATION TEACHER I

Learning Area	Functional Academics – Math (Advanced)	**Grade Level**	Non-Graded
Quarter	Third-Fourth	**Date**	
I. LESSON TITLE		Money Recognition	
II. LEARNING COMPETENCY		Identifying coins and bills, matching amounts to items, computing totals and change, using calculator. CODE: TP_MFP-FO-NG-1-14 TP_MFP-fr-NG-1-4.1	
III. CONTENT STANDARD		The learners demonstrate understanding of applying mathematics to everyday transactions, as well as build independence in money handling and proper use of calculator.	
IV. PERFORMANCE STANDARD		The learners perform applying mathematics to everyday-life and be capable of handling money with/without using calculator.	

OBJECTIVES:

1. Identify Philippine coins and bills by their denominations and physical characteristics.
2. Match amounts of coins and bills to given items or prices.
3. Compute total amounts and change using actual money or play money.
4. Use a calculator to assist in solving addition and subtraction of money amounts.
5. Demonstrate responsible handling and basic decision-making in money-related situations.

I. LEARNING PHASES AND LEARNING ACTIVITIES

PRELIMINARIES:

- Prayer
- Checking attendance
- Motivation

Game: Let's Go Shopping (Identifying Coins and Bills)

Present real or play money (coins and bills) on the table.
Ask: "Can you pick up the ₱1 coin? How about the ₱20 bill?"
Let learners group the same denominations together.
Explain the values and physical features (color, size, symbols) of each coin and bill.

Build the Price (Matching Amounts)
Create a story about a mother having 4 children. If she has 20 chocolates, how many chocolates will each child receive? Introduce the concept of division as repeated subtraction.

Role-Play: Mini Sari-Sari Store (Computing Totals and Change)

Set up a pretend store in class using realia or printed items with prices.
Learners take turns as the "seller" and "buyer."
Buyer chooses two or more items, adds their prices, and pays using play money.
Seller computes the change using subtraction (can use a calculator if needed).

Money Match (Independent Practice)
- Give a worksheet where learners:
  - Match coins/bills to given amounts
  - Solve word problems about buying and giving change
- Let them choose if they want to use a calculato r or not.

LESSON PROPER:

I. Activity:

Group Activity: Market Dash (Money Identification & Addition)

Prepare printed pictures of items (fruits, toys, snacks) with price tags (₱5–₱50).
Scatter play money (coins and bills) around the classroom.
Play music while learners walk around. When the music stops, they will:
Pick an item and collect the exact amount using the scattered play money.
Show and explain how they got the amount (e.g., "₱20 + ₱5 = ₱25").
Those who get it wrong will be guided by their groupmates. Continue until all items are bought.

Group Activity: Change Challenge (Subtraction & Calculator Use)

Pair or group the learners and give them a scenario card (e.g., "You bought bread for ₱35. You gave ₱50. How much is your change?").

They will solve using mental math first, then check using a calculator.

They must show their solution and the correct change using play money.

First group to complete all scenarios correctly wins.

II. Analysis:

Answer the following questions:
1. What do we call the process of combining or putting together amounts of money? *(Addition)*
2. What do we call the process of finding how much money is left after paying? *(Subtraction / Getting the change)*
3. What tool can we use to help us add and subtract money amounts faster and check our answers? *(Calculator)*
4. How can identifying coins and bills help us in real-life situations?
5. Can you give examples of situations where you use money (spending, buying, saving, giving change)?

III. Abstraction: Unlocking of terms:

- Discuss the concept of money recognition — identifying coins and bills by their appearance and values.
- Demonstrate how to add amounts of money by using illustrations or pictures of coins and bills (e.g., ₱5 + ₱10 + ₱20 = ₱35).
- Demonstrate how to subtract to find change using illustrations (e.g., ₱50 - ₱35 = ₱15).
- Show how to use a calculator to add or subtract money and check the answers.
- Present illustrated word problems involving buying items and giving change and solve them together as a class.

IV. Application:

Differentiated activities are as follows:

Level 1 (Easy Level):
Know Your Money: Connect each bill or coin to its correct value.
₱1, ₱5, ₱10, ₱100, ₱500, ₱1000, etc.

Level 2 (Average Level):

Adding Up Your Order: Compute the total of the items. Use the calculator if needed.

Example:

1. Spaghetti ₱35 + Juice ₱15 = **₱50**
2. Sandwich ₱25 + Juice ₱15 + Cookie ₱10 = **₱50**
3. Rice Meal ₱45 + Juice ₱15 = **₱60**
4. Burger ₱40 + Fries ₱25 + Water ₱15 = **₱80**
5. Pancit ₱35 + Banana ₱10 + Juice ₱15 = **₱60**

Level 3 (Difficult Level):

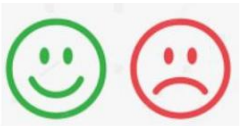
Count the Change: Solve the correct change. Use the calculator if needed.

Example:

1. Total=₱50, Money=₱100, **Change=₱50**
2. Total=₱60, Money=₱100, **Change=₱40**
3. Total=₱80, Money=₱100, **Change=₱20**
4. Total=₱50, Money=₱200, **Change=₱150**
5. Total=₱35, Money=₱50, **Change=₱15**

Reflection:

How do you feel after the lesson. Encircle it and explain. Learners can be guided in answering this part.



Complete the following:

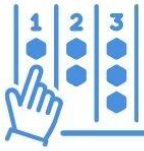
I am HAPPY/ SAD because _____
_____
_____
_____.

Remarks:

_____
_____
_____
_____
_____
_____.

Prepared by:

RHEA P. BARBOZA
SPECIAL EDUCATION TEACHER I



LESSON 1:

Counting and Recognizing Numbers

I. Learning Objectives

- Define the basic terminologies used in counting.
- Recognize numbers.
- Count objects correctly.
- Arrange numbers correctly in order, either ascending or descending.
- Use grouping or skip counting for large quantities.

II. Key Concepts

- **Counting**
  - It is the act of determining the quantity or the exact number of objects in a set or a group.
  - It is the cardinal number that tells how many.
    - **Cardinal numbers** or also known as natural numbers or cardinals, are numbers that are used for counting; they are numbers that are used for counting, representing how many of something are present.

1 **ONE**	6 **SIX**	11 **ELEVEN**	16 **SIXTEEN**	21 **TWENTY-ONE**
2 **TWO**	7 **SEVEN**	12 **TWELVE**	17 **SEVENTEEN**	22 **TWENTY-TWO**
3 **THREE**	8 **EIGHT**	13 **THIRTEEN**	18 **EIGHTEEN**	23 **TWENTY-THREE**
4 **FOUR**	9 **NINE**	14 **FOURTEEN**	19 **NINETEEN**	24 **TWENTY-FOUR**
5 **FIVE**	10 **TEN**	15 **FIFTEEN**	20 **TWENTY**	..... *and so on.*

- **Number**
  - It is an arithmetic value used to represent quantity.
  - It is a mathematical concept used to count and form the basis of mathematics.

- **One-to-One Correspondence**
  - It is the ability to match one object with one number and vice versa.
  - It is considered as an early math skill that involves the ability to count a group of objects by assigning one number to each object and only counting each object once.

- **Sequence**
  - It is an ordered list of numbers that follows a specific pattern or rule.
    - **Ascending Order** is when numbers are ordered from least to greatest.
    - **Descending Order** is when numbers are ordered from greatest to least.

- **Grouping**
  - In terms of counting, it is a technique where a collection of objects is organized into smaller, equal sets to simplify the process.

- **Skip Counting**
  - It refers to the technique of counting by adding the same number each time to the previous number.

III. Understanding the Concept

- **Recognizing Symbols**
  - Numbers may be single-digit or multi-digit.

SINGLE-DIGIT	MULTI-DIGIT
Numbers from **0 - 9**	Numbers **10 and above**

REMEMBER:
- It is important to connect quantities with written numerals because it helps us understand that numbers are not just symbols but representations of real quantities.

- **Counting Objects**
  - To count properly and correctly, assign one number to each object.

Five balloons are counted as **1, 2, 3, 4, 5**

REMEMBER:
- Each object must have a unique count; skipping or repeating results in errors.

- **One-to-One Correspondence**
  - Every item must have one count.

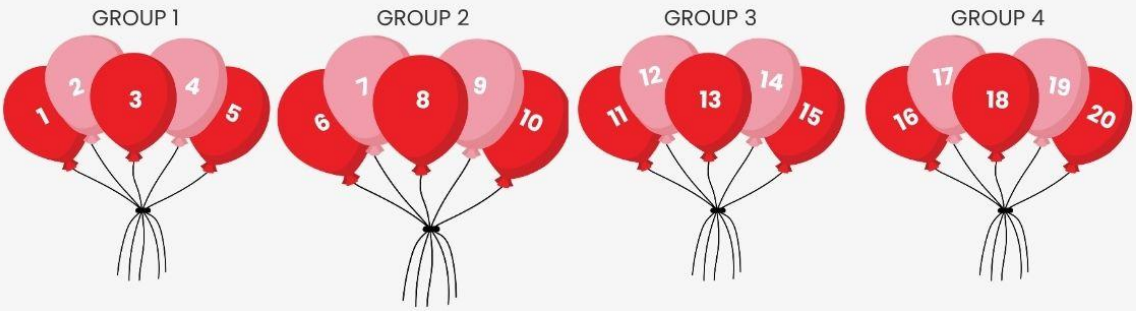
If there are 10 cupcakes, **count each** cupcake **exactly once** to reach 10.

REMEMBER:
- Do not skip. Do not double count.
  1 OBJECT = 1 COUNT

- **Number Sequencing**
  - It helps in understanding order and comparison.

ASCENDING ORDER	DESCENDING ORDER
from **smallest to largest value**	from **largest to smallest value**
1 → 2 → 3 → 4 → 5	**5 → 4 → 3 → 2 → 1 .....**

- **Grouping** and **Skip Counting**
  - When it comes to large quantity, it is easier to count by grouping, which results in skip counting.



20 balloons can be grouped into sets of 5.

Skip count by fives: → **5, 10, 15, 20**

REMEMBER:

- By using the strategy of grouping and skip counting, it is faster to determine the total number of objects since it is counted by sets instead of individually, forming a foundation for multiplication and division.

IV. Key Takeaways

- Counting tells us how many.
- A number is an idea of quantity, while a numeral is the written symbol.
- Each object is counted once, one-to-one rule.
- Numbers follow a sequence, either ascending or descending.
- Grouping and skip counting help us count faster when numbers are large.

LESSON 2:
BASIC OPERATIONS

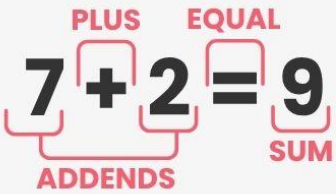
I. Learning Objectives

- Understand the meaning and importance of basic operations in daily life.
- Recognize and apply the symbols for addition (+) and subtraction (−).
- Solve simple addition and subtraction problems using real-life objects.
- Use repeated addition and subtraction to understand multiplication and division concepts.

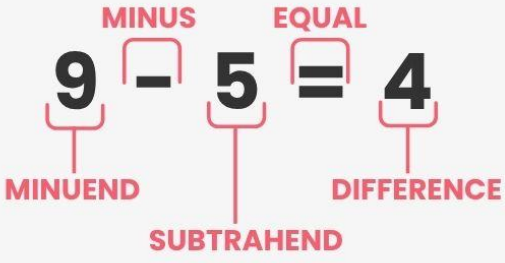
II. Key Concepts

- **Operation**
  - In mathematics, it refers to the processes or rules that combine two or more inputs to produce a single output.

- **Addition**
  - It is the process of combining two or more numbers to find their total.
    - **Addends** are the values that are added.
    - **Plus (+) sign** is the operation symbol used.
    - **Sum** is the result obtained when we add numbers.



- **Subtraction**
  - It is the process of taking away one number from another, making the total smaller.
    - **Minuend** is the number from which the other number is subtracted.
    - **Subtrahend** is the number which is to be subtracted from the minuend.
    - **Minus (-) sign** is the operation symbol used.
    - **Difference** is the result obtained when we subtract numbers.



REMEMBER:
- Always subtract the smaller number from the larger number to get the correct answer.

- **Repeated Addition**
  - It refers to the process of adding numbers together multiple times.
  - It is a foundational concept for understanding multiplication.

- **Repeated Subtraction**
  - It refers the process of subtracting the same number repeatedly from a large number.
  - It demonstrates the concept of division by breaking a total into equal groups.

III. Understanding the Concept

- **Symbols** and **Words**

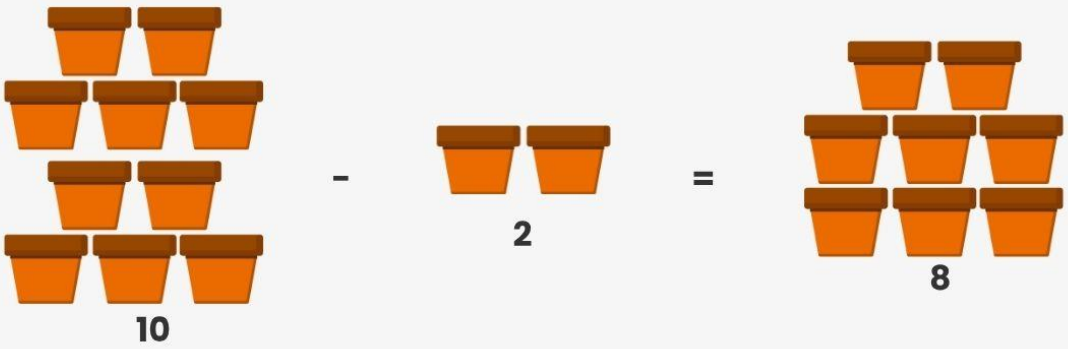
ADDITION	SUBTRACTION
Symbol: **PLUS (+)**	Symbol: **MINUS (-)**
Words: ***plus, add, total, altogether***	Words: ***minus, takeaway, less, left***

- **Addition in Daily Life**
  - When we put together objects, we are adding.

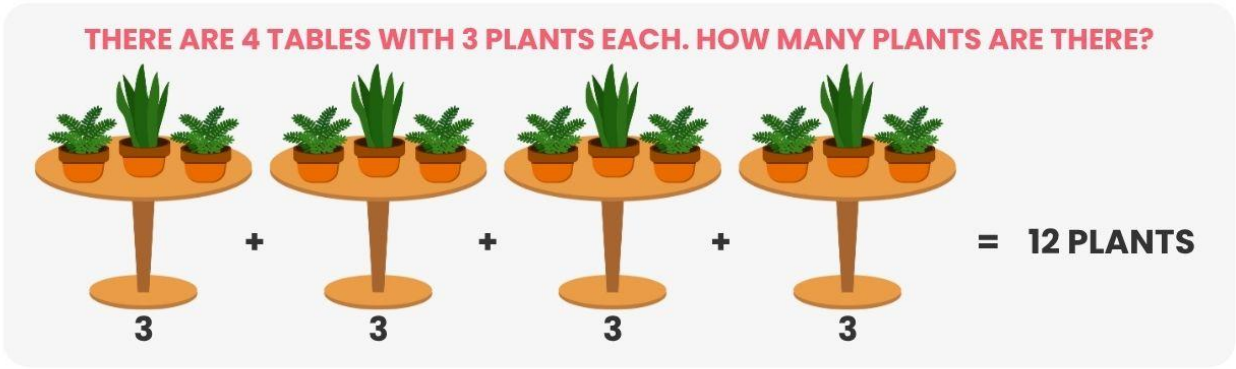
There are **4 plants** on one table and **3 plants** on another. How many plants are there in total?

- **Subtraction in Daily Life**
  - When we remove or take away objects, we are subtracting.

There are **10 pots** but **2** were removed. How many pots are left?



- **Repeated Addition**



- **Repeated Subtraction**

IV. Key Takeaways

- Addition means combining, subtraction means taking away.
- Addition's answer is called "sum"; subtraction's answer is called "difference."
- We use + for addition and − for subtraction.
- Solving mathematical problems can show how math is used in daily life.
- Repeated addition introduces multiplication, while repeated subtraction introduces division.

Math is more than just a subject—it's a functional life skill you will carry forever.

LESSON 3:
MONEY RECOGNITION

I. Learning Objectives

- Identify and recognize Philippine coins and bills of different values.
- Compute the total cost of items in everyday transactions.
- Calculate change correctly when given different amounts of money.
- Apply basic operations in solving money-related problems.
- Use a calculator properly as an option for checking totals and change.

II. Key Concepts

- **Peso (₱)**
  - The Philippine peso, also called by its Filipino name "piso," is the official currency of the Philippines.
    - **Centavo (¢)** or "sentimo" in Filipino, is the official currency used for smaller units.

- **Coin**
  - It is a piece of money used as an official currency, in the Philippines, it is set in peso and centavo.
    - Each coin is a different size, color, and design so it is easy to tell them apart.

- **Bill** or **Banknote**
  - The Philippine paper bill or banknote is a type of paper money used for larger amounts.
    - Like coins, each bill has its own color and design.

- **Total**
  - It is the term used for the entire quantity when all parts are combined into a single value.
    - In some mathematical context, "total" is the same with "sum."

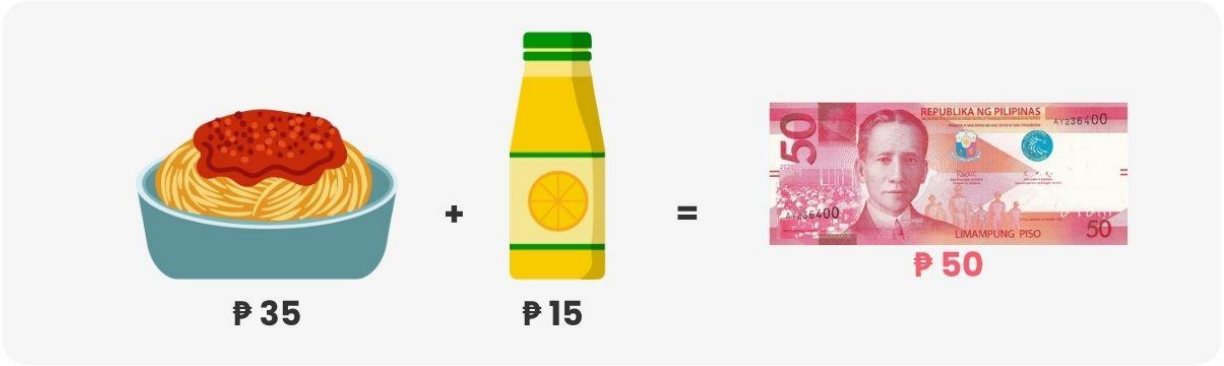
- **Change**
  - It is what you receive after paying more than the total cost.
    - To find it, subtract the total from the amount of money you paid. Making the "change" somehow the same with "difference."

- **Calculator**
  - It is an electronic hardware device or software, capable of performing mathematical calculations.

- **Mental Math**
  - It is a skill that allows individuals to do "math" or calculations "in their head" without using a pen and paper or a calculator.

III. Understanding the Concept

- **Computing the Total**
  - When you buy more than one item, add the prices together to determine the total.



- **Computing the Change**
  - Change is obtained by subtracting the total cost from the amount of money given.

- **Proper Usage of Calculator**
  - The calculator helps confirm the total or change, but students should also practice solving without it to strengthen money-handling skills.

REMEMBER:

- Keep in mind that we should also practice mental math or mental calculation because it builds independence and confidence, particularly in situations where a calculator is not available or necessary, like in markets or jeepney rides.

IV. Key Takeaways

- Recognize coins and bills by their size, color, and value.
- Add prices of items to get the total cost.
- Subtract the total from the money given to find the change.
- A calculator can be a tool for checking answers, but learning to solve without it is a must.
- Money recognition and handling are essential life skills for everyday independence.

Math is more than just a subject—it's a functional life skill you will carry forever.

Accessibility Feature: Questions and instructions can be read aloud (text-to-speech)

Beginner Level: Counting and Recognizing Numbers
- Foundation Skills: Quantity recognition and matching
- Activity Skills: Counting objects, matching quantities to numbers, arranging numbers in order
- Goal: To strengthen the fundamental number sense using pure visual and mental counting

Proficient Level: Basic Operations
- Foundation Skills: Introduction to symbolic mathematics and real-life computation
- Activity Skills: Addition and subtraction in context, mixed addition/subtraction for multiple numbers
- Goal: To teach fundamental number operations

Advanced Level: Money Recognition
- Foundation Skills: Life skill focus
- Activity Skills: Identifying coins and bills, matching amounts to items, computing totals and change, using calculator
- Goal: To apply mathematics to everyday transactions, as well as build independence in money handling and proper use of calculator

Beginner Level:

Opening Skit

Kai is helping prepare for his friend's birthday party. He sees boxes filled with balloons, cupcakes, and chairs that need to be counted.

Narrator: Today is the birthday of Kai's friend! But before the party starts, Kai needs to help count all the things they prepared.

Kai: (looking around) Woah, there are so many things! Balloons, chairs, cupcakes... Where should I start?

Friend: Maybe, you can start by counting them individually, then later we will see if there's a faster way.

Narrator: Let us help Kai count everything for the party.

- INTERACTIVE PROMPT - BUTTON PROCEEDING TO ACTIVITY 1 -

Activity 1: Count It Right Mode of Answer: Click-to-Answer

Instructions: Look at the objects. Count them carefully. Click the number that matches the total.

5 Balloons | 2, 3, **5**
7 Cupcakes | 9, **7,** 6
3 Birthday Cakes | 4, **3,** 6
10 Party Hats | **10,** 7, 8
15 Plastic Cups | **15,** 11, 9

Activity 2: Set the Table Mode of Answer: Drag-and-Drop

Instructions: Move the items to the table. Match the number given.

3 Presents
9 Plates
12 Spoons
2 Cakes
7 Balloons

Activity 3: Counting by Groups Mode of Answer: Encode/Speech

Instructions: Look at the objects. Count how many in each group, then find the total.

20 Balloons in groups of 5 | 5+5+5+5 = 20 Balloons
15 Chairs in groups of 3 | 3+3+3+3+3 = 15 Chairs
25 Cupcakes in groups of 5 | 5+5+5+5+5 = 25 Cupcakes
30 Plates in groups of 10 | 10+10+10 = 30 Plates
50 Party Hats in groups of 10 | 10+10+10+10+10 = 50 Party Hats

Feedback: Correct — Confetti; Wrong — "Try Again" and gentle shake animation (per item if possible or other execution)

Ending Skit

Kai and his friend have counted everything, the tables are set, and the party looks amazing.

Narrator: Thanks to your counting skills and for helping Kai! Everything is set!

Kai: Wow! I learned that I can count one-by-one or even faster by groups.

Friend: Now, we can enjoy the party without missing anything!

Narrator: Great job! You're ready for the next challenge!

- INTERACTIVE PROMPT - BUTTON PROCEEDING TO NEXT LEVEL (OR BACK) -

Proficient Level:

Opening Skit

Kai is helping his classmates in taking care of the classroom plants. Their teacher gives them a list of how many plants each table has and asks Kai to count the total of the items, as well as how many more will be needed.

Narrator: Today, Kai's class is taking care of the plants in their classroom. Along with that, they need to know exactly how many plants they have in total and how many more they need.

Teacher: Kai, can you count the plants from each table and check if we have enough for our project?

Kai: Sure! But, I will need someone's help.

Narrator: Let's start helping Kai!

- INTERACTIVE PROMPT - BUTTON PROCEEDING TO ACTIVITY 1 -

Activity 1: Counting Challenge Mode of Answer: Encode/Speech
Instructions: Read each problem carefully. Solve to find the answer.
There are 4 plants on one table and 3 on another. How many plants are there? 7
There are 10 plants in the window, 2 were removed. How many are left? 8
There are 5 small pots and 6 big pots. How many pots are there? 11
There are 9 plants in the classroom, 4 were given to another class. How many are left? 5
There are 3 cacti and 7 flowers. How many plants are there? 10

Activity 2: Find the Total Mode of Answer: Encode/Speech
Instructions: Look at the objects. You can count them by groups to find the answer.
There are 3 plants on each of 4 tables. How many plants are there? | 3+3+3+3=12
There are 5 seeds in each of 3 pots. How many seeds are there? | 5+5+5=15
There are 16 flowers, each vase holds 4 flowers. How many vases are there? | 4

Activity 3: Think and Solve Mode of Answer: Encode/Speech
Instructions: Read each question. Think carefully and solve the problems.
8 seedlings were planted in the morning, while 5 were planted in the afternoon. How many seedlings were planted? 13
There are 12 plants inside the classroom. If 4 were moved outside, how many are left? 8
A total of 18 leaves were counted. If each plant has 6 leaves, how many plants are there? 3
There are 3 pots on each of 5 shelves. How many pots are there? 15
There are 7 big plants and 7 small plants. How many plants are there? 14

Feedback: Correct — Confetti; Wrong — "Try Again" and gentle shake animation (per item if possible or other execution)

Ending Skit
Kai's class presents their plant project properly, knowing exactly how many plants they have.
Narrator: Thanks to you! Kai's class is ready to present their plant project.
Kai: That's quite a lot of thinking. But at least, we did a great job in organizing our plant project!
Teacher: Good job, Kai!
Narrator: You're onto your next challenge!
-INTERACTIVE PROMPT - BUTTON PROCEEDING TO NEXT LEVEL (OR BACK) -

Advanced Level:
Opening Skit
Kai goes to the school canteen to buy his lunch. He is given money by his mother and is encouraged to pay and count the change himself.
Narrator: It's lunch time and Kai is hungry, so his mother gave him money to buy lunch at the school canteen.
Mother: Here's 100 pesos for your lunch. Can you handle paying and checking your change by yourself?
Kai: I'll try my best! My money will be safe with me!
Narrator: Let's join Kai as he learns how to handle his money himself.

- INTERACTIVE PROMPT - BUTTON PROCEEDING TO ACTIVITY 1 -

Activity 1: Know Your Money Mode of Answer: Drag-and-drop
Instructions: Drag each bill or coin to its correct value.
₱1 coin ₱1, ₱5, ₱10
₱5 coin ₱1, ₱5, ₱10
₱10 coin ₱1, ₱5, ₱10
₱20 coin ₱1, ₱20, ₱10
₱50 bill ₱1, ₱50, ₱10

Activity 2: Adding Up Your Order Answer: Encode/Speech/Calculator
Instructions: Compute the total of the items. Use the calculator if needed.
Spaghetti ₱35 + **Juice ₱15** = ₱50
Sandwich ₱25 + **Juice ₱15** + **Cookie ₱10** = ₱50
Rice Meal ₱45 + **Juice ₱15** = ₱60
Burger ₱40 + **Fries ₱25** + **Water ₱15** = ₱80
Pancit ₱35 + **Banana ₱10** + **Juice ₱15** = ₱60

Activity 3: Count the Change Answer: Encode/Speech/Calculator
Instructions: Solve the correct change. Use the calculator if needed.
Total=₱50, Money=₱100, Change=₱50
Total=₱60, Money=₱100, Change=₱40
Total=₱80, Money=₱100, Change=₱20
Total=₱50, Money=₱200, Change=₱150
Total=₱35, Money=₱50, Change=₱15

Feedback: Correct — Confetti; Wrong — "Try Again" and gentle shake animation (per item if possible or other execution)

Ending Skit
Kai returned home after school and shared his lunch experience. He still has some change in his pocket.
Narrator: After school, Kai went home with a smile, holding the change from his lunch money.
Mother: Hello there, Kai! How was school? Did you eat well?
Kai: Yes! I bought spaghetti for 35 pesos, juice for 15 pesos, and cookie for 10 pesos. That's 60 in total, so I got 40 pesos change because you gave me 100 pesos.
Mother: That's great, Kai! You know how to properly handle your money.
Narrator: Kai can now handle money confidently and properly. Not just in the canteen, but anywhere he goes.

- INTERACTIVE PROMPT - BUTTON PROCEEDING TO BACK TO LEVELS OR TRY AGAIN -

- LEARNING WRAPUP BUTTON ENABLED ONCE ALL THE ACITIVITIES WERE COMPLETED -

Epilogue (Learning Wrap-Up): Math in Everyday Life
Scene 1:
·Setting: At the classroom, the mathematics class just ended. Kai and classmates are packing up their things.
Teacher: Great job, everyone! Remember, math is not just all about numbers, it's something you'll use every day.
Kai: (nods, agreeing with their teacher, smiling)

Scene 2:
·Kai will realize that math is everywhere *(thinking of the happenings earlier)*
·Short clip/montage of real-life scenarios

- Shot 1: At home – Kai stands against a wall height chart while his mother marks his height
  Kai: (happy and proud) Wow! It seems like I'm growing taller every day!
- Shot 2: Inside a jeepney – Kai hands money and checks the fare change
  Kai: (nervous but smiling) Glad the change was exact.
- Shot 3: At the canteen – Kai buys snacks and counts change
  Kai: (happy and proud) Yes! I still have 20 pesos left.
- Shot 4: In the classroom – Kai and his friend play scrabble and add up the scores
  Kai: (happy and proud) Cool, I got the highest score!

Scene 3:
·Setting: Near the door, Kai just arrives home. His mother was waiting on the couch.
Mother: Hello, Kai! So, how was your day?
Kai: It was great! I learned that mathematics helps me in so many ways, counting, solving problems, handling my money, and even when I'm just about to eat or play. Math is literally everywhere.
Mother: That's right, Kai! Math is more than just a subject—it's a functional life skill you will carry forever.

FUNCTIONAL MATHEMATICS: AN INTERACTIVE LEARNING AID FOR STUDENTS WITH SPECIAL NEEDS
SCRIPT AND FLOW OF LEARNING MATERIALS (– BASED FROM LPs)

Lesson 1: Counting and Recognizing Numbers
VO: Welcome to Lesson 1: Counting and Recognizing Numbers. In this lesson, we will learn the basics of counting, understand numbers, and arrange them in order. These are the foundation skills we need to become confident in mathematics.

I. Learning Objectives
- Define the basic terminologies used in counting.
- Recognize numbers.
- Count objects correctly.
- Arrange numbers correctly in order, either ascending or descending.
- Use grouping or skip counting for large quantities.

VO: Our goals are simple: to define the basic terminologies used in counting, to recognize numbers and their symbols, to practice counting objects correctly, to arrange numbers in the correct order, and explore faster ways to count large groups of objects using grouping and skip counting.

II.Key Concepts
Counting
- the act of determining the quantity or the exact number of objects in a set or a group.
- it is the cardinal number that tells how many.

Cardinal numbers or also known as natural numbers or cardinals, are numbers that are used for counting; they are numbers that are used for counting, representing how many of something are present.

VO: Counting is the act of determining the quantity or the exact number of objects in a set or a group.
VO: When we count, we use cardinal numbers or also known as natural numbers or cardinals. Those are the numbers that are used for counting that represents how many of something are present. For example, if we have five balloons, we count one, two, three, four, five —so there are five balloons in total.

Number
- it is an arithmetic value used to represent quantity.
- it is a mathematical concept used to count and form the basis of mathematics.

VO: A number is an arithmetic value used to represent quantity. It is also described as a mathematical concept used to count and form the basis of mathematics.
VO: A number is more than just a symbol—it represents quantity. For example, the numeral '5' is a written symbol, but it stands for the idea of five objects. That's why it is important to connect quantities with written numerals because it helps us understand that numbers are not just symbols but representations of real quantities.

One-to-One Correspondence
- it is the ability to match one object with one number and vice versa.
- it is considered as an early math skill that involves the ability to count a group of objects by assigning one number to each object and only counting each object once.

VO: One-to-One Correspondence is an early math skill that refers to the ability to count a group of objects by assigning one number to each object and only counting each object once.
VO: This means that each object gets counted only once. For example, if there are ten cupcakes, we count each cupcake once: one, two, three, until we reach ten. We don't skip, and we don't double count.

Sequence
- it is an ordered list of numbers that follows a specific pattern or rule.

Ascending Order is when numbers are ordered from least to greatest.
Descending Order is when numbers are ordered from greatest to least.

VO: The term "sequence" refers to the ordered list of numbers that follows a specific pattern or rule.
VO: Numbers can be arranged in ascending order, which means from SMALLEST to LARGEST: one, two, three, four, five. Or in descending order, which means from LARGEST to SMALLEST VALUE: five, four, three, two, one.

Grouping
- in terms of counting, grouping is a technique where a collection of objects is organized into smaller, equal sets to simplify the process.

VO: In terms of counting, grouping is a technique where a collection of objects is organized into smaller, equal sets to simplify the process.

Skip Counting
- it refers to the technique of counting by adding the same number each time to the previous number.

VO: and Skip Counting refers to the technique of counting by adding the same number each time to the previous number.

VO: When the numbers are large, counting one-by-one takes a lot of time. By using the strategy of grouping and skip counting, it is faster to determine the total number of objects since it is counted by sets instead of individually, forming a foundation for multiplication and division. For example, there are 20 balloons. If we group them into sets of five, we can skip count: five, ten, fifteen, twenty.

III. Explanation & Discussion

1. Counting Objects
- To count properly and correctly, assign one number to each object.
  e.g. Five ballons are counted as 1, 2, 3, 4, 5

NOTE: Each object must have a unique count; skipping or repeating results in errors.

2.Recognizing Symbols
- Numbers may be single-digit (0-9) or multi-digit (10 and above).

NOTE: It is important to connect quantities with written numerals because it helps us understand that numbers are not just symbols but representations of real quantities.

3.One-to-one Correspondence
- Every item must have one count. Do not skip. Do not double count (1 OBJECT = 1 COUNT)
  e.g. If there are 10 cupcakes, we must count each cupcake exactly once to reach 10

4.Number Sequencing
- It helps in understanding order and comparison.

Ascending means from least to greatest
e.g. 1, 2, 3, 4, 5
Descending means from greatest to least
e.g. 5, 4, 3, 2, 1

5.Grouping and Skip Counting
- When it comes to large quantity, it is easier to count by grouping, which results in skip counting.
  e.g. 20 balloons can be grouped into sets of 5. Skip count by fives → 5, 10, 15, 20

NOTE: By using the strategy of grouping and skip counting, it is faster to determine the total number of objects since it is counted by sets instead of individually, forming a foundation for multiplication and division.

IV.Key Takeaways
- Counting tells us how many.
- A number is an idea of quantity, while a numeral is the written symbol.
- Each object is counted once, one-to-one rule.
- Numbers follow a sequence, either ascending or descending.
- Grouping and skip counting help us count faster when numbers are large.

VO: Excellent! Now, we know how to count objects, recognize numbers, and arrange them in order. And we also learned that grouping and skip counting can help us count large numbers more quickly.

Visualization
Show animations of objects being counted
Display numerals as they appear with objects
Animate ascending and descending order on a number line
Show skip counting

Lesson 2: Basic Operations
VO: Welcome to Lesson 2: Basic Operations. In this lesson, we will explore addition and subtraction, the two most important building blocks of mathematics. These skills help us solve everyday problems.

I.Learning Objectives
- Understand the meaning and importance of basic operations in daily life.
- Recognize and apply the symbols for addition (+) and subtraction (–).
- Solve simple addition and subtraction problems using real-life objects.
- Use repeated addition and subtraction to understand multiplication and division concepts.

VO: By the end of this lesson, you will understand the meaning and importance of basic operations, learn the symbols used for addition and subtraction, solve real-life problems by adding and subtracting, and see how repeated addition and subtraction lead to multiplication and division.

II.Key Concepts
Operation
- it refers to the mathematical processes or rules that combine two or more inputs to produce a single output.

VO: An operation is a mathematical process or rule that combines two or more inputs to produce a single output. For our lesson, we will be focusing on two operations which are addition and subtraction.

Addition
- *it* is the process of combining two or more numbers to find their total.

Addends are the values that are added.
Operation Symbol + plus
Sum is the result obtained when we add numbers.

VO: Addition is the process of combining two or more numbers to find their total, the values that are added are called “addends”, and the result obtained is called the “sum.” For example, four plants on one table plus three on the others equals seven plants. The addends are four and three are the addends, the plus sign is the operation symbol, and seven is the sum.

Subtraction
- it is the process of taking away one number from another, making the total smaller.

Minuend is the number from which the other number is subtracted.
Subtrahend is the number which is to be subtracted from the minuend.
Operation Symbol - minus
Difference is the result obtained when we subtract numbers.

VO: Subtraction is the process of taking away one number from another, making the total smaller. The number from which the other number is subtracted is called the "minuend", the number which is to be subtracted from the minuend is called the "subtrahend", and the result obtained is called the "difference." For example, if we have ten pots and remove two, we are left with eight pots. The minuend is ten, the subtrahend is two, the minus sign is the operation symbol, and eight is the difference.
VO: Remember that in subtraction, we always subtract the smaller number from the larger number to get the correct answer.

VO: Now, let us proceed with repeated addition and repeated subtraction.

Repeated Addition
- it refers to the process of adding numbers together multiple times.
- it is a foundational concept for understanding multiplication.

VO: Repeated addition is the process of adding numbers together multiple times, gradually introducing the concept of multiplication. For example, if there are three plants on each of four tables, instead of counting one by one, we add: three plus three plus three plus three equals twelve.

Repeated Subtraction
- it refers the process of subtracting the same number repeatedly from a large number.
- it demonstrates the concept of division by breaking a total into equal groups.

VO: On the other hand, repeated subtraction is the process of subtracting the same number repeatedly from a large number, gradually demonstrating the concept of division by breaking a total into equal groups. For example, if we have sixteen flowers and each vase holds four, we keep subtracting groups of four: sixteen minus four minus four minus four minus four, until nothing is left.

III.Explanations & Discussions
Symbols and Words
- Addition uses: + , plus, add, total, altogether
- Subtraction uses: – , minus, take away, less, left

Addition in Daily Life
- When we put together objects, we are adding.
  e.g. 4 plants on one table and 3 on another, 4 + 3 = 7 plants

Subtraction in Daily Life
- When we remove or take away objects, we are subtracting.
  e.g. 10 pots, but 2 were removed. 10 – 2 = 8 pots left

Repeated Addition and Subtraction
- Instead of adding one by one, we can group:
  3 plants on each of 4 tables → 3 + 3 + 3 + 3 = 12.

Subtraction can also be grouped:
16 flowers in vases, each vase has 4 → 16 ÷ 4 = 4 vases.

IV.Key Takeaways
- Addition means combining, subtraction means taking away.
- Addition's answer is called the sum; subtraction's answer is the difference.
- We use + for addition and – for subtraction.
- Solving mathematical problems show how math is used in daily life.
- Repeated addition introduces multiplication, while repeated subtraction introduces division.

VO: Great Job!, We learned that addition means combining and subtraction means taking away. And when we repeat addition or subtraction, we start to understand multiplication and division, making these primary operations helpful for us in solving real-life problems.

Visualization
Addition: Animate two groups of plants joining into one bigger group.
Subtraction: Show 10 plants, then animate 2 being removed, leaving 8.
Symbols: Show "+" and "–" appearing with real examples.
Repeated Addition: Display 3 plants on each of 4 tables, highlight the grouping.
Word Problems: Illustrate classrooms, shelves, pots, and plants to represent each example.

Lesson 3: Money Recognition
VO: Welcome to Lesson 3: Money Recognition. This is one of the most important lessons because it connects math directly to daily life. Here, we will learn how to identify coins and bills, compute totals, calculate change, and practice using a calculator.

I.Learning Objectives
- Identify and recognize Philippine coins and bills of different values.
- Compute the total cost of items in everyday transactions.
- Calculate change correctly when given different amounts of money.
- Apply basic operations in solving money-related problems.
- Use a calculator properly as an option for checking totals and change

VO: Our goals are to recognize Philippine moneys, know how to compute the total cost, learn how to calculate change, apply basic operations in solving money-related problems, and understand when and how to use a calculator.

I.Key Concepts

Peso

- the Philippine peso, also called by its Filipino name “piso,” is the official currency of the Philippines.

VO: The Philippine peso, also called by its Filipino term “piso,” is the official currency of the Philippines, while the smaller units are referred to as “centavo.”

VO: The money comes in two forms which are coins and bills or banknotes.

Coin

- in the Philippines, coin is a piece of money used as official currency in the country, set in pesos and centavos.

VO: Coins are used for smaller amounts. In the Philippines, they are set in peso and centavo. Each coin is a different size, color, and design so it is easy to tell them apart. Nowadays, we commonly use one peso, five pesos, ten pesos, and twenty pesos coins.

Bill

- the Philippine paper bill or banknote is a type of paper money that is made and issued by the Central Bank of the Philippines or Bangko Sentral ng Pilipinas.

VO: Bills or banknotes are type of paper money used for larger amounts. Like coins, each bill has its own color and design. Today, we can see twenty peso, fifty peso, one hundred peso, two hundred peso, five hundred peso, and one thousand peso bills.

Total

- is the term used for the entire quantity when all parts are combined into a single value.
- In some mathematical context, “total” is the same with “sum.”

VO: When you buy items, you need to compute the total by adding the prices together. Total (or also the same with “sum” in some mathematical context) is the term used for the entire quantity when all parts are combined into a single value. For example, if spaghetti costs thirty-five pesos and juice costs fifteen pesos, we add thirty-five plus fifteen to get fifty pesos in total.

VO: Next is counting the change. Change is what you receive after paying more than the total cost. To find it, subtract the total from the amount of money you paid. For example, if you pay one hundred pesos for a meal that costs sixty pesos, your change is forty pesos.

Calculator

- it is an electronic hardware device or software, capable of performing mathematical calculations.

VO: Now, let’s talk about the calculator. It is an electronic hardware device or software, capable of performing mathematical calculations.

VO:
Let’s learn how to use it step by step, for now let’s just focus on addition and subtraction since these are the primary operations for calculating total and change:

For the total or addition:

1. Press the number of the first item’s price. For example, press ‘3’, ‘5’ for thirty-five.
2. Press the plus button (+).
3. Press the number of the second item’s price. For example, press ‘1’, ‘5’ for fifteen.
4. Press the equal button (=).
5. The total, fifty, will appear on the screen.

For the change or subtraction:

1. Press the amount you paid, for example ‘1’, ‘0’, ‘0’ for one hundred.
2. Press the minus button (–).
3. Press the total cost, for example ‘6’, ‘0’ for sixty.
4. Press the equal button (=).
5. The answer, forty, is the change you should receive.

VO: Though using calculator is indeed helpful, it is also important to practice solving without them.

Mental Math

- is a skill that allows individuals to do “math” or calculations “in their head” without using a pen and paper or a calculator.

VO: Keep in mind that we should also practice mental math or mental calculation. It is the skill that allows us to do “math” or calculations “in our head” without using a pen and paper or a calculator. Practicing mental math builds independence and confidence, particularly in situations where a calculator is not available, like in markets or jeepney rides.

III.Explanations & Discussions
Recognizing Coins and Bills
- Coins: ₱1, ₱5, ₱10, and ₱20 are used for smaller purchases.
- Bills: ₱20, ₱50, ₱100, ₱200, ₱500, ₱1000 are used for larger amounts.

NOTE: Each coin and bill have its own design, size, and color for easy recognition.

Computing the Total
- When you buy more than one item, add the prices together to determine the total.
  e.g. Spaghetti ₱35 + Juice ₱15 = ₱50

NOTE: Total can be checked mentally, by writing down, or using a calculator.

Counting the Change
- Change is found by subtracting the total cost from the amount of money given.
  e.g. If the total is ₱60 and you pay ₱100, the change is ₱40

NOTE: Correctly checking the change is important to avoid mistakes in real-life purchases.

Calculator Use
- The calculator helps confirm the total or change, but students should also practice solving without it to strengthen money-handling skills.

NOTE: Though using calculator is indeed helpful, keep in mind that we should also practice mental math or mental calculation to build independence.

IV.Key Takeaways
- Recognize coins and bills by their size, color, and value.
- Add prices of items to get the total cost.
- Subtract the total from the money given to find the change.
- Calculators are tools for checking answers, but learning to count without them builds independence.
- Money recognition and handling are essential life skills for everyday independence.

VO: Well done! We can now recognize coins and bills, add prices to find totals, subtract to find change, and even check with a calculator or mental math. These are life skills we can use anywhere—at the canteen, in stores, or when we are riding a jeepney.

Visualization
Coins and Bills: Show animations of actual coins and bills with their values.
Total: Animate food items with prices being added together on screen.
Change: Show money being given and the correct change being returned.
Calculator: Display how to input simple additions or subtractions on a digital calculator.

FUNCTIONAL MATHEMATICS: AN INTERACTIVE LEARNING AID FOR STUDENTS WITH SPECIAL NEEDS
SCRIPT AND FLOW OF TUTORIAL VIDEOS (– BASED FROM THE FLOW OF THE INTERACTIVE LEARNING AID)

Account Creation and Login:
- In this part, we'll learn how to create an account and log in to the interactive learning aid.
- On the Login Page, click 'Create an Account'. A registration form will appear.
- Fill in your Name, Email, and Password. Make sure to remember your password, because you'll need it every time you log in.
- When you're done, click 'Register'. You'll see a message that says your account has been successfully created.
- Now, click 'Log in Here'. On the text fields, enter your name and password, then click 'Log In'. Once logged in, you will be directed to the dashboard.

Navigating the Dashboard:
- In this part, let's explore your Dashboard. This is your main home page inside the interactive learning aid.
- Once you log in, you'll be greeted with a message that says 'Welcome, [your Name]', just like this. You'll also see your current level, which may be Beginner, Proficient, or Advanced.
- Below that, you'll find your progress bar. It shows how many activities you have completed out of the total. This helps you keep track of your learning journey.
- Under the progress bar, there are five main buttons. Each one will take you to a different section.
  - Start Activity
  - Learning Materials
  - Tutorials
  - Edit Profile; and
  - Settings
- All these buttons are clearly labeled, and you can click any of them to move to that section easily.

'Start Activity' Button:
- Now, let's learn how to start your activities. This is where you'll practice everything you've learned in each lesson.
- After clicking Start Activity from your dashboard, you will be brought to the Choose Your Level screen. Here, you'll see three levels: Beginner, Proficient, and Advanced.
- Each level focuses on a specific skill: Beginner is for Counting and Recognizing Numbers; Proficient is for Basic Operations; and Advanced is for Money Recognition.
- You can only open the next level once you've finished the previous one. Levels that are not yet available will have a lock icon. This helps you follow the lessons in order, step by step.
- After completing all three levels, a special section will unlock called Learning Wrap-Up. This part can only be accessed once everything is finished. It will review all the skills you've learned from Beginner to Advanced, to help you remember and apply them in real life.

Answering Activities:
- In this part, we'll learn how to answer the activities using different modes of answering.
- Each activity comes with clear instructions and a Text-to-Speech Button that can read the directions for you.
- You may encounter different types of answering modes:
  - Click-to-Answer — Simply click the correct answer from the choices.
  - Drag-and-Drop — Drag the item and drop it to its correct place.
  - Type-in/Encode — Type your answer in the blank space.
  - Speech-to-Text — You may speak your answer instead of typing. Make sure your environment is quiet and your speech is clear. Keep in mind that the speech recognition might sometimes be inaccurate or may not detect your answer properly, especially in noisy places or if the speech is unclear.
  - Calculator — For some activities, you may use the built-in calculator to check your answers.
- Choose the mode that fits you and the activity best, also make sure to follow the instructions carefully to complete each item.
- There is also a sidebar here in the corner for you to keep track of your progress in answering the activities.

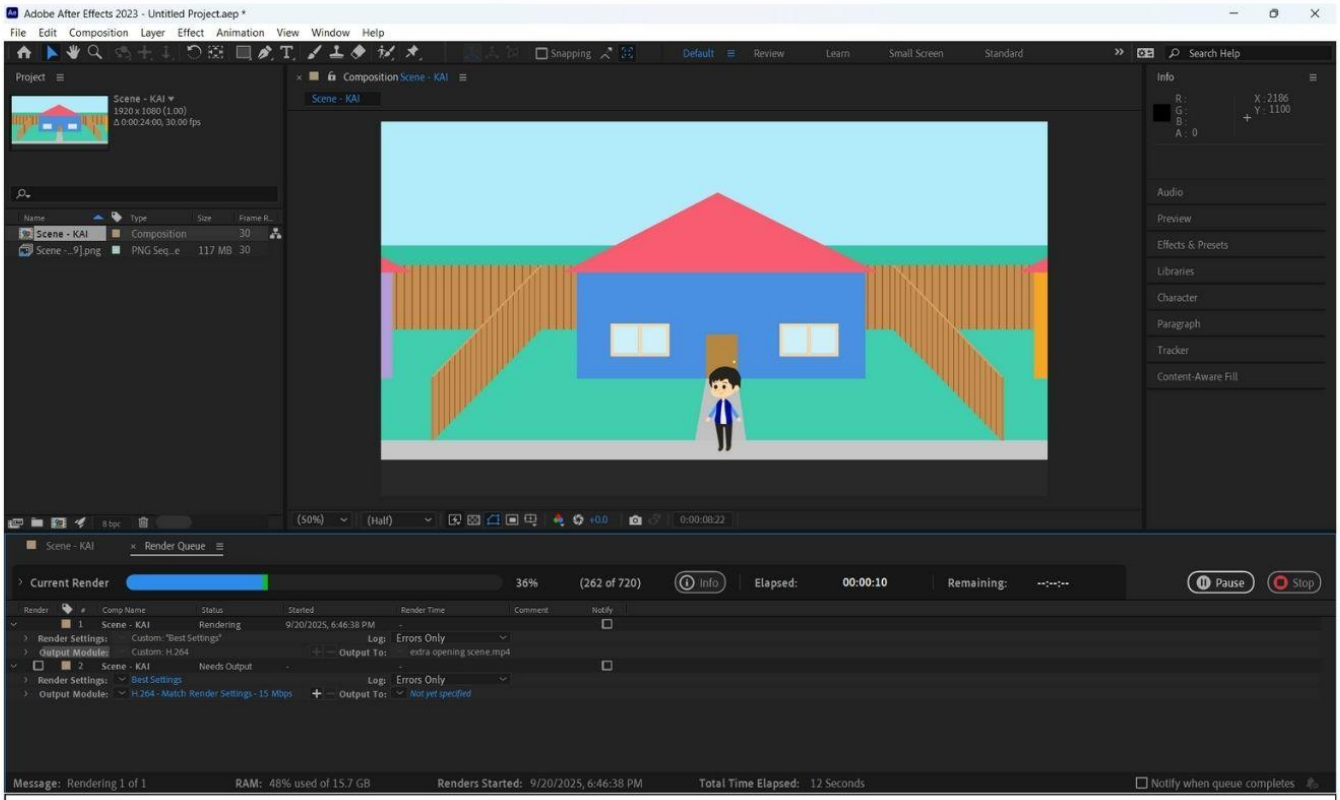
'Learning Materials' Button:
- Now let's explore the Learning Materials section. Here, you can study and review the lessons at your own pace. On your screen, you will see three boxes or buttons: Counting & Recognizing Numbers, Basic Operations, and Money Recognition. Simply click the lesson you want to study.
- Once you click, you'll be directed to the lesson page. On the left side, you'll see the video lesson, and on the right side, there are two buttons: Download Video and Download Notes. You can watch the videos to understand the lesson before answering the activities, and if you want to study offline or review anytime, you can download both the video and the notes.

'Edit Profile' Button:
- Let's learn how to edit your profile. This is where you can update your name, email, or password whenever needed.
- If you want to change your name, simply click the name box, erase the old one, and type your new name.
- You can also update your email. Click the email box and type your new email address carefully.
- If you want to change your password, type your new password in the box. But if you don't want to change it, just leave it blank.
- When you're done, click Save Changes. This will update all the information you entered.
- If you change your mind, click Back to return to the Dashboard without saving.

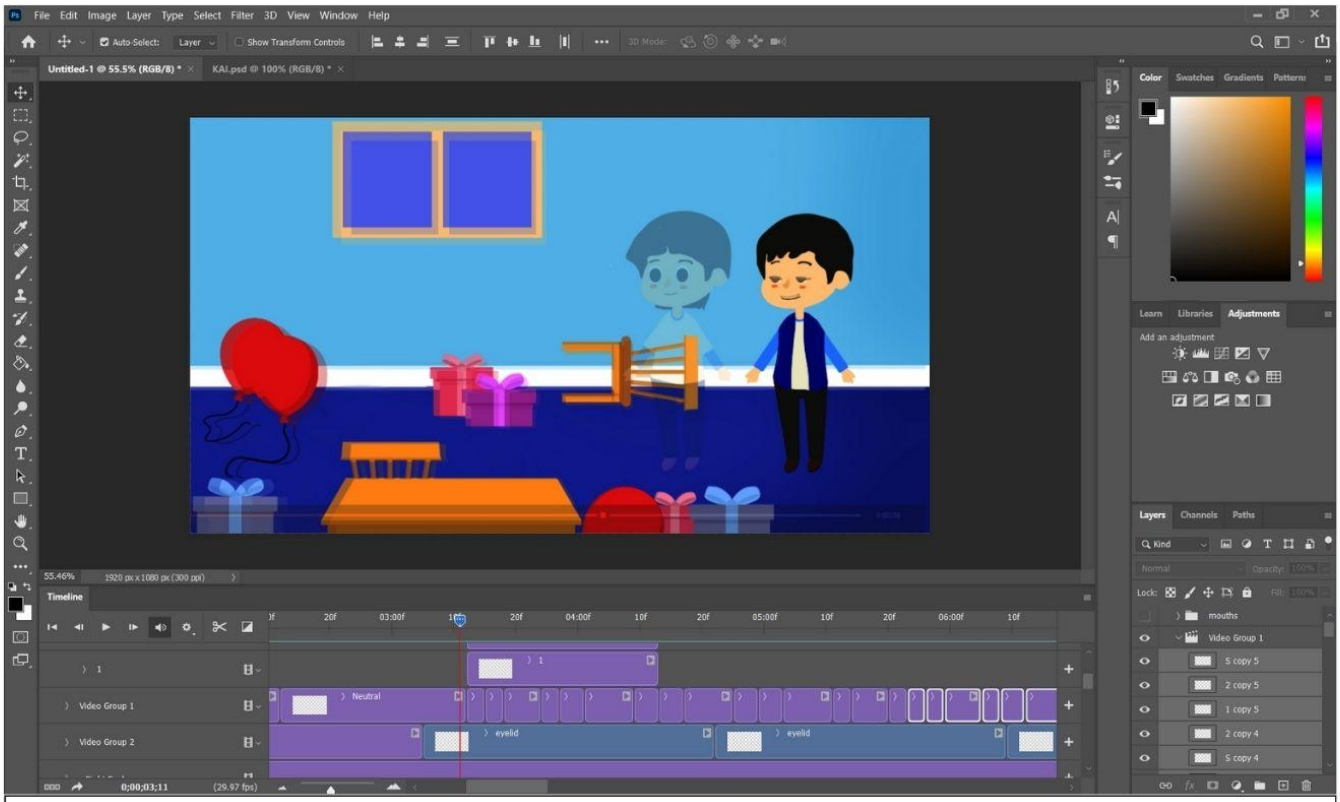
'Settings' Button:
- Let's go to the settings button. This is where you can adjust and manage your account and how the learning aid looks and sounds.
- Next is the Sound button. You can turn the sound on or off by clicking this button, depending on your preference or if your environment is noisy.
- The Delete Account button is used if you want to permanently remove your account. Remember, if you click this, your account and progress will be deleted. So only use it if, you're sure.
- You can click Logout if you're done using the learning aid. This will return you to the login page.
- And finally, click Back to return to the Dashboard without making any changes.

Appendix D

Animation and Interactive Design

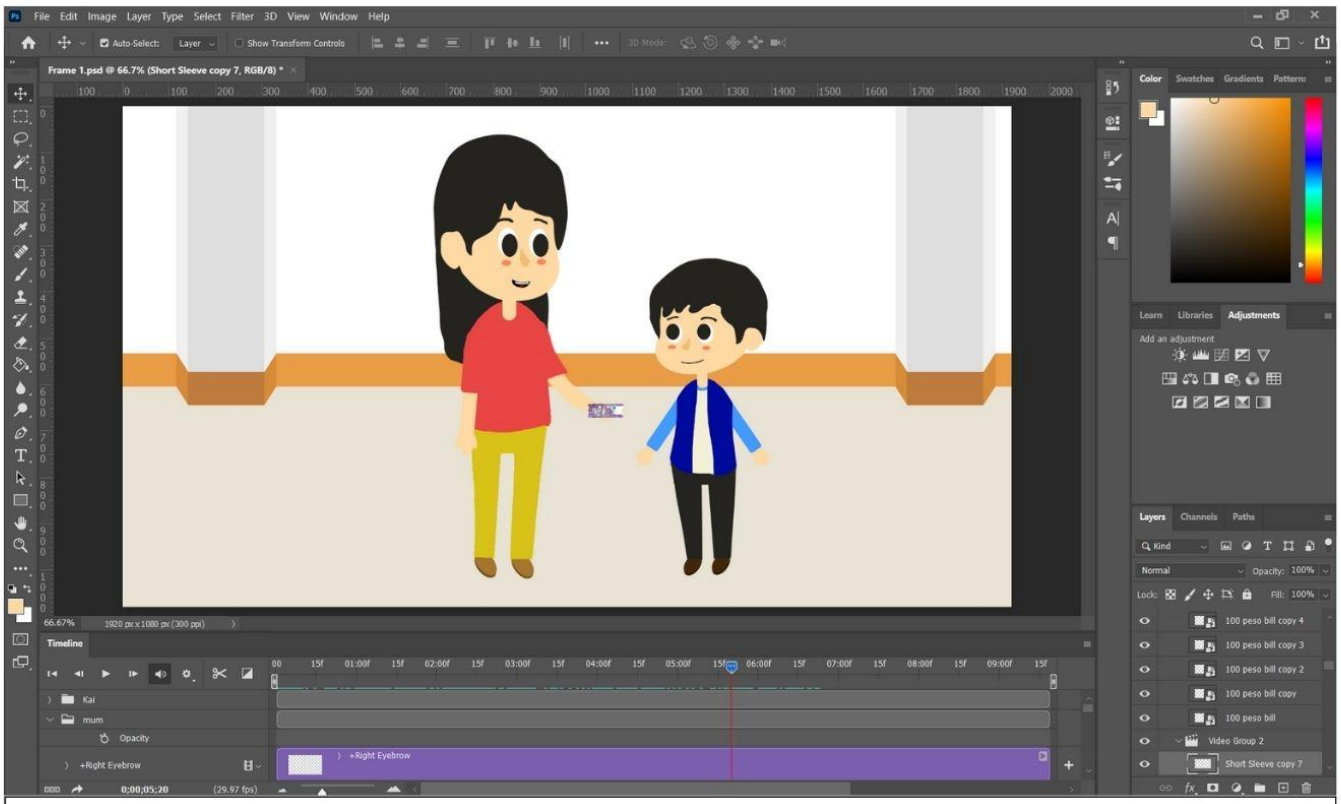


SAMPLE ANIMATION DESIGN 1 RENDERED IN ADOBE AFTER EFFECTS – OPENING SCENE FOR SHORT SKIT

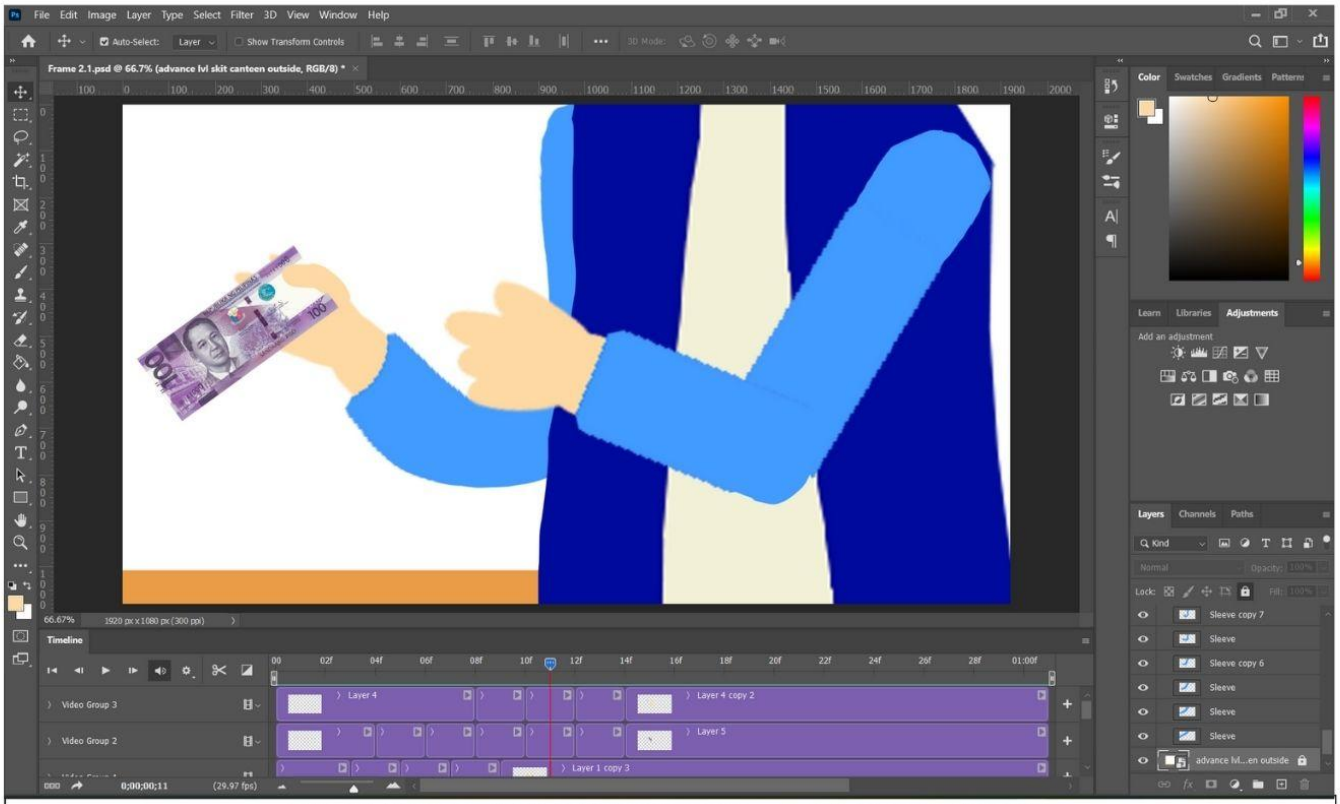


SAMPLE ANIMATION DESIGN 2 IN ADOBE PHOTOSHOP – SAMPLE SCENE FOR SHORT SKIT

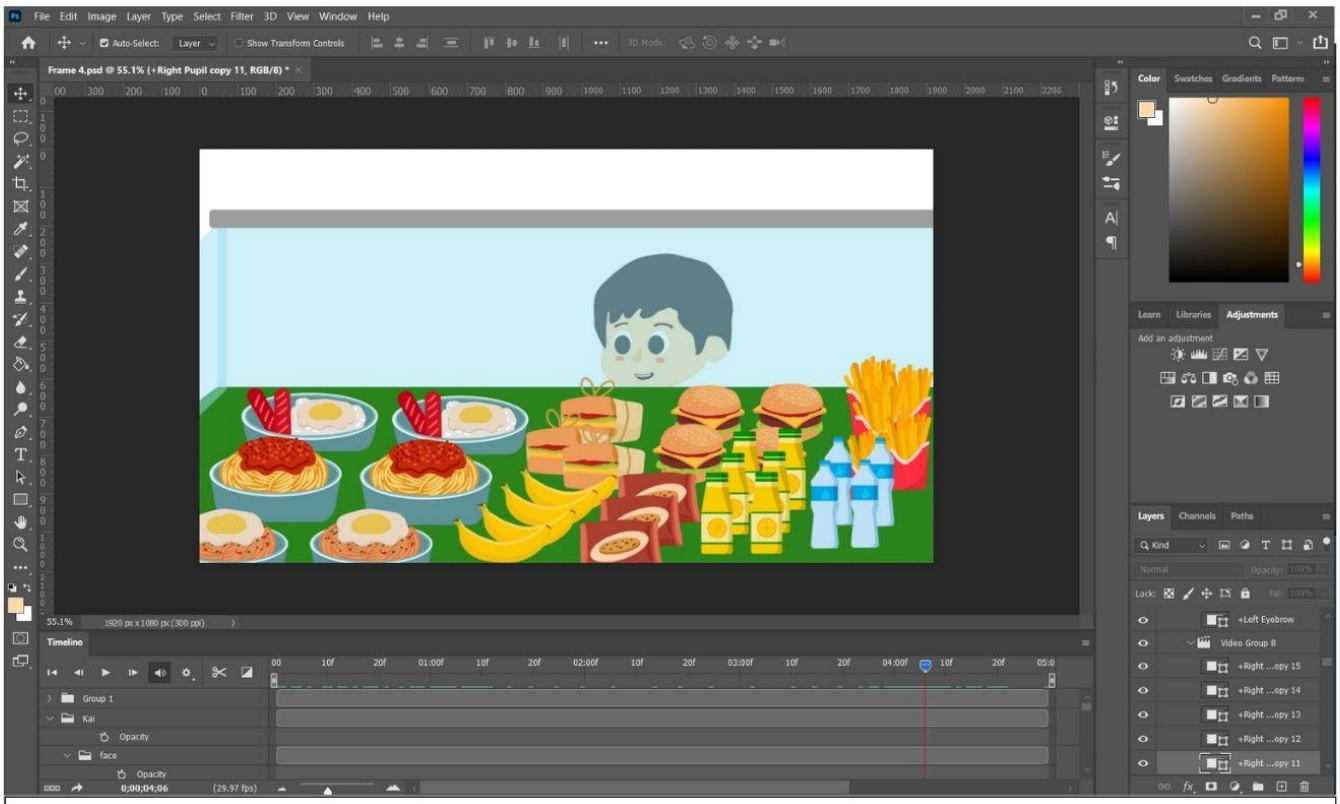
SAMPLE ANIMATION DESIGN 3 IN ADOBE PHOTOSHOP – SAMPLE SCENE FOR SHORT SKIT



SAMPLE ANIMATION DESIGN 4 IN ADOBE PHOTOSHOP – SAMPLE SCENE FOR SHORT SKIT

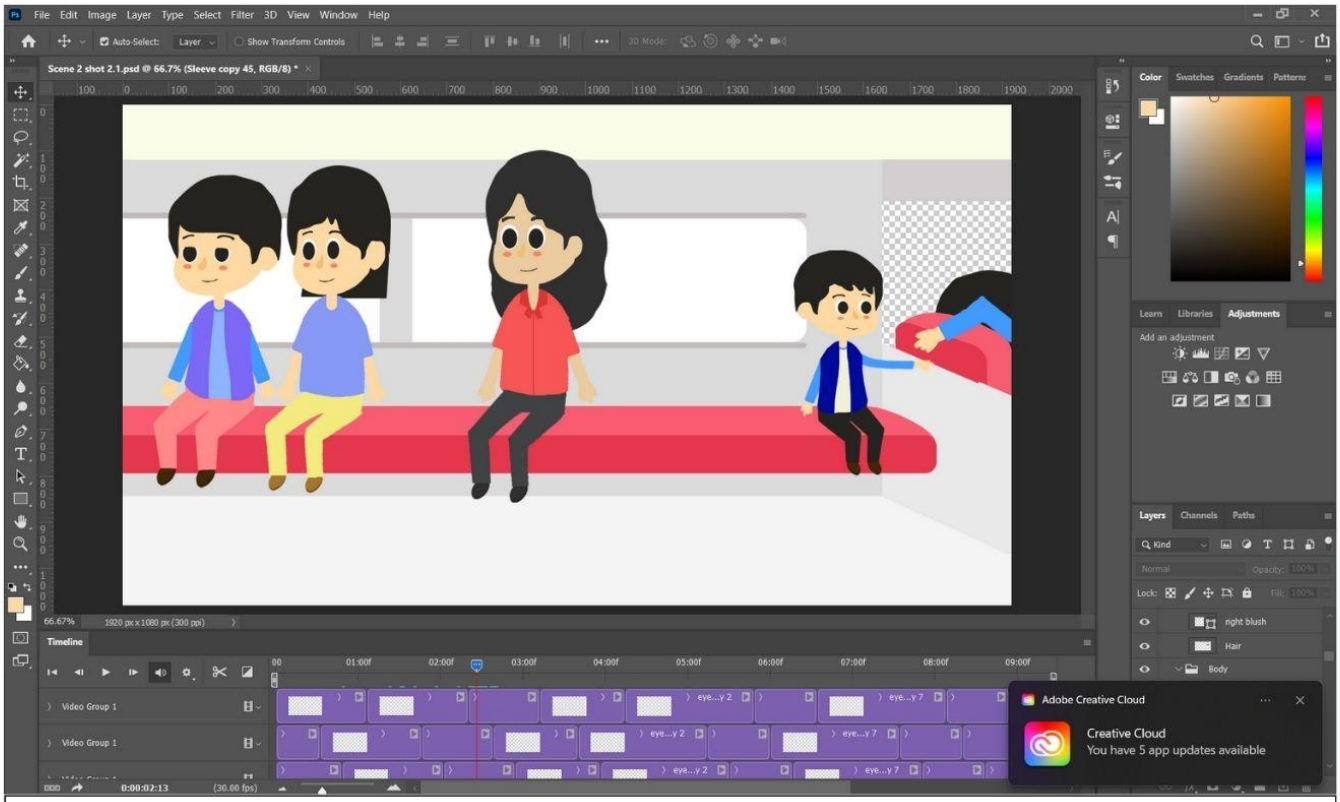


SAMPLE ANIMATION DESIGN 4 IN ADOBE PHOTOSHOP – SAMPLE SCENE FOR SHORT SKIT



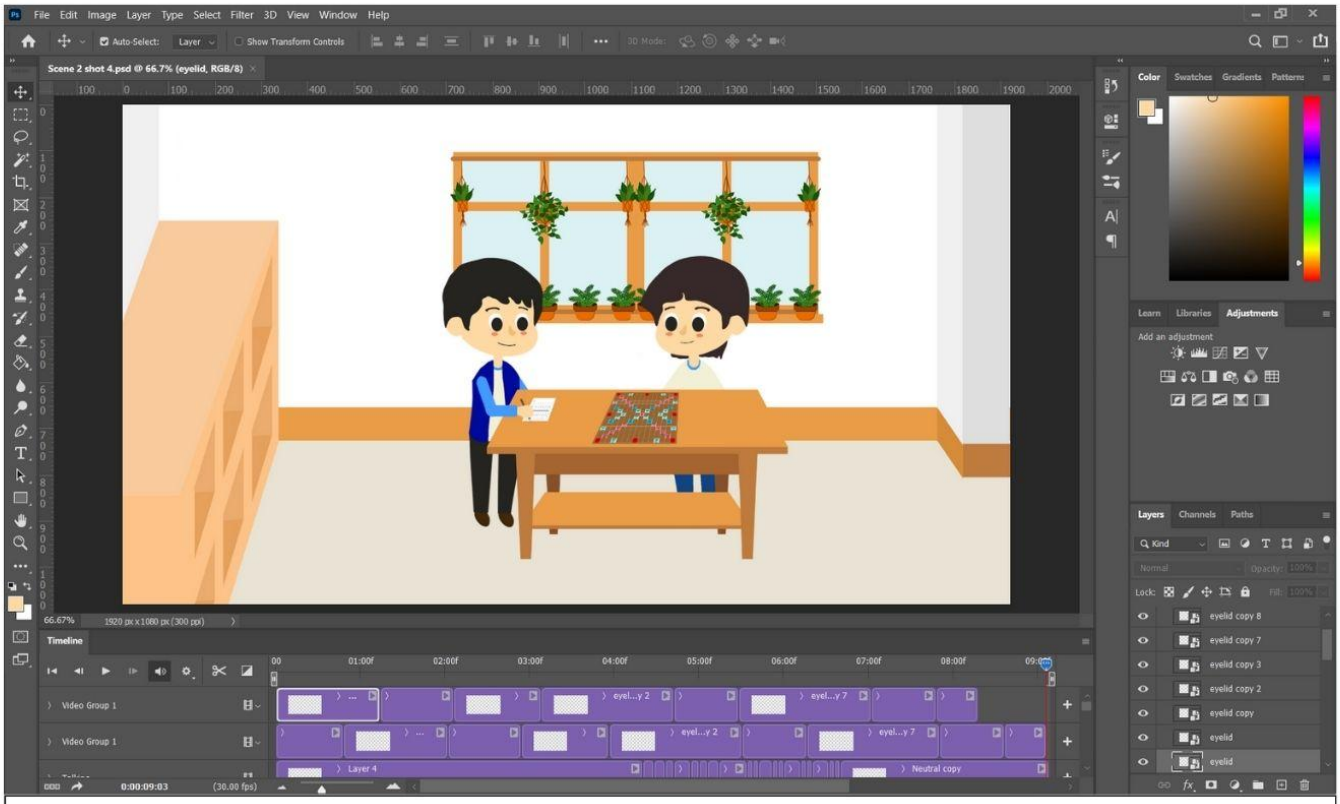
SAMPLE ANIMATION DESIGN 6 IN ADOBE PHOTOSHOP – SAMPLE SCENE FOR SHORT SKIT

SAMPLE ANIMATION DESIGN 7 IN ADOBE PHOTOSHOP – SAMPLE SCENE FOR SHORT SKIT

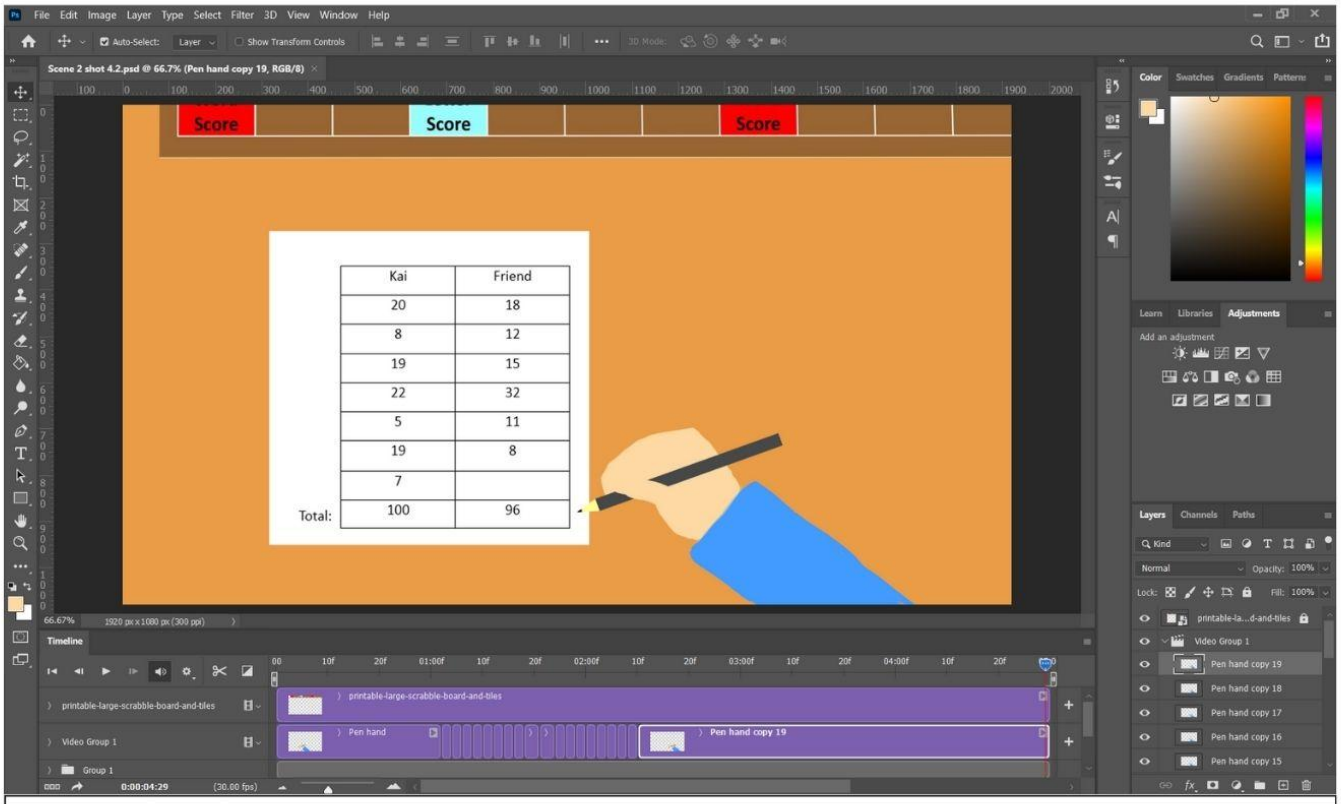


SAMPLE ANIMATION DESIGN 8 IN ADOBE PHOTOSHOP – SAMPLE SCENE FOR SHORT SKIT

SAMPLE ANIMATION DESIGN 9 IN ADOBE PHOTOSHOP – SAMPLE SCENE FOR SHORT SKIT

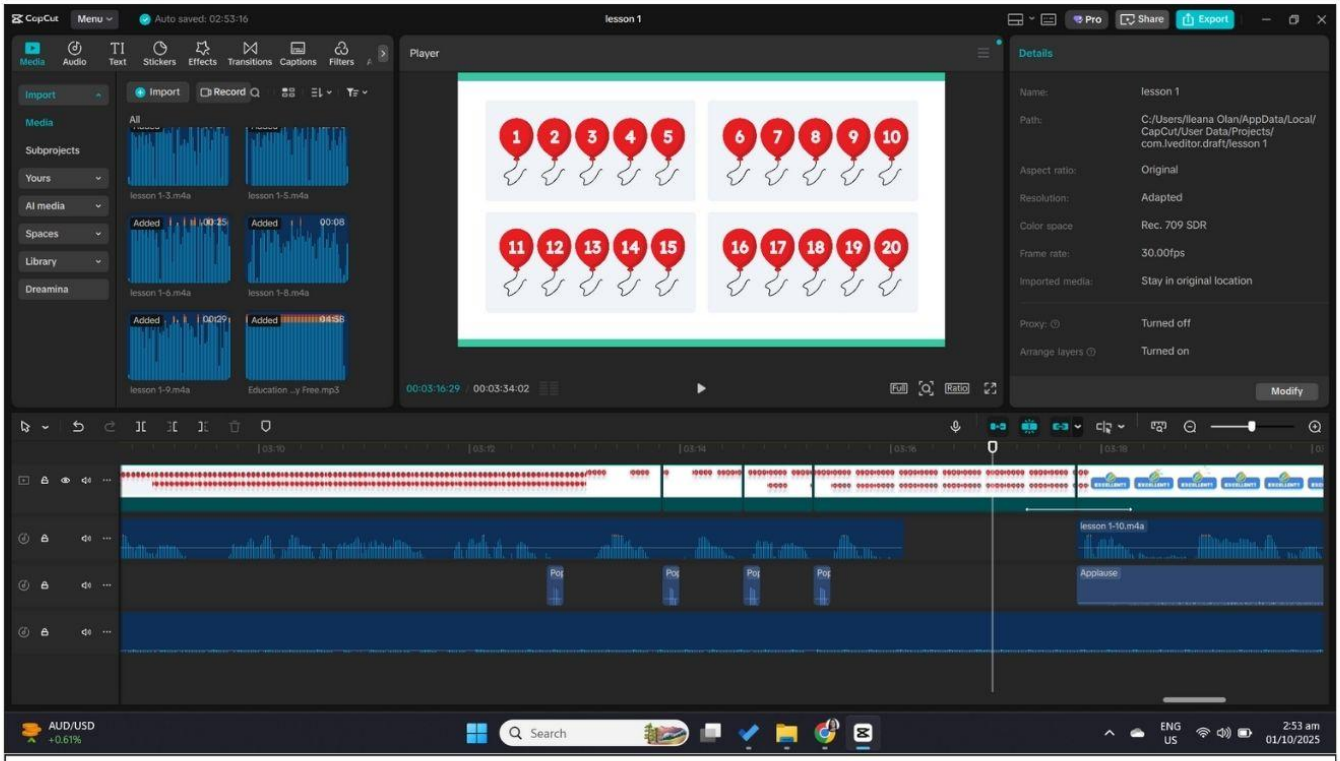


SAMPLE ANIMATION DESIGN 10 IN ADOBE PHOTOSHOP – SAMPLE SCENE FOR SHORT SKIT

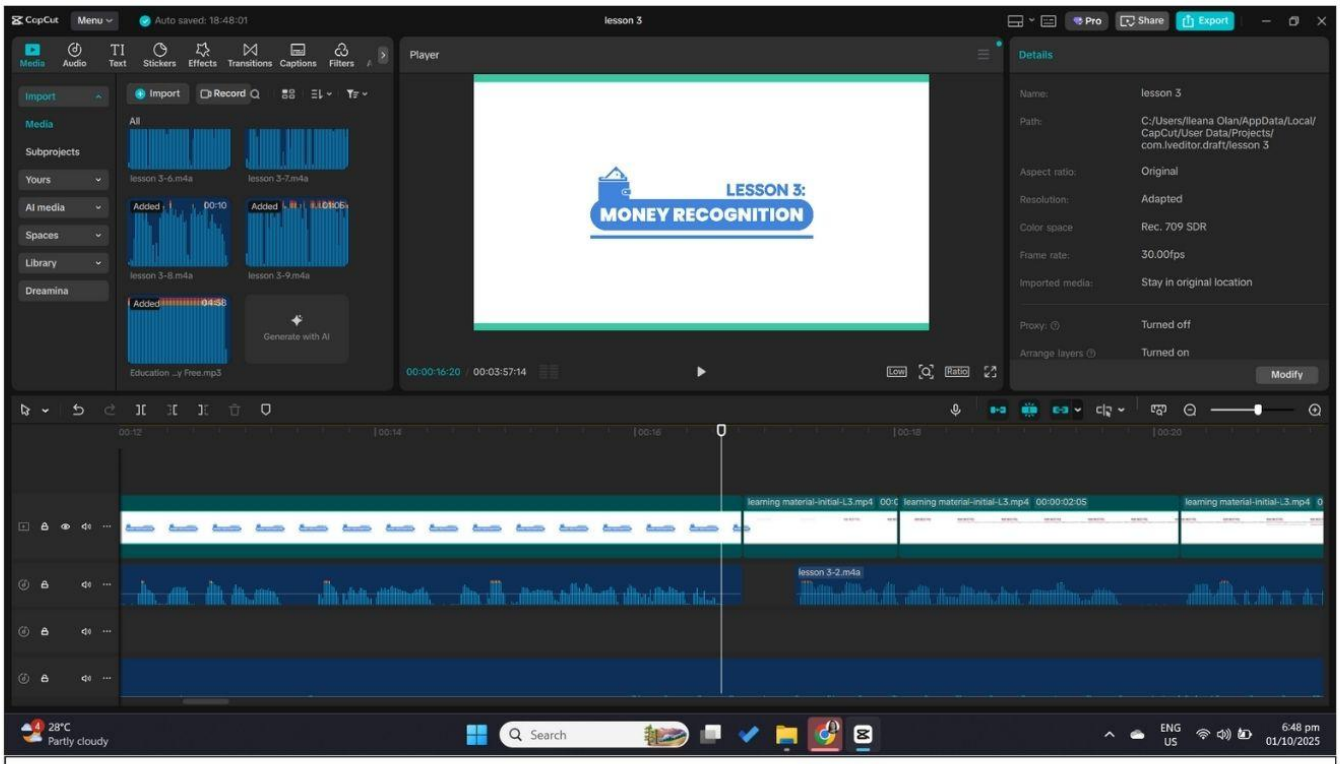


SAMPLE ANIMATION DESIGN 11 IN ADOBE PHOTOSHOP – SAMPLE SCENE FOR SHORT SKIT

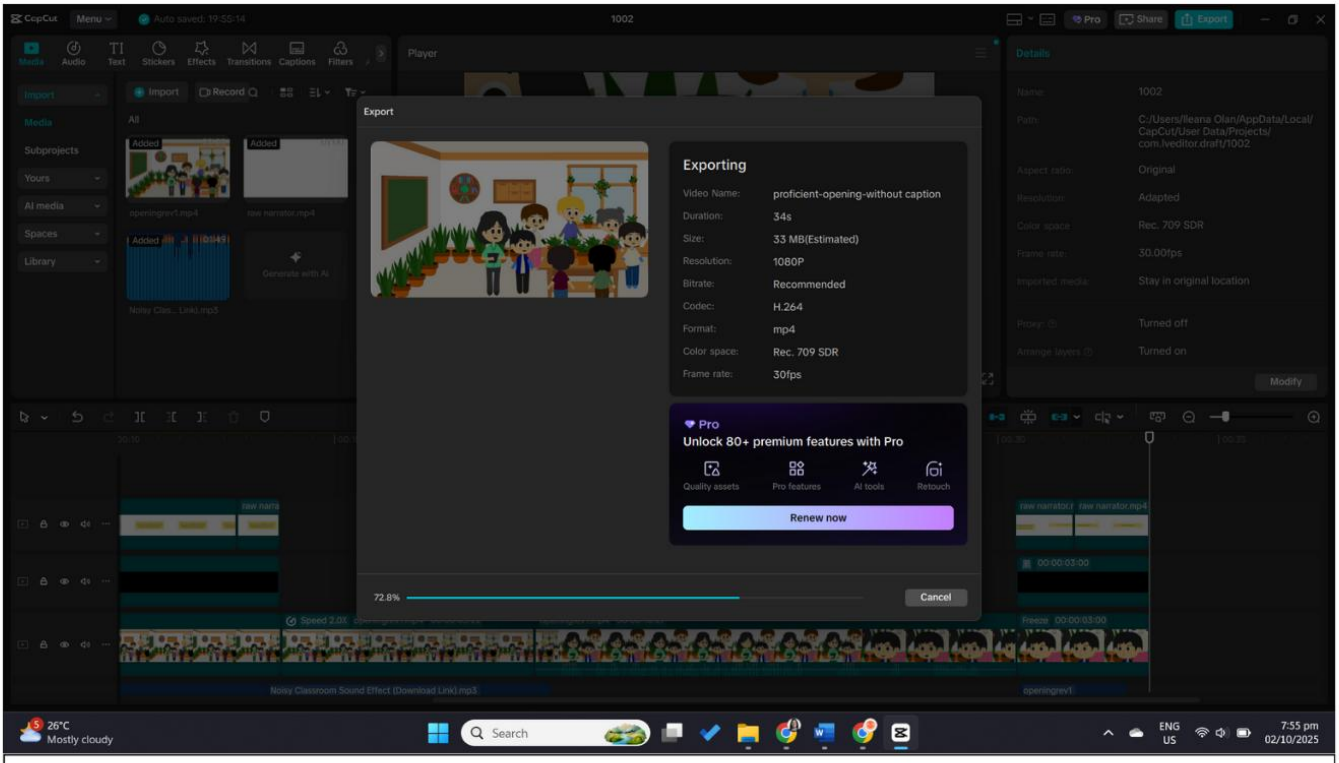
SAMPLE ANIMATION DESIGN 12 IN ADOBE PHOTOSHOP – SAMPLE SCENE FOR SHORT SKIT



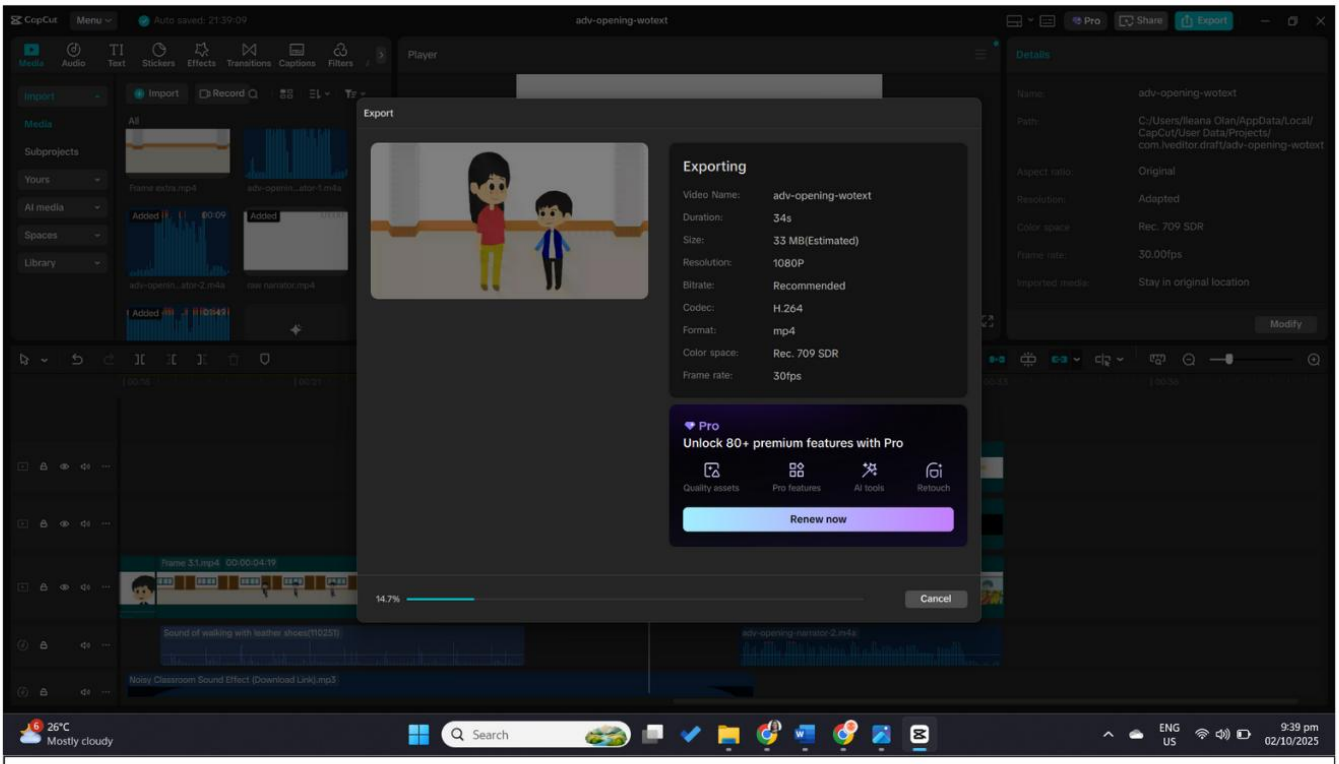
SAMPLE MOTION GRAPHICS & VIDEO EDITING IN CAPCUT PRO 1 – SAMPLE CLIP OF LEARNING MATERIAL



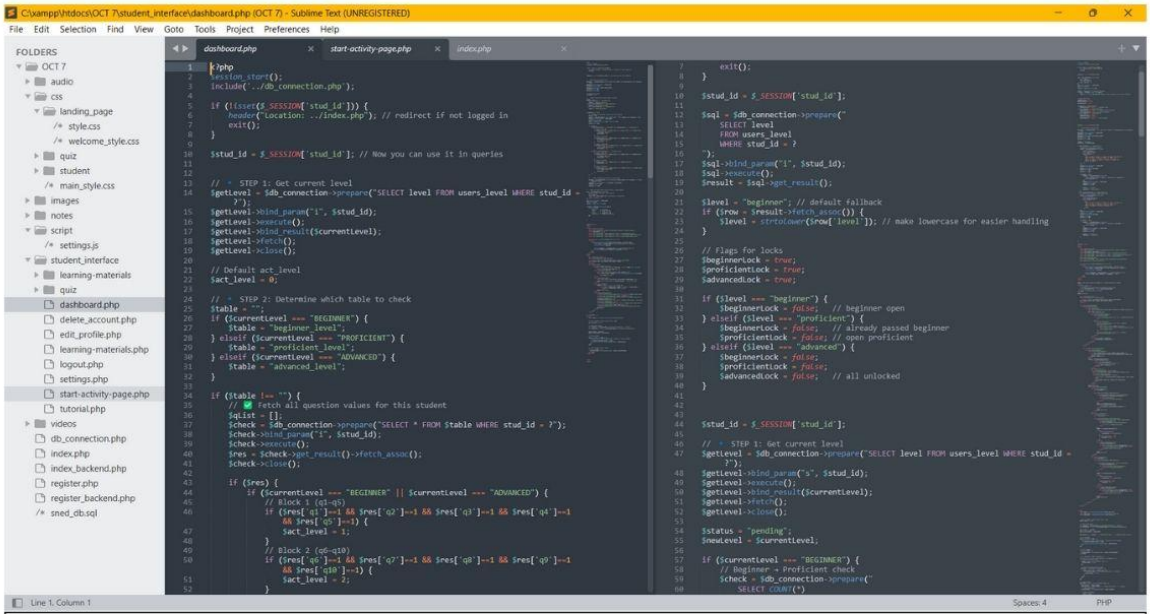
SAMPLE MOTION GRAPHICS & VIDEO EDITING IN CAPCUT PRO 2 – SAMPLE CLIP OF LEARNING MATERIAL



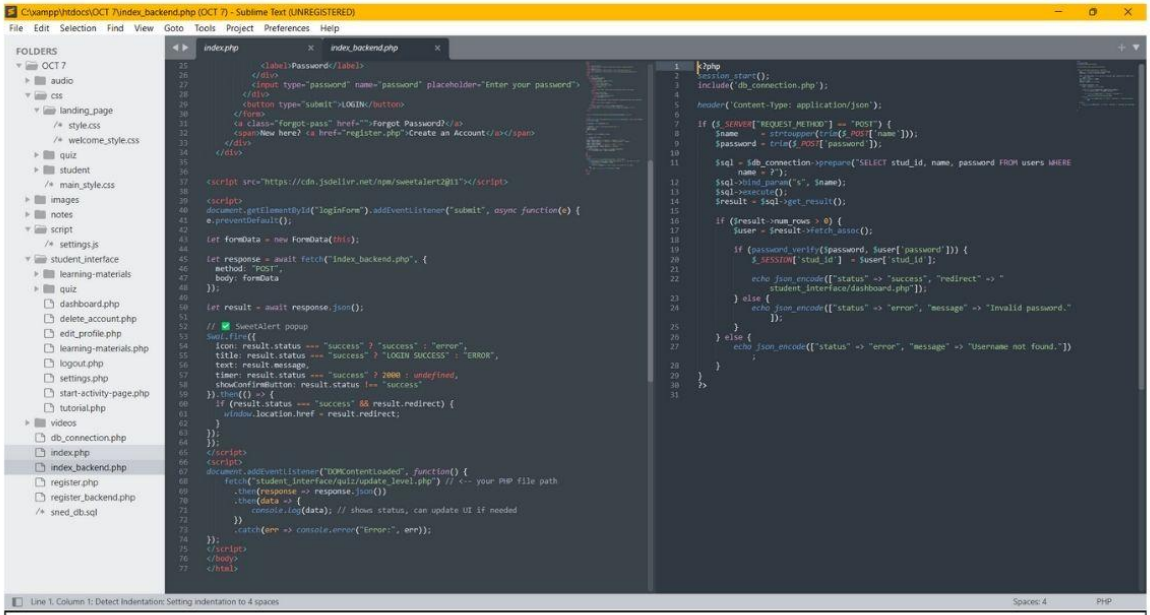
SAMPLE RENDERING AND EXPORTING IN CAPCUT PRO 1



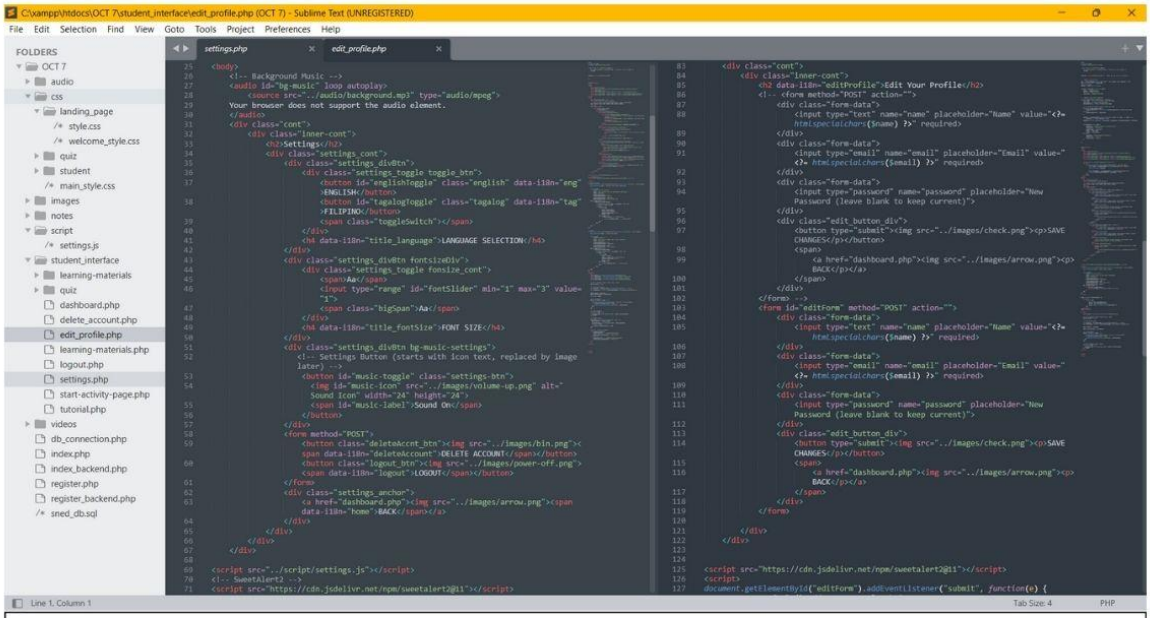
SAMPLE RENDERING AND EXPORTING IN CAPCUT PRO 1



SAMPLE 1 : SUBLIME TEXT EDITOR INTERFACE : DEVELOPMENT OF DESKTOP WEB APPLICATION



SAMPLE 2 : SUBLIME TEXT EDITOR INTERFACE : DEVELOPMENT OF DESKTOP WEB APPLICATION



SAMPLE 3 : SUBLIME TEXT EDITOR INTERFACE : DEVELOPMENT OF DESKTOP WEB APPLICATION

Appendix E

Approval Letters and Parent Consent

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

March 11, 2025

AIDA M. BEJO, Ed.D.
Principal IV
Los Baños National High School-Poblacion
Brgy. Timugan, Los Baños, Laguna

Dear Ma'am Bejo,

I hope this letter finds you well. We are **third-year Bachelor of Science in Information Technology (BSIT) students specializing in Animation and Motion Graphics at Laguna State Polytechnic University – Los Baños Campus.** *As part of our Capstone Project, we are interested in Special Needs Education (SNED) as our research focus.*

In line with this, we would like to request permission to conduct an interview with relevant personnel at your esteemed institution. The purpose of this interview is to gain insights into the teaching methodologies, tools, and challenges faced in Special Needs Education (SNED), which will greatly contribute to the development of our project.

The key points we aim to cover during the interview include:

- Teaching methodologies used in SNED
- Challenges faced by SNED educators
- Technological tools and resources utilized in SNED
- Adaptations and modifications in instructional materials
- Insights on how animation and motion graphics can support SNED learning

We assure you that the interview will be conducted at a convenient date and time for the school, and we will adhere to all ethical and confidentiality guidelines. Should you require any additional information or documentation regarding our study, we would be happy to provide them.

We sincerely hope for your positive response to this request. Kindly let us know a suitable schedule for the interview at your earliest convenience. You may reach us at **ileanaolan17gmail.com** or **0966-603-0416** for any further discussions.

Thank you very much for your time and consideration. We look forward to your kind approval and support.

Sincerely,

ILEANA MARIE P. OLAN

AUBREY MAE R. SUAREZ

JOHN REI L. DOMALANTA

SCHEDULE OF INTERVIEW
MARCH 13, 2025
11 - 12 P.M.

Noted by:

MERARDO ARENAS CAMBA, JR.
Adviser, Capstone Project
LSPU-LBC-CCS
merardo.camba@lspu.edu.ph

Received by:
MARIA ANGELICA MENDOZA
LONHS POBLACION
3/12/2025

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

BAGONG PILIPINAS

October 7, 2025

Ms. Emma L. Izon
Principal IV
Los Baños National High School – Poblacion
Los Baños, Laguna

Dear Ms. Izon,

Good day!

We are fourth-year **BSIT students** from the **College of Computer Studies, Laguna State Polytechnic University – Los Baños Campus**, currently enrolled in the **Capstone Project** course.

We have developed an **Interactive Learning Material Tool** to assist **Students with Special Needs** in learning **Functional Mathematics** through engaging digital activities.

In line with this, we respectfully request permission to **conduct an evaluation** of our developed tool among selected students and teachers at **Los Baños National High School – Poblacion**. The activity will be a **guided evaluation** following school protocols and will be supervised by our research team and designated school personnel.

For transparency, **attached is the Parent/Guardian Consent Form** for distribution to the parents or guardians of participating students.

We sincerely hope for your kind approval of our request. Thank you very much for your time and support.

Respectfully yours,

ILLEANA MARIE P. OLAN.
Group Leader, BSIT-Capstone Project Group, LSPU-LBC-CCS

Noted by:

MERARDO ARENAS CAMBA, JR.
Capstone Project Adviser, LSPU-LBC-CCS

RECEIVED
Date: 10/8/25
LBNHS POBLACION
(049) 536-5573

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

PARENT/GUARDIAN CONSENT FORM

Project Title: *Functional Mathematics: An Interactive Learning Aid for Students with Special Needs* **Researchers:** Ileana Marie P. Olan, Aubrey Mae R. Suarez, John Rei L. Domalanta **Adviser:** Merardo Arenas Camba, Jr.

Purpose

We are *fourth-year students under the program Bachelor of Science in Information Technology specialized in Animation and Motion Graphics from Laguna State Polytechnic University-Los Baños Campus* conducting an evaluation of our developed **Interactive Learning Material Tool** intended to help **Students with Special Needs** learn **Functional Mathematics** through interactive activities.

Evaluation Details

- The activity will be a **guided evaluation**, **facilitated by the students' teacher** and supervised by our research team.
- Selected students will use the learning tool and provide simple feedback.
- The session will not affect their grades or academic standing.

Confidentiality & Voluntary Participation

All information will be kept **confidential** and used **only for academic purposes**. Participation is **voluntary**, and students may withdraw anytime without consequence.

Consent

I, the undersigned parent/guardian, have read and understood the information above and **give my consent** for my child to join the guided evaluation of the interactive learning material under the supervision of the teacher and LSPU Student-researchers.

Name of Student: _____
Grade/Section: _____
Parent/Guardian Name: _____
Signature: _____ **Date:** _____
Contact No.: _____

For School Use Only

Approved by: _____
(Signature over Printed Name of Teacher/Coordinator)
Date: _____

Appendix F

Survey Questionnaire

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
College of Computer Studies

To the Respondent,

We are students of the Laguna State Polytechnic University – Los Baños Campus under the College of Computer Studies conducting research on the project entitled **FUNCTIONAL MATHEMATICS: AN INTERACTIVE LEARNING AID FOR STUDENTS WITH SPECIAL NEEDS.**

We would like to seek your opinion on the acceptability of the said project by scoring the statements below. Rest assured that any information that you provide will be treated with utmost confidentiality (RA 10173. Data Privacy Act) and will be used for research purposes only. Thank you for your cooperation.

Respectfully yours,
Olan, Ileana Marie P. **Suarez, Aubrey Mae R.** **Domalanta, John Rei L.**

Noted by:

Merardo Arenas Camba, Jr.
Capstone Project Adviser

Name *(optional)*: _____ **Date:** _____

Direction: Please rate the level based on your experience on the interactive learning aid (web application) by putting check mark (✓) on the rows column. Use the scale below:
4 – Highly Acceptable ***(Lubos na Katanggap-tanggap)***
3 – Moderately Acceptable ***(Katanggap-tanggap)***
2 – Slightly Acceptable ***(Bahagyang Katanggap-tanggap)***
1 – Not Acceptable ***(Hindi Katanggap-tanggap)***

Technology Acceptance Model (TAM) Questionnaire				
Perceived Usefulness *(Pakikinabang)*	**4**	**3**	**2**	**1**
1. The animations and visuals help students better understand mathematical concepts. ***Nakatutulong ang mga 'animation' at 'visual' upang mas maintindihan ang mga konsepto sa 'Mathematics.'***				
2. The visual elements support the lessons and make explanations clearer. ***Napalilinaw ang pagpapaliwanag ng mga aralin sa tulong ng mga 'visual.'***				
3. The animations make learning more meaningful and engaging for students. ***Sa tulong ng mga 'animation,' nagiging mas makabuluhan at mas nakaka-engganyo ang pag-aaral.***				
4. The visuals help simplify abstract or difficult topics for every learner. ***Mas napapadali ng paggamit ng 'visual' ang mga mahihirap o mga komplikadong paksa para sa mga mag-aaral.***				
5. Overall, animations and visuals are useful for achieving the learning objectives of each lesson. ***Sa kabuuan, nakatutulong ang paggamit mga 'animation' at 'visual' sa pagkamit ng mga itinakdang layunin sa bawat aralin.***				
Perceived Ease of Use *(Kadalian ng Paggamit)*	**4**	**3**	**2**	**1**
1. The animations and visual illustrations are easy to follow and understand. ***Madaling sundan at maintindihan ang mga ginamit na 'animations' at 'visual.'***				
2. The pacing, transitions, and movements are appropriate for every learner. ***Angkop ang nakatalagang bilis ng galaw at paglipat ng mga 'animation' para sa lahat ng mga mag-aaral.***				
3. Text, symbols, and visuals are presented clearly and consistently. ***Malinaw, nasa ayos, at organisado ang pagkakapakita ng mga teksto, simbolo, at 'visual.'***				

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
College of Computer Studies

4. The visuals support comprehension without causing confusion or overload. ***Sa paggamit ng mga 'visual,' mas nauunawaan ang mga aralin nang hindi nakakalito o patong-patong sa isip ng mga mag-aaral.***				
5. Students can easily focus on and interact with the animations without difficulty. ***Madaling makatututok at makasasabay ang mga mag-aaral sa mga 'animation.'***				
Attitude toward Using *(Saloobin o Pananaw sa Paggamit)*	**4**	**3**	**2**	**1**
1. The animations and visuals make the lessons more enjoyable to watch and learn from. ***Mas nakakatuwang panoorin at pag-aralan ang mga aralin dahil sa 'animation' at 'visual.'***				
2. I believe using animations is an effective way to teach functional mathematics. ***Naniniwala ako na ang paggamit ng 'animation' ay isang epektibong paraan sa pagtuturo ng 'Mathematics.'***				
3. I have a positive impression of the animations and visual presentations used in the learning aid. ***Maganda at maayos ang impresyon ko sa 'animation' at 'visual' na presentasyon ng nasabing proyekto.***				
4. I would recommend this type of animated and visual learning approach for other lessons. ***Ire-rekomenda ko ang ganitong paraan at uri ng pag-aaral at pagtuturo para sa ibang aralin kung saan ginagamitan ito ng mga 'animation' and 'visual' na presentasyon.***				
5. The animations make the learning experience feel more dynamic and engaging. ***Dahil sa 'animationm,' nagiging mas masigla at nakaka-engganyo ang pag-aaral.***				
Behavioral Intention to Use *(Intensyon na Gamitin)*	**4**	**3**	**2**	**1**
1. I intend to continue using animated and visual learning materials for teaching/learning mathematics. ***Balak kong ipagpatuloy ang paggamit ng 'animation' at 'visual' sa pagtuturo/pag-aaral ng 'Mathematics.'***				
2. I plan to recommend the use of these animations to other teachers or students. ***Plano kong irekomenda ang paggamit ng mga 'animation' sa ibang guro o mag-aaral.***				
3. I am likely to use these kinds of visual materials in future lessons. ***Mataas ang posibilidad na gagamitin ko ang ganitong klaseng materyales sa mga susunod na aralin.***				
4. I will integrate similar animation-based materials in my teaching/learning strategies. ***Gagamitin ko ang ganitong uri ng 'animation-based' na materyales sa aking paraan ng pagtuturo/pag-aaral.***				
5. I support the continued use of these visuals in classrooms. ***Sinusuportahan ko ang tuloy-tuloy na paggamit ng ganitong uri ng materyales sa klase.***				
Perceived Cost *(Gastos o Konsumo)*	**4**	**3**	**2**	**1**
1. The animations can be used without requiring costly resources. ***Magagamit ang proyekto nang hindi kakailanganin ang malaking gastos.***				
2. The time needed to use or integrate the visuals is manageable. ***Madaling paglaanan ng oras ang paggamit ng 'visual' na presentasyon.***				
3. The benefits of the animations outweigh any cost of production or access. ***Higit na malaki ang benepisyo ng 'animation' kumpara sa gastos para magamit ito.***				
4. Schools can adopt these materials without heavy financial requirements. ***Kayang gamitin ng mga paaralan ang materyales na ito nang hindi maglalabas ng malaking halaga.***				
5. Overall, the animations and visuals are cost-effective educational tools. ***Sa kabuuan, matipid at sulit gamitin ang 'animation' at 'visuals' bilang learning tools.***				
Information Awareness *(Pag-unawa o Kaalaman)*	**4**	**3**	**2**	**1**
1. The animations provide new insights or reinforce important mathematics concepts. ***Nagbibigay ng bagong kaalaman o nagpapatibay ng mga aralin at konsepto sa 'Mathematics' ang mga ginamit na 'animation.'***				
2. The content of the visuals clearly shows what the lesson is about. ***Malinaw na ipinapakita ng 'visuals' kung ano at tungkol saan ang paksa na aaralin.***				
3. The visual materials increased awareness and understanding of mathematical skills used in daily life. ***Mas naunawaan ko ang mga 'mathematical skills' na ginagamit sa totoong buhay dahil sa mga 'visuals' na ginamit.***				

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
College of Computer Studies

4. The animations help clarify common misunderstandings in the topic. ***Natutulungan ng 'animation' na maitama ang mga maling pagkakaunawa tungkol sa paksa.***				
5. I gained a better understanding of the lesson through the visuals and animations. ***Mas naintindihan ko ang mga aralin dahil sa paggamit ng mga 'animation' at 'visuals.'***				
Social Influence *(Impluwensya ng Iba)*	**4**	**3**	**2**	**1**
1. Positive feedback from teachers or peers influences my willingness to use the animations. ***Naeengganyo akong gamitin ang 'animation' (proyekto) kapag may positibong tugon mula sa guro o kapwa mag-aaral.***				
2. I am more inclined to use these materials if others recommend them. ***Mas gusto kong gamitin ang materyales kapag inirerekomenda ito ng iba.***				
3. Endorsement by teachers or schools increases my trust in the animations. ***Mas nagtitiwala ako sa mga nilalaman ng 'animation' kapag ineendorso ito ng mga guro o paaralan.***				
4. Seeing other students engaged with the visuals motivates me to use them too. ***Nae-engganyo akong gamitin ang proyekto kapag nakikita kong ginagamit din ito ng ibang mag-aaral.***				
5. I am more likely to participate in learning activities that use these kinds of visuals. ***Mas mataas ang posibilidad na makilahok ako sa mga gawain kung ganitong uri ng 'visuals' ang ginagamit.***				

_____
Signature of the Respondent

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
College of Computer Studies

To the Respondent,

We are students of the Laguna State Polytechnic University – Los Baños Campus under the College of Computer Studies conducting research on the project entitled **FUNCTIONAL MATHEMATICS: AN INTERACTIVE LEARNING AID FOR STUDENTS WITH SPECIAL NEEDS.**

We would like to seek your opinion on the acceptability of the said project by scoring the statements below. Rest assured that any information that you provide will be treated with utmost confidentiality (RA 10173. Data Privacy Act) and will be used for research purposes only. Thank you for your cooperation.

Respectfully yours,
Olan, Ileana Marie P. **Suarez, Aubrey Mae R.** **Domalanta, John Rei L.**

Noted by:

Merardo Arenas Camba, Jr.
Capstone Project Adviser

Name *(optional)*: _____ **Date:** _____

Direction: Please rate the level based on your experience on the interactive learning aid (web application) by putting check mark (✓) on the rows column. Use the scale below:
4 – Highly Acceptable
3 – Moderately Acceptable
2 – Slightly Acceptable
1 – Not Acceptable

Software Quality Model adapted from ISO 25010				
Functional Suitability	**4**	**3**	**2**	**1**
1. The learning aid provides accurate and appropriate lessons.				
2. All interactive features *(e.g., click-to-answer, speech-to-text, animations)* work as intended.				
3. The learning aid meets the educational objectives of the students.				
Performance Efficiency	**4**	**3**	**2**	**1**
1. The learning aid loads quickly and responds without noticeable delay.				
2. Activities run smoothly on standard school laptops, desktops, and similar devices.				
3. The interactive learning aid performs consistently even with extended use.				
Compatibility	**4**	**3**	**2**	**1**
1. The learning aid works properly across different devices (e.g., laptops, desktops, smart TVs).				
2. The interface and functions remain consistent on different browsers.				
3. The learning aid can be accessed by multiple users simultaneously without issues.				
Interaction Capability	**4**	**3**	**2**	**1**
1. The interface is clear, easy to understand, and accessible for every student.				
2. Instructions and navigation help learners use the web application independently.				
3. Interactive elements are responsive and support student engagement.				

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
College of Computer Studies

Reliability	**4**	**3**	**2**	**1**
1. The learning aid operates without crashes or major errors during use.				
2. Lessons and activities can be completed without data loss or interruptions.				
3. The learning aid remains stable and dependable during extended periods of use.				
Security	**4**	**3**	**2**	**1**
1. User accounts and information (e.g., name, email) are protected within the web application.				
2. Only authorized users can access their profiles and activities.				
3. The web app ensures that personal data is not exposed to unauthorized users.				
Maintainability	**4**	**3**	**2**	**1**
1. The interactive learning aid's structure allows for easy updates or improvements without altering the familiar interface used by students.				
2. Bugs and errors can be fixed without affecting other parts of the learning aid or causing confusion for student users.				
3. The project can be maintained smoothly over time, ensuring stable performance and accessibility for learners.				
Flexibility	**4**	**3**	**2**	**1**
1. The learning aid can adapt to different learning setups *(e.g., classroom, computer lab*) and supports individual pacing and varied teaching strategies.				
2. The web app allows adjustments to meet varying student needs *(e.g., font size, audio instructions, or alternative interaction methods)* or school requirements.				
3. The learning aid remains usable and effective across different environments and teaching contexts without requiring major web application changes.				
Safety	**4**	**3**	**2**	**1**
1. The web-app operates in a way that does not expose learners or devices to harm or inappropriate content.				
2. The learning aid functions properly under defined conditions without compromising data or device security.				
3. The platform promotes a safe and controlled digital learning environment.				

_____
Signature of the Respondent

Appendix G

Level of Acceptance

Appendix H

Publishable Paper

FUNCTIONAL MATHEMATICS: AN INTERACTIVE LEARNING AID FOR STUDENTS WITH SPECIAL NEEDS

John Rei L. Domalanta, Aubrey Mae R. Suarez, Ileana Marie P. Olan
College of Computer Studies, Laguna State Polytechnic University, Los Baños, Laguna, Philippines

Article Info:

Article History:
Received by March 2025
Revised by November 2025
Accepted by December 2025

Keywords:
Functional Mathematics, Special Needs Education (SNED), Interactive Learning Aid, Animation and Motion Graphics, Speech Recognition, Click-to-Answer, Web-Based Application, Technology Acceptance Model (TAM), ISO 25010

ABSTRACT

This study encompasses the development of a desktop-based web application that will help students develop a deeper understanding of the subject Functional Mathematics through visual learning and interactivity. The project aims to address the learning difficulties of students enrolled in the Special Needs Education (SNED) program by incorporating interactive and accessible features, along with animated skits, into a platform aligned with the Functional Mathematics curriculum. The assessment was conducted using two frameworks which are: the Technology Acceptance Model (TAM) for the evaluation of user acceptability, perceived usefulness, and participation, and the ISO 25010 Software Quality Model for the assessment of system characteristics like functionality, usability, and reliability. In general, the results from teachers and students obtained an acceptability rate of 3.76, which is verbally interpreted as "highly acceptable," meaning the learning aid is both effective and easy to use by the target users. These findings confirm that integrating animation and interactivity enhances comprehension, focus, and retention among learners. This study concludes that the learning aid satisfies educational and software quality standards and is therefore suitable for SNED instruction. It is recommended that future enhancements include extending accessibility for other types of disabilities and consideration of its potential as a standalone mobile application.

Corresponding Author:

Olan, Ileana Marie P.
College of Computer Studies
Laguna State Polytechnic University,
Malinta, Los Baños, Laguna 4030
Email: ileanaolan17@gmail.com

1. PROJECT CONTEXT

Providing quality education for students with special needs in the Philippines has always proved to be a challenging goal despite the initiatives by the government under the program called SPED. Students with special needs face difficulties in the field of Mathematics, specifically in numeracy skills, with only 45% of students with disability completing basic education (National Center for Education Statistics, 2022; UNESCO, 2022). Research shows that interactive learning with animation in multimedia learning environments facilitates learning for more students with learning difficulties; hence, the adoption of technology in special learning applications can be considered effective (Mayer, 2021; UNESCO, 2023).

Nevertheless, in spite of this, the quality of education being provided in the SPED schools is still being influenced (Espeño et al., 2024). The said particular environment has been considered as the setting for the capstone project entitled Functional Mathematics: An Interactive Learning Aid for Students with Special Needs, which tries to come up with an interactive learning aid that focuses on concepts for Functional Mathematics such as concepts for operations and identifying money..

RESEARCH OBJECTIVES

Generally, the capstone project aimed to produce an interactive learning aid that effectively taught Functional Mathematics to Special Needs Education (SNED) students. Specifically, the project sought to:

1. Identify the core competencies in Functional Mathematics essential for SNED students, such as:
    a. Curriculum standards
    b. Learning challenges faced by the students
    c. Skills and knowledge areas that should be addressed
2. Develop an interactive learning aid aligned with the lessons and curriculum requirements of the SNED program for Functional Mathematics.
3. Test and evaluate the effectiveness, efficiency, and user acceptance of the interactive learning aid using the Technology Acceptance Model (TAM) and ISO 25010.

2. THEORETICAL FRAMEWORK OF THE PROJECT

This chapter reviews various existing studies and literature that provide the theoretical foundation for the project. The researchers have gathered relevant concepts and ideas regarding the difficulties faced by students with special needs in learning Functional Mathematics, as well as the potential of interactive learning aids to alleviate comprehension and engagement.

Teaching Philosophy for Special Needs Education (SNED)

Differentiated teaching plays an important role in special education classes as it allows tailoring teaching activities to the needs of students and is supported by collaborative learning spaces (Ugalde et al., 2021; Rasnawati et al., 2019). Educational technologies such as speech-to-text software, screen readers, text-to-speech software, and adaptive learning platforms support differentiated teaching and allow students with intellectual and sensory disabilities to learn flexibly and independently (Benedict et al., 2020). As technology advances, it further supports special education teachings that are student-oriented and provide inclusiveness (Stingo, 2024).

Classification of Students with Special Needs

Statistics on Autism Spectrum Disorder further indicate the presence of the condition based on difficulties in communication, social interaction, behavior, and variability of individual differences among affected persons; (National Institute of Mental Health, 2024). Effective interventions, such as individualized education programs, visual supports, sensory interventions, social skills interventions, and behavior management, have demonstrated their significance in the development of children affected by ASD," (McDougal et al., 2023).

Attention Deficit Hyperactivity Disorder (ADHD) refers to inattention, hyperactivity, and impulsiveness that impact concentration and social functioning (Cleveland Clinic, 2022). Studies point out that interactive and participative learning methodologies like using visual aids, physical exercises, and feedback are highly effective in improving the concentration and learning abilities of children suffering from ADHD (Ugalde et al., 2021).

Teaching Mathematics to Students with Special Needs

Visual-spatial skills also present difficulties to most learners with special needs, and these challenges are more pronounced in mathematics, which involves abstract concepts hard to comprehend. It was observed that the use of interactive multimedia, for example, motion graphics, can improve spatial skills and solve problems (Lei et al., 2020; Lambert et al., 2020). Although autistic persons tend to be more gifted in mathematics, it is imperative to understand that instruction should be presented in properly sequenced, specially designed material considering their psychological needs (Sabaruddin et al., 2020).

Numerical Comprehension of Students with Special Needs

Learners with special needs also have difficulties with-numeracy skills, hence low performance in Mathematics. Research demonstrates the use of interactive learning tools with audio-visual synchronization techniques, kinetic typography technology, and multi-modal learning and teaching to improve learning outcomes and abilities in numeracy and overall academic performance (Chen et al., 2020; Ugalde et al., 2021). In addition to learning tools and technology, customized learning plans incorporating simple language and visual learning and interactive techniques can improve teaching and learning mathematics to learners with special needs (Upa & Mbato, 2020)..

Implementing Interactive Learning Aids in Special Education

Current trends in special education underscore the emerging trend of the use of interactive learning media, simulations, and motion graphics to make learning more engaging and effective (Fajriani et al., 2023). Assistive technologies demonstrate a positive impact on persons with special educational needs (Erdem, 2017). Also, learning aids that promote interaction improve engagement, motivation, concentration, and retention of content by learners who suffer from cognitive and linguistic disorders (Smith and Jones, 2020; Garcia et al., 2021).

Impact of Interactive Learning Aids on Cognitive and Behavioral Growth

Interactive and graphical learning materials are important for providing support to students with special needs because they help improve the skills of language acquisition, expression, receptivity, and socialization (Lee et al., 2022; O'Neill & Topping, 2019; Kucirkova & Cremin, 2020). These materials include animated videos and graphical representations that help simplify complicated information into manageable pieces, provide students with sustained focus, increase academic engagement, and enhance performance (Otero et al., 2020).

SYNTHESIS AND RELEVANCE OF THE STUDY

The literature underlines the challenges faced by students with special needs in understanding Functional Mathematics, as the conventional learning methods fail to satisfy the different learning patterns. It emphasizes the use of interactive learning tools, especially those that use pictorial and animated features, which help in a better understanding of complex concepts in mathematics, thereby increasing numerical proficiency. As a solution to this challenge, the current study suggests the design of an interactive learning tool that incorporates animation and interactivity, which will cater to an effective and efficient learning process among SNED students for Functional Mathematics.

CONCEPTUAL FRAMEWORK

Figure 1 illustrates the conceptual framework developed for designing the interactive learning tool for students with special needs. It not only describes the planning stage of designing the learning content but also the implementation phase of the learning content to ensure the final product is appropriate and efficient in teaching Functional Mathematics.

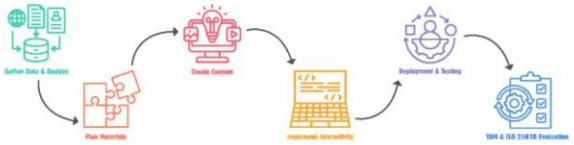


Figure 1. Conceptual Framework

3. METHODOLOGY

This chapter presents the research design to be used in the project, the animation development model, the study population, data collection instruments, and the statistical treatment used throughout the research process.

RESEARCH DESIGN

The learning aid for Functional Mathematics was designed to be interactive and specifically designed to cater to the different learning styles of Special Needs Education (SNED) students. The project will incorporate animated visual elements and interactive features into the learning concept of SNED students and aims to improve engagement, retention, and comprehension in Mathematics.

3.1. Software Development Life Cycle

The Software Development Life Cycle (SDLC) adapted from the sequential model approach (Priya Sr., 2018) guided the development of the interactive educational tool from the initial concept to the final output. This model is disciplined and systematic structure, which is well-suited for producing high-quality animated content and integrating interactive features through a clearly defined and linear development process.

Figure 2. Software Development Life Cycle

Planning

In the planning phase of developing the interactive learning aid, an interview was conducted

with Mrs. Rhea Barboza, a Special Needs Education (SNED) Teacher at Los Baños National High School-Poblacion (LBNHS-Poblacion), to take her input regarding the needs, capabilities, and learning styles of SNED students. In addition, the researchers studied existing literature on SNED and educational technology tools to conceptualize the structure and features of the learning aid. To ensure accessibility, engagement, and appropriateness of learning materials for students with special needs, recent references regarding animation practices and visual tools were also studied.

Defining

After gathering information, the researchers focused on the significant findings that can support the development of the interactive learning aid. The insights obtained from the interview, along with existing studies on special needs education and visual educational tools, were incorporated to serve as the foundation for the project. In addition, based on the consultation with the SNED teacher, it is recommended to concentrate on developing a learning aid that can teach Functional Mathematics to students with special needs.

Designing

The designing phase converts the defined concepts into visual and interactive components. At this stage, character designs, script, storyboard, animations, audio elements, and interactive features are developed to ensure that each individual component of the project is aligned with the needs and capabilities of SNED students.

Character Design

In this phase, the researchers aimed for the characters in the interactive learning aid to be visually engaging, age-appropriate, and relevant to students with special needs. The character designs were digitally completed using iBis Paint, Adobe Illustrator, and Adobe Photoshop to ensure visual consistency and clarity.

Figure 3. Sample Character Design

Scriptwriting

This stage required the creation of scripts for the visual, auditory, and interactive components of the learning support. Using the needs determined in the content analysis, these scripts included simple explanations in a friendly voice, making use of interactions and feedback to ensure alignment with visual components, objectives, and engagement with the target audience.

Storyboard and Interface

In this stage, the script was developed into a storyboard design, as well as a user interface, which aided the animation of interactions involving the learning aid. Each module had scene layouts, graphical designs, as well as interactions relevant to lessons, quizzes, and activities. The design of modules facilitated smooth transitions, and engaging content.

Figure 4. Sample Interface

Figure 5. Sample Storyboard

Illustration and Animations

The researchers have managed to create graphical assets and motion effects using software such as iBis Paint, Adobe Illustrator, and Photoshop. This was done in order to produce illustrations that are simple, clear, and expressive. The illustrations were animated through software such as Adobe Animate, Character CC, and After Effects. This is due to the inclusion of skits at the beginning and end of each lecture.

Figure 6. Sample Animation Process

Voiceover and Sound Design

The voice-over and sound design stage improved audio interaction by employing clear, amicable, and age-friendly narration during lessons, skits, quizzes, and feedback. Background music and audio effects were included to enhance immersion by combining audio with visuals during SNED learning sessions.

Interactive Features

During this phase, the addition of functional, accessible, and engaging elements to aid the different learning styles in SNED students was the key focus. By applying the skills gained in HTML5 Canvas, JavaScript, and the Web Speech API, the study tool included interactive lessons incorporating clickable

options for sound and visual feedback. The speech-to-text functionality enabled the students to answer in speech form, benefiting students with motor skills or reading problems.

Figure 7. Interactive Features

Building

In the building stage, all components were incorporated into an integrated SNED educational platform. For Functional Mathematics, there were lessons ranging from beginner to advanced, using animation, quiz questions, voice tasks, and skits on a contextual setting, and these were assisted by tutorials. The graphics involved the use of iBis Paint, Illustrator, and Photoshop, while the animations involved Adobe Animate and After Effects, audio synchronization using CapCut, and interactions facilitated by HTML5 Canvas, JavaScript, and Web Speech API, completed in Sublime Text.

Testing

The Testing Phase made it certain that the effectiveness, efficiency, and user acceptance of the learning material, as well as compatibility with normal class computers, had been achieved. The functionality testing included animations, interactions, and sound. The feedback received from the students and instructors of SNED concerning the usefulness, ease of use, and acceptance of the model came from the Technology Acceptance Model. The ISO 25010 model judged the functional suitability, usability, performance, reliability, and maintainability.

Deployment

The deployment phase involved the application of the online learning aid to the SNED students in Los Baños National High School-Poblacion, with the training designed to suit the beginner, proficient, or advanced level. The learning aid is receptive to common browsers on laptops, desktop computers, or Android devices, requiring little storage space with fewer installation requirements. Three SNED teachers were involved in monitoring the online application.

3.2. Materials

The interactive learning tool is built through the combined use of hardware and software. A graphical tablet was used to translate drawn graphical concepts, while laptops are the primary platforms used to build the tool. The graphics are produced using iBis Paint, Illustrator, and Photoshop. Animations are carried out through Adobe Animate, After Effects, as well as CapCut. The interactive functions, including click objects, as well as speech recognition functionality, can be created through the application of HTML5 Canvas, JavaScript, or Web Speech API. Coding is done through Sublime Text.

Figure 8. Materials Used

3.3. Hardware Specification

Two mid-range laptops with quad-core processors, entry-level graphics cards, and ample RAM functionality were used for multitasking during development. Additionally, there was the use of a pen tablet for detailed digital drawing and animation to ensure quality visual assets that are device-compatible.

Table 1. Hardware Specification of Laptop 1

Laptop 1: **HP 15s**	
Hardware	**Specification**
Processor	AMD Ryzen 5 5500U 2.10 GHz
Memory	8GB
Graphics Processing Unit	AMD Radeon ™ Graphics
SSD	120GB
HDD	1TB
OS Version	Windows 11
System Type	64-bit operating system, x64-based processor

Table 2. Hardware Specification of Laptop 2

Laptop 2: **Asus Vivobook 16**	
Hardware	**Specification**
Processor	12th Gen Intel ® Core ™ i5-1235U 1.30 GHz
Memory	16GB
Graphics Processing Unit	Intel ® Iris ® Xe Graphics
SSD	120GB
HDD	1TB
OS Version	Windows 11
System Type	64-bit operating system, x64-based processor

Table 3. Hardware Specification of Pen Tablet

XPPen Artist 12 Pen Display (2nd Gen)	
Hardware	**Specification**
Work Area	263.23 x 148.07 mm
Display Resolution	1920 x 1080
Color Gamut (typical)	90% NTSC, 127% sRGB, 94% Adobe RGB
Stylus	X3 Elite
Resolution	5080 LPI
Accuracy	±0.5 mm (center), ±1 mm (corner)

3.4. Software Specification

The software specifications defined the project's functional and non-functional requirements, guiding development in the direction of user needs. They provided a guideline necessary for structuring visual elements, animations, and interactive features such as clickable content and speech recognition, making sure that the learning aid turns out to be effective and user-friendly in school environments.

Table 4. Software Specification

Specification	Description
Project Name	Functional Mathematics: An Interactive Learning Aid for Students with Special Needs
Platform	Web-based application, compatible with standard laptops and desktop computers, and mobile devices
Development Tools	iBis Paint, Adobe Illustrator, and Adobe Photoshop: These are the tools used for creating and polishing visual elements, illustrations, and other graphics assets for the project. (Graphic design and enhancement) Adobe Animate, Adobe After Effects, and CapCut: These are used to apply the dynamic sequences and edit and complete the animation to make the educational content engaging. (Animation development and video editing) Sublime Text: Sublime Text was used to build the user interface by incorporating HyperText Markup Language (HTML) and Cascading Style Sheet (CSS). It was also used for developing interactive components by implementing JavaScript and the Web Speech API to allow clickable features and speech recognition functions. (Interactivity and platform design)
Programming Language	JavaScript (interactive features and speech recognition)

EVALUATION PROCEDURE

Sampling Design

For determining acceptability, a survey was conducted on forty students and two teachers of SNED using purposive sampling. This helped in getting feedback from people who are best suited for judging the effectiveness and usability of the learning resource.

Data Gathering Instrument and Validation

A Technology Acceptance Model (TAM), as illustrated in Figure 9, is utilized to ensure the acceptability of the learning aid within the target user group. TAM analyzed the following elements in the learning aid: perceived usefulness, ease of use, attitude, behavior, cost, social influence, and information awareness to understand the efficacy of the tool in supporting learning in the subject Functional Mathematics. Feedback from 40 SNED students has been taken into account.

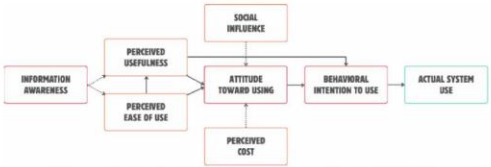


Figure 9. Technology Acceptance Model (TAM)

In Figure 10, the learning aid's effectiveness and sustainability are measured by applying the ISO 25010:2023 Software Quality Model. The software quality model is based on evaluating six characteristics: Functional Suitability, Usability, Performance Efficiency, Reliability, Maintainability, and Compatibility. These characteristics were validated by applying human observations and system test results to determine if it served SNED students' needs. They are based on software quality assessments. Figure 10: ISO 25010:2023 Software Quality Model

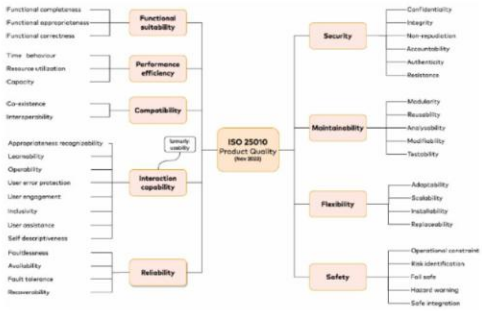


Figure 10. ISO 25010:2023 Software Quality Model

Population of the Study

The project entailed the SNED students of Los Baños National High School-Poblacion, grouped according to their beginner, proficient, and advanced levels, for matching the learning aid with the competencies of the learners. Three SNED teachers also rendered their services to see the appropriateness of the tool for the learners.

Statistical Treatment of Data

Descriptive statistics were used to analyze and summarize data on the overall acceptability of the interactive learning aid. A four-point Likert scale provided an easy avenue for SNED students to provide feedback on the said tool.

Table 5. Likert Scale

Rating	Verbal Interpretation
4	Highly Acceptable
3	Moderately Acceptable
2	Slightly Acceptable
1	Not Acceptable

The data was compiled in Microsoft Excel and further analyzed with the assistance of IBM's SPSS tool. Median functions were utilized for determining the central tendencies, while the reliability of the responses for the factors of both the TAM and ISO 25010 scales was analyzed by Cronbach's Alpha. This would facilitate the assessment of the efficiency and usability of the learning tool for SNED students and faculty.

Table 6. Reliability Test Results for Technology Acceptance Model

Categories	Cronbach's Alpha	Verbal Interpretation
Perceived Usefulness	0.832	Good
Perceived Ease of Use	0.827	Good
Attitude toward Using	0.882	Good
Behavioral Intention to Use	0.711	Acceptable
Perceived Cost	0.749	Acceptable
Information Awareness	0.709	Acceptable
Social Influence	0.831	Good

Legend: above .90 = Excellent, 0.80-0.89 = Good, 0.70-0.79 = Acceptable, 0.60-0.69 = Questionable, 0.50-0.59 = Poor, below 0.50 = Unacceptable

Table 7. Reliability Test Results for ISO 25010 Software Quality Model

Categories	Cronbach's Alpha	Verbal Interpretation
Functional Suitability	0.750	Acceptable
Performance Efficiency	0.750	Acceptable
Compatibility	0.923	Excellent
Interaction Capability	0.857	Good
Reliability	0.857	Good
Security	0.857	Good
Maintainability	0.857	Good
Flexibility	0.808	Good
Safety	0.750	Acceptable

Legend: above .90 = Excellent, 0.80-0.89 = Good, 0.70-0.79 = Acceptable, 0.60-0.69 = Questionable, 0.50-0.59 = Poor, below 0.50 = Unacceptable

Ethical Consideration

Approvals were acquired from the school principal and SNED department, while informed consent was sought from the parents to ensure that participation in this study was on a purely voluntary basis. Ethical issues of respect, equity, and privacy were ensured through the right of the participants to withdraw at any time, as well as giving open responses. These procedures assured the protection of welfare for the SNED students, integrity, and transparency of the investigation.

4. RESULTS AND DISCUSSION

This chapter presents the findings and analysis obtained from the development and evaluation of Functional Mathematics: An Interactive Learning Aid for Students with Special Needs. This section aims to assess the project's effectiveness, usability, and quality using user feedback and a systematic software evaluation. The results and discussions are concluded based on the project objectives, software development process, and software evaluation results.

Results by Objectives of the Study

The capstone project mainly focuses on creating an interactive learning aid that will effectively teach Functional Mathematics to Special Needs Education (SNED) students by integrating animation, motion graphics, visual elements, and interactive components to support their various learning needs and participation. Overall, the objectives were attained through planning, alignment to the curriculum, software creation, and testing in alignment with both the Technology Acceptance Model (TAM) and the ISO 25010 Software Quality Model.

1. **Identify the core competencies in Functional Mathematics essential for SNED students.**

To identify key learning competencies, the researchers interviewed a Special Needs Education (SNED) teacher at Los Baños National High School-Poblacion and analyzed curriculum-based lesson plans. This process highlighted foundational skills, such as number recognition, basic calculations, and money management, as essential for developing both mathematical competency and daily living skills. These competencies were validated through teacher interviews to ensure that the project's instructional goals remained directly aligned with the primary focus of Functional Mathematics for special education learners.

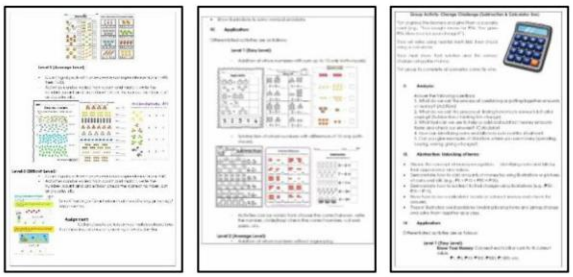


Figure 11. Sample Activities from Curriculum-Based Lesson Plans

2. **Develop an interactive learning aid aligned with the lessons and curriculum requirements of the SNED program for Functional Mathematics.**

Based on the identified competencies, the researchers developed an interactive learning aid utilizing animation and motion graphics as the primary instructional medium. The project was built using HTML5 Canvas, JavaScript, and the Web Speech API to implement interactive features such as click-to-answer, drag-and-drop, speech-to-text, and text-to-speech. Each lesson includes animated video discussions, downloadable PDF notes, and activities tailored to diverse learning styles. Testing confirmed that the application functions effectively on standard hardware, ensuring its feasibility and sustainability for public school settings.

Figure 12. Sample Interface of Interactive Learning Aid

3. **Evaluate and test the effectiveness, efficiency, and user acceptance of the interactive learning aid using the Technology Acceptance Model (TAM) and ISO 25010.**

The researchers used two frameworks, which are the Technology Acceptance Model (TAM) and ISO 25010:2023, in evaluating the overall quality, usability,

and user acceptance of the interactive learning aid. Surveys and testing sessions were conducted with students and teachers, wherein the project gained high levels of acceptance, which indicates that the animations and interactive features effectively support the various learning needs of the learners. In addition, the system met the required software quality standards, which demonstrate proper execution and ease of use on standard devices.

Animated Content Effectiveness Evaluation

Table 8 presents the overall summarized results of the Technology Acceptance Model (TAM) evaluation. With an overall composite mean of 3.97, verbally interpreted as "highly acceptable," the interactive learning aid was well-received by the users, thus underscoring its effectiveness, ease of use, and relevance for an engaging and accessible learning experience.

Table 8. Summary Mean Table of TAM

STATEMENTS	MEAN	SD	VERBAL INTERPRETATION
Perceived Usefulness	3.98	0.10	Highly Acceptable
Perceived Ease of Use	3.95	0.07	Highly Acceptable
Attitude toward Using	3.98	0.08	Highly Acceptable
Behavioral Intention to Use	3.98	0.08	Highly Acceptable
Perceived Cost	3.98	0.11	Highly Acceptable
Information Awareness	3.95	0.21	Highly Acceptable
Social Influence	3.97	0.16	Highly Acceptable
Composite Mean	**3.97**	**0.12**	**Highly Acceptable**

Legend: 1.0-1.75: Not Acceptable; 1.76-2.50: Slightly Acceptable; 2.51-3.25: Moderately Acceptable; 3.26-4.0: Highly Acceptable

Software Quality Evaluation

The quality assessment of the software using the ISO 25010 model is shown in Table 9. The Safety attribute attained the highest mean value of 3.78, "highly acceptable," reflecting a safe learning environment. The attribute Safety, together with the remaining attributes such as Functional Suitability, Performance Efficiency, Interaction Capability, Reliability, Maintainability, and Flexibility, attained a mean score of 3.44 ("highly acceptable") in terms of supporting learning and establishing long-use functionality in accordance with the ISO 25010 model.

Table 9. Summary Mean Table of ISO 25010

STATEMENTS	MEAN	SD	VERBAL INTERPRETATION
Functional Suitability	3.44	0.33	Highly Acceptable
Performance Efficiency	3.44	0.33	Highly Acceptable
Compatibility	3.11	0.24	Moderately Acceptable
Interaction Capability	3.44	0.58	Highly Acceptable
Reliability	3.44	0.58	Highly Acceptable
Security	3.44	0.58	Highly Acceptable
Maintainability	3.44	0.58	Highly Acceptable
Flexibility	3.44	0.33	Highly Acceptable
Safety	3.78	0.33	Highly Acceptable
Composite Mean	**3.44**	**0.43**	**Highly Acceptable**

Legend: 1.0-1.75: Not Acceptable; 1.76-2.50: Slightly Acceptable; 2.51-3.25: Moderately Acceptable; 3.26-4.0: Highly Acceptable

5. SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

In this chapter, the plans, methodologies, developments, testing, and implementations involved in the study will be viewed. Moreover, essential research findings arising from the evaluation sessions by the target group regarding the application, effectiveness, functionality, and sustainability of the tool, Functional Mathematics: An Interactive Learning Aid for Students with Special Needs, will be outlined.

5.1. Summary of Findings

Functional Mathematics: An Interactive Learning Aid for Students with Special Needs was developed to enhance instruction for SNED learners through a systematic process which took into consideration core competencies, curriculum standards, and students' challenges. Feedback from expert teachers and lesson plans based on the curriculum guided the development of animated lessons and interactive activities.

It includes three modules, namely counting and number recognition, basic operations, and money recognition at beginner, proficient, and advanced levels. The interaction styles that can be utilized in the software are drag and drop, Speech-To-Text, and Text-To-Speech, while videos and animated skits are used to reinforce

Based on the ISO 25010:2023 standard and the Technology Acceptance Model, the results with the SNED learners and faculty members of the Los Baños National High School-Poblacion showed a high degree of acceptance that demonstrated the usability and compatibility of the software with the hardware used in the facility. The data indicates that inclusive, animated, and interactive learning content has the ability to provide quality education for the learners.

5.2. Conclusion

The project employed interactive elements, 2D animation techniques, and desktop web technology to provide an optimal learning tool delivery for students at SNED through the creation of an effective learning tool based on Functional Mathematics. The activities and animations were based on SNED curriculum standards to meet content relevance and support students' learning goals.

5.3. Recommendations

On the basis of findings of this research, some recommendations have been framed, which tend to increase efficiency, user friendliness, and learning efficiency of Functional Mathematics: An Interactive Learning Aid for Students with Special Needs. The recommendations are framed in a way so as to improve efficiency, availability, and sustainability of this tool, which helps in creating a efficient learning environment for students with special education needs:

1. **Increased Accessibility Features:** Future improvements may further include among the accessibility features the sign language

interpretation for the benefit of deaf or hard-of-hearing students.

2. **Develop Mobile or Standalone Versions:** The development of a mobile/IoT version and an offline version of this learning aid would ensure that students are free to learn anytime and anywhere, thus enhancing self-learning capabilities.
3. **Regular content review and technical maintenance:** Periodic verification should confirm content integrity, alignment to the curriculum, removal of bugs, and browser or device portability.
4. **Teacher Training:** Workshops/training can prepare teachers on how to properly utilize visual aids, animation, or interactive objects in a classroom while incorporating the learning object into the process.
5. **Expand to Other Subjects:** Future expansion can ensure that the system can be applied to other disciplines for studies in addition to Functional Mathematics.

REFERENCES

[1] Alanazi, A. S., Almulla, A. A., & Khasawneh, M. A. S. (2023). Exploring the E-learning Supporting Systems for Students with Special Needs. Revista de Gestão Social e Ambiental, retrieved from proquest.com: https://www.proquest.com/openview/9bb51e6b5fea7c5bd147313fd1b18f46/1?cbl=2031968&pq-origsite=gscholar

[2] Ballin, A., Davidson, E., Caron, J., & Drago, M. (2022). Making math add up for students receiving special education. International Journal of Whole Schooling, Retrieved from eric.ed,gov: https://eric.ed.gov/?id=EJ1339890

[3] Chen, L., et al. (2020). Peer language resources and language development in inclusive classrooms. Journal of Inclusive Education, 22(4), 345–360.

[4] Cleveland Clinic. (2025). Attention-Deficit/Hyperactivity Disorder. ADHD. Retrieved from https://my.clevelandclinic.org/health/diseases/4784-attention-deficithyperactivity-disorder-adhd

[5] Erdem, R. (2017). Students with Special Educational Needs and Assistive Technologies: A Literature Review. TOJET: The Turkish Online Journal of Educational Technology, retrieved from eric.ed.gov: https://eric.ed.gov/?id=EJ1124910

[6] Espeño, S.R., Babiano, E.H., Bucoy, M.R., et al. (2024). Issues and Challenges of Implementing Special Education (SPED) Curriculum in the Philippines: A Systematic Literature Review. Retrieved from https://edukhasi.org/index.php/jip/article/view/390?utm_source

[7] Fajriani, A., Adnan, A., & Faisal, F. (2023). The effect of interactive multimedia on students' cognitive learning outcomes. Biodidaktika: Jurnal Biologi dan Pembelajarannya, 18(1), 9–23. https://doi.org/10.30870/biodidaktika.v18i1.17808

[8] Gernot Starke. (2023). Arc42 Quality Model: System and Product Quality Made Easy. Retrieved from https://quality.arc42.org/

[9] Harahap, L. W., & Surya, E. (2017). Development of Learning Media in Mathematics for Students with Special Needs. International Journal of Sciences: Basic and Applied Research. Retrieved from https://core.ac.uk/download/pdf/249335803.pdf

[10] International Organization for Standardization. (2023). Systems and Software Engineering — Systems and Software Quality Requirements and Evaluation (SQuaRE) — Product Quality Model. Retrieved from https://www.iso.org/standard/78176.html

[11] James R. Lewis. (2021). Comparison of Four TAM Item Format: Response Option Labels and Order. Retrieved from https://uxpajournal.org/tam-formats-effect-response-labels-order/

[12] Kassim, A., & Nordin, M. N. (2024). AN EFFECTIVE TEACHING AIDS USING VISUAL, AUDITORY AND KINESTHETIC LEARNING STYLES FOR STUDENTS WITH SPECIAL NEEDS. Special Education. Retrieved from: https://prademicpress.com/index.php/specialeducation/article/view/9

[13] Kucirkova, N., & Cremin, T. (2020). Improving engagement in learning through animated tools: A case study in primary education. Journal of Computer-Assisted Learning, 36(4), 489-498. https://doi.org/10.1111/jcal.12402

[14] Lambert, M., et al. (2020). Enhancing mathematical thinking in students with autism. Journal of Special Education Technology, 35(3), 215–229.

[15] Luo, Y., Weibman, D., Halperin, J. M., & Li, X. (2019). A Review of Heterogeneity in Attention Deficit/Hyperactivity Disorder (ADHD). Frontiers in Human Neuroscience, retrieved from: https://www.frontiersin.org/journals/human-neuroscience/articles/10.3389/fnhum.2019.00042/full

[16] Mayer, R. (2024). The Past, Present, and Future of the Cognitive Theory of Multimedia Learning. Educational and Psychology Review. Retrieved from https://link.springer.com/article/10.1007/s10648-023-09842-1#:~:text=Abstract,based%20principles%20of%20multimedia%20design

[17] McDougal, E., Tai, C., Stewart, T. M., Booth, J. N., & Rhodes, S. M. (2023). Understanding and Supporting Attention Deficit Hyperactivity

Disorder (ADHD) in the Primary School Classroom: Perspectives of Children with ADHD and their Teachers. Journal of Autism and Developmental Disorders, retrieved from https://link.springer.com/article/10.1007/s10803-022-05639-3

[18] NIMH. (2024). Autism Spectrum Disorder. National Institute of Mental Health. NIMH Information Resource Center. Retrieved from https://www.nimh.nih.gov/health/topics/autism-spectrum-disorders-asd

[19] O'Neill, J., & Topping, K. (2019). Using visual aids to improve academic performance for students with ADHD: A review. Journal of Special Education Technology, 34(4), 273-285. https://doi.org/10.1177/0162643419849264

[20] Pocaan, J.M. (2022). Exploring Teaching Strategies and Challenges towards a Holistic Context-Based Special Education Teaching Strategies Program. The Normal Lights. Retrieved from https://www.researchgate.net/publication/361487607_Exploring_teaching_strategies_and_challenges_towards_a_holistic_context-based_special_education_teaching_strategies_program_The_Normal_Lights

[21] Priya, Sr. (2018). SDLC – Sequential Model. H2K Information Systems. Retrieved from https://www.h2kinfosys.com/blog/sdlc-sequential-model/?fbclid=IwY2xjawOEhtNleHRuA2FlbQIxMABicmlkETFYek9VVENOUjdTZXpVZnROc3J0YwZhcHBfaWQQMjIyMDM5MTc4ODIwMDg5MgABHp1FIQZwha2A2TpwT_5pHTYZQ_Q-dTLLGXsp951xGphLPjlADDXNMPr3s7Am_aem_qrhPgshFERJckVMJCDeaBw

[22] Rasnawati, A., Rahmawati, W., Akbar, P., & Putra, H. D. (2019). Analisis kemampuan berfikir kreatif matematis siswa SMK pada materi sistem persamaan linier dua variabel (SPLDV) di kota Cimahi. Jurnal Cendekia: Jurnal Pendidikan Matematika, 3(1), 164–177. Retrieved from https://d1wqtxts1xzle7.cloudfront.net/111997966/71-libre.pdf?1709329224=&response-content-disposition=inline%3B+filename%3DAnalisis_Kemampuan_BerfikiraUVxDjGQkOHN1P5RMEJdWb4g__&Key-Pair-Id=APKAJLOHF5GGSLRBV4ZA

[23] Sabaruddin, S., Mansor, R., Rusmar, I., & Husna, F. (2020). Student with Special Needs and Mathematics Learning: A Case Study of an Autistic Student. Journal of Research and Advances in Mathematics Education. Retrieved from https://files.eric.ed.gov/fulltext/EJ1280943.pdf

[24] Sanchhaya Education Private Limited. (2023). Software Development Life Cycle (SDLC). Retrieved from https://www.geeksforgeeks.org/software-engineering/software-development-life-cycle-sdlc/

[25] Smith, A., & Jones, R. (2020). Interactive learning tools for students with cognitive impairments: A longitudinal study. Journal of Learning Disabilities Research, 35(3), 287-305.

[26] Stingo, J. (2024). The Effects of Differentiated Instruction and Individualized Instruction on Special Education Students. Retrieved from https://www.researchgate.net/publication/380392960_The_Effects_of_Differentiated_Instruction_and_Individualized_Instruction_on_Special_Education_Students

[27] Subarna, M. T. N., Masud, N. A., Mensah, J., San, S. S. S., Hasan, M., & Tania, J. S. (2022). Teaching Strategies for Students with Disabilities in Regular Classes. Creative Education, retrieved from https://www.scirp.org/journal/paperinformation?paperid=117809

[28] Ugalde, L., Santiago-Garabieta, M., Villarejo-Carballido, B., & Puigvert, L. (2021). Impact of interactive learning environments on learning and cognitive development of children with special educational needs: A literature review. Frontiers in Psychology, 12, 674033. https://doi.org/10.3389/fpsyg.2021.674033

[29] United Nations Educational, Scientific, and Cultural Organization. UNESCO. (2023). Technology in Education. Good, Impartial Evidence Report. Retrieved from https://gem-report-2023.unesco.org/technology-in-education/

[30] Upa, Y., & Mbato, C. L. (2020). Teacher Identity Construction: Indonesian Teachers' Motivation and Strategies in Teaching Mathematics for Special Needs Students. PROJECT (Professional Journal of Education), retrieved from https://core.ac.uk/download/pdf/323115397.pdf

[31] Wilsa, A. W. (2022). The Effect of Interactive Multimedia-Based Learning on Students' Mathematical Critical Thinking Ability in the Course of Structure Algebra. Jurnal Didaktik Matematika. Retrieved from https://www.researchgate.net/publication/365796415_The_Effect_of_Interactive_Multimedia-Based_Learning_on_Students'_Mathematical_Critical_Thinking_Ability_in_the_Course_of_Structure_Algebra/fulltext/6384abe0c2cb154d2937ae2c/The-Effect-of-Interactive-Multimedia-Based-Learning-on-Students-Mathematical-Critical-Thinking-Ability-in-the-Course-of-Structure-Algebra.pdf?origin=journalDetail

Appendix I

Student Curriculum Vitae

CONTACT

ileanaolan17@gmail.com

0966-603-0416

Mayondon, Los Baños, Laguna

EDUCATION

- **Laguna State Polytechnic University – Los Baños Campus**
  Bachelor of Science in Information Technology major in Animation and Motion Graphics
  2022 - Present
- **Laguna State Polytechnic University – Los Baños Campus**
  Information and Communications Technology
  2020 - 2022

  Los Baños National High School – Poblacion
  Science, Technology, and Engineering Program
  2016 - 2020
- **Malinta Elementary School**
  2010 - 2016

AWARDS

LSPU-LBC | College of Computer Studies
Dean's Lister (2022-2025)

Regional Higher Education Press Conference 2025 | **Head Layout Artist**
5th Place – Best Concept
6th Place – Best Page Design

LSPU-LBC | Senior High School
High Honors (2020-2022)

Inter-Senior High School Acad-Arts Festival, Laguna State Polytechnic University 2022
2nd Place – Essay Writing (English)

Los Baños National High School-Poblacion, Junior High School
With Honors (2016-2020)
Best in Communication Arts (English)
Special Award in Campus Journalism

Ileana Marie P. Olan

Graphic Designer & Layout Artist

PROFILE

To apply my skills in animation, motion graphics, and digital media to contribute to creative and technical projects, while further enhancing my expertise in IT.

With experience as a student leader and publication artist, I aim to bring analytical thinking, adaptability, and teamwork to a dynamic organization that values innovation, collaboration, and growth.

ORGANIZATION/AFFILIATION(S)

2025 - 2026
CACE Publication Representative, CALABARZON Association of College Editors
The student-journalist counterpart of the Association of Tertiary Press Advisers of Southern Tagalog

2024 - 2026
Managing Editor, Bay Breeze Group of Publications
The Official Student Publication of LSPU-Los Baños Campus

2024 - 2026
Chief Graphics and Layout Artist, Bay Breeze Group of Publications
The Official Student Publication of LSPU-Los Baños Campus

2023 - 2024
Junior Graphics and Layout Artist, Bay Breeze Group of Publications
The Official Student Publication of LSPU-Los Baños Campus

2022 - 2023
Graphics Artist and Illustrator, Bay Breeze Group of Publications
The Official Student Publication of LSPU-Los Baños Campus

January 2023
Marketing and Publication Committee, Silakbay, LSPU System
Silakbo ng Bayanihan para sa mga Pag-asa ng Bayan: LSPU Concert for a Cause

SKILL

- Analytical and Logical Thinking
- Problem Solving
- Adaptability and Willingness to Learn
- Teamwork and Collaboration
- Leadership Skills
- Communication Skills
- Journalistic Skills
- Computer Literate

REFERENCES

MRS. DANA GHIA L. CAOAGDAN
School Paper Adviser
LBNHS-Poblacion

MRS. SHINE M. AGUILA
School Paper Adviser
LBNHS-Poblacion

CONTACT

aubreysuarez6@gmail.com

0935-643-2495

Purok 1 Mapagong Calamba City, Laguna

EDUCATION

- **Laguna State Polytechnic University – Los Baños Campus**
  Bachelor of Science in Information Technology major in Animation and Motion Graphics
  2022 - Present
- **Camp Vicente Lim Integrated School - Mayapa Calamba City, Laguna**
  Senior High School
  Accounting Business Management
  2019 - 2021

  Camp Vicente Lim Integrated School - Mayapa Calamba City, Laguna
  Junior High School
  2015- 2019
- **Mapagong Elementary School**
  2009 - 2015

REFERENCES

Ms. Camille Joy A. Ramos
Certified Human Resource Associate - CHRA

Mrs. Camille Jhed Mamasu
Teacher I - Camp Vicente Lim Integrated School

Aubrey Mae R. Suarez

PROFILE

A motivated and enthusiastic IT student seeks an internship to apply technical knowledge and gain practical experience in the technology industry. passionate about problem-solving and learning new things to fully engage in group work.

CERTIFICATIONS

SEPTEMBER 2025

- **Enterprise Design Thinking Practitioner**
- **Cultivate a Digital Mindset**
- **Navigating Ambiguity and Change**
- **Collaboration and Networking**

WORK EXPERIENCE

Service Crew
2021 - 2025
Mcdonald's Batino

Crew Trainer
2023 - 2025
Mcdonald's Batino

SKILL

- Adaptability and Willingness to Learn
- Teamwork and Collaboration
- Computer Literate
- Communication Skills

John Rei L. Domalanta

Visual Artist

CONTACT

johnrei.domalanta@gmail.com

0994-743-8653

Barandal, Calamba, Laguna

PROFILE

To contribute to technical and creative projects using my animation, motion graphics, and digital media talents while developing my IT knowledge.

With my background as an artist, I hope to contribute my critical thinking, flexibility, and inventiveness to a vibrant company that promotes growth, innovation, and teamwork.

EDUCATION

- **Laguna State Polytechnic University – Los Baños Campus**
  Bachelor of Science in Information Technology major in Animation and Motion Graphics
  2022 - Present
- **Calamba City Science Integrated School**
  Science, Technology, Engineering, and Mathematics
  2020 - 2022

  Asian Computer College
  Grade 7 - Grade 10
  2016 - 2020
- **Asian Computer College**
  Grade 3 - Grade 6
  2012 - 2016
- **Lazare Academy**
  Kinder - Grade 2
  2008 - 2012

AWARDS

LSPU-LBC | College of Computer Studies
3rd Place – Quiz Bee

CCSIS | Senior High School
With Honors (2020-2022)

Asian Computer College
With Honors
Scholastic Highest Lexile Growth
Scholastic 100+lexile Growth Awards
Special Citation in Mathematics
Constant in Attendance
Potential in Visual Arts
Loyalty Award

CERTIFICATIONS

SEPTEMBER 2025

- **Enterprise Design Thinking Practitioner**
- **Cultivate a Digital Mindset**
- **Navigating Ambiguity and Change**
- **Collaboration and Networking**

SKILL

- Analytical and Logical Thinking
- Problem Solving
- Adaptability and Willingness to Learn
- Teamwork and Collaboration
- Computer Literate

REFERENCES

MRS. BERNARDITA O. SALAZAR
PRINCIPAL II, Calamba City Science Integrated School

MR. JUAN PAOLO R. GUERRERO
Teacher, Asian Computer College

Appendix J

Grammar Rate

Report: FUNC-MATH-C1-3

FUNC-MATH-C1-3

by Ileana Marie Olan

General metrics

54,292	7,565	513	30 min 15 sec	58 min 11 sec
characters	words	sentences	reading time	speaking time

Score

99

This text scores better than 99% of all texts checked by Grammarly

Writing Issues

12	2	10
Issues left	Critical	Advanced

Report: FUNC-MATH-C4-5

FUNC-MATH-C4-5

by Ileana Marie Olan

General metrics

49,509	6,946	682	27 min 47 sec	53 min 25 sec
characters	words	sentences	reading time	speaking time

Score

98

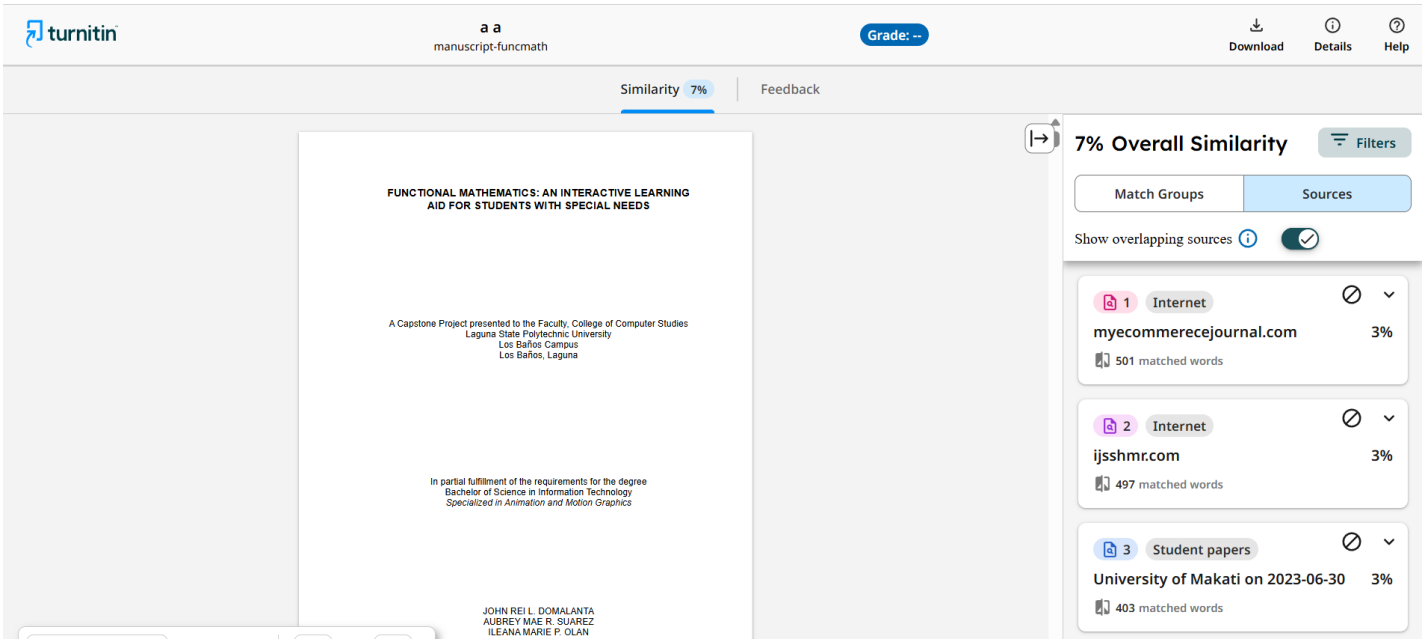
This text scores better than 98% of all texts checked by Grammarly

Writing Issues

38	✓	38
Issues left	Critical	Advanced

Appendix K

Similarity Report


turnitin
a a
manuscript-funcmath
Grade: --
Download
Details
Help
Similarity 7%
Feedback
FUNCTIONAL MATHEMATICS: AN INTERACTIVE LEARNING AID FOR STUDENTS WITH SPECIAL NEEDS
A Capstone Project presented to the Faculty, College of Computer Studies
Laguna State Polytechnic University
Los Baños Campus
Los Baños, Laguna
In partial fulfillment of the requirements for the degree
Bachelor of Science in Information Technology
Specialized in Animation and Motion Graphics
JOHN REI L. DOMALANTA
AUBREY MAE R. SUAREZ
ILEANA MARIE P. OLAN
7% Overall Similarity
Filters
Match Groups
Sources
Show overlapping sources
1 Internet
myecommerecejournal.com
3%
501 matched words
2 Internet
ijsshmr.com
3%
497 matched words
3 Student papers
University of Makati on 2023-06-30
3%
403 matched words